GLENCOE MATHEMATICS

Algebra 2

Chapter 10
Resource Masters

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Consumable Workbooks

Many of the worksheets contained in the Chapter Resource Masters booklets are available as consumable workbooks.

Study Guide and Intervention Workbook	0-07-828029-X
Skills Practice Workbook	0-07-828023-0
Practice Workbook	0-07-828024-9

ANSWERS FOR WORKBOOKS The answers for Chapter 10 of these workbooks can be found in the back of this Chapter Resource Masters booklet.

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Algebra 2
Chapter 10 Resource Masters

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Teacher's Guide to Using the Chapter 10 Resource Masters

The ***Fast File*** Chapter Resource system allows you to conveniently file the resources you use most often. The *Chapter 10 Resource Masters* includes the core materials needed for Chapter 10. These materials include worksheets, extensions, and assessment options. The answers for these pages appear at the back of this booklet.

All of the materials found in this booklet are included for viewing and printing in the *Algebra 2 TeacherWorks* CD-ROM.

Vocabulary Builder Pages vii–viii include a student study tool that presents up to twenty of the key vocabulary terms from the chapter. Students are to record definitions and/or examples for each term. You may suggest that students highlight or star the terms with which they are not familiar.

WHEN TO USE Give these pages to students before beginning Lesson 10-1. Encourage them to add these pages to their Algebra 2 Study Notebook. Remind them to add definitions and examples as they complete each lesson.

Study Guide and Intervention
Each lesson in *Algebra 2* addresses two objectives. There is one Study Guide and Intervention master for each objective.

WHEN TO USE Use these masters as reteaching activities for students who need additional reinforcement. These pages can also be used in conjunction with the Student Edition as an instructional tool for students who have been absent.

Skills Practice There is one master for each lesson. These provide computational practice at a basic level.

WHEN TO USE These masters can be used with students who have weaker mathematics backgrounds or need additional reinforcement.

Practice There is one master for each lesson. These problems more closely follow the structure of the Practice and Apply section of the Student Edition exercises. These exercises are of average difficulty.

WHEN TO USE These provide additional practice options or may be used as homework for second day teaching of the lesson.

Reading to Learn Mathematics
One master is included for each lesson. The first section of each master asks questions about the opening paragraph of the lesson in the Student Edition. Additional questions ask students to interpret the context of and relationships among terms in the lesson. Finally, students are asked to summarize what they have learned using various representation techniques.

WHEN TO USE This master can be used as a study tool when presenting the lesson or as an informal reading assessment after presenting the lesson. It is also a helpful tool for ELL (English Language Learner) students.

Enrichment There is one extension master for each lesson. These activities may extend the concepts in the lesson, offer an historical or multicultural look at the concepts, or widen students' perspectives on the mathematics they are learning. These are not written exclusively for honors students, but are accessible for use with all levels of students.

WHEN TO USE These may be used as extra credit, short-term projects, or as activities for days when class periods are shortened.

Assessment Options

The assessment masters in the *Chapter 10 Resource Masters* offer a wide range of assessment tools for intermediate and final assessment. The following lists describe each assessment master and its intended use.

Chapter Assessment

CHAPTER TESTS

- *Form 1* contains multiple-choice questions and is intended for use with basic level students.
- *Forms 2A and 2B* contain multiple-choice questions aimed at the average level student. These tests are similar in format to offer comparable testing situations.
- *Forms 2C and 2D* are composed of free-response questions aimed at the average level student. These tests are similar in format to offer comparable testing situations. Grids with axes are provided for questions assessing graphing skills.
- *Form 3* is an advanced level test with free-response questions. Grids without axes are provided for questions assessing graphing skills.

  All of the above tests include a free-response Bonus question.
- The **Open-Ended Assessment** includes performance assessment tasks that are suitable for all students. A scoring rubric is included for evaluation guidelines. Sample answers are provided for assessment.
- A **Vocabulary Test**, suitable for all students, includes a list of the vocabulary words in the chapter and ten questions assessing students' knowledge of those terms. This can also be used in conjunction with one of the chapter tests or as a review worksheet.

Intermediate Assessment

- Four free-response **quizzes** are included to offer assessment at appropriate intervals in the chapter.
- A **Mid-Chapter Test** provides an option to assess the first half of the chapter. It is composed of both multiple-choice and free-response questions.

Continuing Assessment

- The **Cumulative Review** provides students an opportunity to reinforce and retain skills as they proceed through their study of Algebra 2. It can also be used as a test. This master includes free-response questions.
- The **Standardized Test Practice** offers continuing review of algebra concepts in various formats, which may appear on the standardized tests that they may encounter. This practice includes multiple-choice, grid-in, and quantitative-comparison questions. Bubble-in and grid-in answer sections are provided on the master.

Answers

- Page A1 is an answer sheet for the Standardized Test Practice questions that appear in the Student Edition on pages 572–573. This improves students' familiarity with the answer formats they may encounter in test taking.
- The answers for the lesson-by-lesson masters are provided as reduced pages with answers appearing in red.
- Full-size answer keys are provided for the assessment masters in this booklet.

NAME _____ DATE _____ PERIOD _____

10 Reading to Learn Mathematics

Vocabulary Builder

This is an alphabetical list of the key vocabulary terms you will learn in Chapter 10. As you study the chapter, complete each term's definition or description. Remember to add the page number where you found the term. Add these pages to your Algebra Study Notebook to review vocabulary at the end of the chapter.

Vocabulary Term	Found on Page	Definition/Description/Example
Change of Base Formula		
common logarithm LAW·guh·RIH·thuhm		
exponential decay EHK·spuh·NEHN·chuhl		
exponential equation		
exponential function		
exponential growth		
exponential inequality		

(continued on the next page)

10 Reading to Learn Mathematics

Vocabulary Builder *(continued)*

Vocabulary Term	Found on Page	Definition/Description/Example
logarithm		
logarithmic function LAW·guh·RIHTH·mihk		
natural base, *e*		
natural base exponential function		
natural logarithm		
natural logarithmic function		
rate of decay		
rate of growth		

10-1 Study Guide and Intervention

Exponential Functions

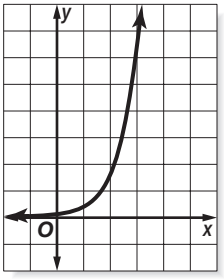
Exponential Functions An **exponential function** has the form $y = ab^x$, where $a \neq 0$, $b > 0$, and $b \neq 1$.

Properties of an Exponential Function	**1.** The function is continuous and one-to-one. **2.** The domain is the set of all real numbers. **3.** The x-axis is the asymptote of the graph. **4.** The range is the set of all positive numbers if $a > 0$ and all negative numbers if $a < 0$. **5.** The graph contains the point $(0, a)$.
Exponential Growth and Decay	If $a > 0$ and $b > 1$, the function $y = ab^x$ represents exponential growth. If $a > 0$ and $0 < b < 1$, the function $y = ab^x$ represents exponential decay.

Example 1 **Sketch the graph of $y = 0.1(4)^x$. Then state the function's domain and range.**

Make a table of values. Connect the points to form a smooth curve.

x	-1	0	1	2	3
y	0.025	0.1	0.4	1.6	6.4



The domain of the function is all real numbers, while the range is the set of all positive real numbers.

Example 2 **Determine whether each function represents exponential *growth* or *decay*.**

a. $y = 0.5(2)^x$

exponential growth, since the base, 2, is greater than 1

b. $y = -2.8(2)^x$

neither, since -2.8, the value of a is less than 0.

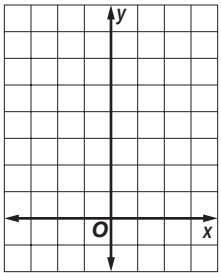
c. $y = 1.1(0.5)^x$

exponential decay, since the base, 0.5, is between 0 and 1

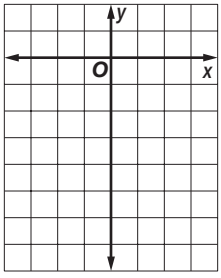
Exercises

Sketch the graph of each function. Then state the function's domain and range.

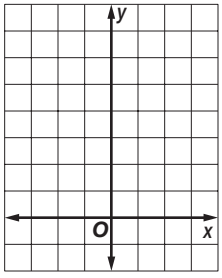
1. $y = 3(2)^x$



2. $y = -2\left(\frac{1}{4}\right)^x$



3. $y = 0.25(5)^x$



Determine whether each function represents exponential growth or decay.

4. $y = 0.3(1.2)^x$

5. $y = -5\left(\frac{4}{5}\right)^x$

6. $y = 3(10)^{-x}$

10-1 Study Guide and Intervention *(continued)*

Exponential Functions

Exponential Equations and Inequalities All the properties of rational exponents that you know also apply to real exponents. Remember that $a^m \cdot a^n = a^{m+n}$, $(a^m)^n = a^{mn}$, and $a^m \div a^n = a^{m-n}$.

Property of Equality for Exponential Functions	If b is a positive number other than 1, then $b^x = b^y$ if and only if $x = y$.
Property of Inequality for Exponential Functions	If $b > 1$ then $b^x > b^y$ if and only if $x > y$ and $b^x < b^y$ if and only if $x < y$.

Example 1 **Solve $4^{x-1} = 2^{x+5}$.**

$4^{x-1} = 2^{x+5}$	Original equation
$(2^2)^{x-1} = 2^{x+5}$	Rewrite 4 as 2^2.
$2(x-1) = x+5$	Prop. of Inequality for Exponential Functions
$2x - 2 = x + 5$	Distributive Property
$x = 7$	Subtract x and add 2 to each side.

Example 2 **Solve $5^{2x-1} > \frac{1}{125}$.**

$5^{2x-1} > \frac{1}{125}$	Original inequality
$5^{2x-1} > 5^{-3}$	Rewrite $\frac{1}{125}$ as 5^{-3}.
$2x - 1 > -3$	Prop. of Inequality for Exponential Functions
$2x > -2$	Add 1 to each side.
$x > -1$	Divide each side by 2.

The solution set is $\{x | x > -1\}$.

Exercises

Simplify each expression.

1. $(3^{\sqrt{2}})^{\sqrt{2}}$

2. $25^{\sqrt{2}} \cdot 125^{\sqrt{2}}$

3. $(x^{\sqrt{2}}y^{3\sqrt{2}})^{\sqrt{2}}$

4. $(x^{\sqrt{6}})(x^{\sqrt{5}})$

5. $(x^{\sqrt{6}})^{\sqrt{5}}$

6. $(2x^{\pi})(5x^{3\pi})$

Solve each equation or inequality. Check your solution.

7. $3^{2x-1} = 3^{x+2}$

8. $2^{3x} = 4^{x+2}$

9. $3^{2x-1} = \frac{1}{9}$

10. $4^{x+1} = 8^{2x+3}$

11. $8^{x-2} = \frac{1}{16}$

12. $25^{2x} = 125^{x+2}$

13. $4^{\sqrt{x}} = 16^{\sqrt{5}}$

14. $x^{\sqrt{3}} = 36^{\sqrt{\frac{3}{4}}}$

15. $x^{\sqrt{2}} = 81^{\frac{1}{\sqrt{8}}}$

16. $3^{x-4} < \frac{1}{27}$

17. $4^{2x-2} > 2^{x+1}$

18. $5^{2x} < 125^{x-5}$

19. $10^{4x+1} > 100^{x-2}$

20. $7^{3x} < 49^{x^2}$

21. $8^{2x-5} < 4^{x+8}$

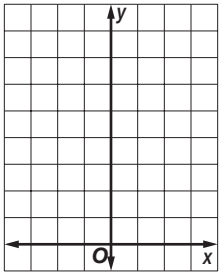
NAME _____ DATE _____ PERIOD _____

10-1 Skills Practice

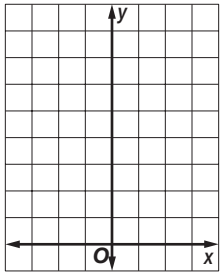
Exponential Functions

Sketch the graph of each function. Then state the function's domain and range.

1. $y = 3(2)^x$



2. $y = 2\left(\frac{1}{2}\right)^x$



Determine whether each function represents exponential *growth* or *decay*.

3. $y = 3(6)^x$

4. $y = 2\left(\frac{9}{10}\right)^x$

5. $y = 10^{-x}$

6. $y = 2(2.5)^x$

Write an exponential function whose graph passes through the given points.

7. (0, 1) and (−1, 3)

8. (0, 4) and (1, 12)

9. (0, 3) and (−1, 6)

10. (0, 5) and (1, 15)

11. (0, 0.1) and (1, 0.5)

12. (0, 0.2) and (1, 1.6)

Simplify each expression.

13. $\left(3^{\sqrt{3}}\right)^{\sqrt{3}}$

14. $\left(x^{\sqrt{2}}\right)^{\sqrt{7}}$

15. $5^{2\sqrt{3}} \cdot 5^{4\sqrt{3}}$

16. $x^{3\pi} \div x^{\pi}$

Solve each equation or inequality. Check your solution.

17. $3^x > 9$

18. $2^{2x+3} = 32$

19. $49^x \leq \frac{1}{7}$

20. $4^{3x-2} = 16$

21. $3^{2x+5} = 27^x$

22. $27^x = 3^{2x+3}$

Lesson 10-1

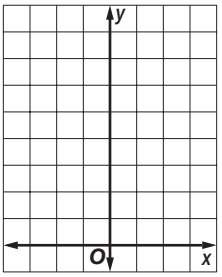
NAME _____ DATE _____ PERIOD _____

10-1 Practice

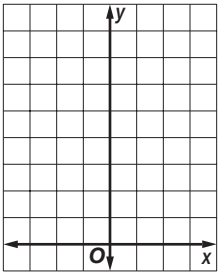
Exponential Functions

Sketch the graph of each function. Then state the function's domain and range.

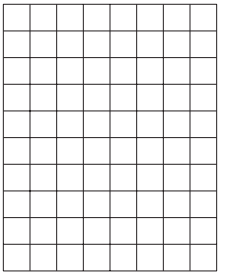
1. $y = 1.5(2)^x$



2. $y = 4(3)^x$



3. $y = 3(0.5)^x$



Determine whether each function represents exponential *growth* or *decay*.

4. $y = 5(0.6)^x$

5. $y = 0.1(2)^x$

6. $y = 5 \cdot 4^{-x}$

Write an exponential function whose graph passes through the given points.

7. (0, 1) and (−1, 4)

8. (0, 2) and (1, 10)

9. (0, −3) and (1, −1.5)

10. (0, 0.8) and (1, 1.6)

11. (0, −0.4) and (2, −10)

12. (0, π) and (3, 8π)

Simplify each expression.

13. $\left(2^{\sqrt{2}}\right)^{\sqrt{8}}$

14. $\left(n^{\sqrt{3}}\right)^{\sqrt{75}}$

15. $y^{\sqrt{6}} \cdot y^{5\sqrt{6}}$

16. $13^{\sqrt{6}} \cdot 13^{\sqrt{24}}$

17. $n^3 \div n^{\pi}$

18. $125^{\sqrt{11}} \div 5^{\sqrt{11}}$

Solve each equation or inequality. Check your solution.

19. $3^{3x-5} > 81$

20. $7^{6x} = 7^{2x-20}$

21. $3^{6n-5} < 9^{4n-3}$

22. $9^{2x-1} = 27^{x+4}$

23. $2^{3n-1} \geq \left(\frac{1}{8}\right)^n$

24. $16^{4n-1} = 128^{2n+1}$

BIOLOGY For Exercises 25 and 26, use the following information.

The initial number of bacteria in a culture is 12,000. The number after 3 days is 96,000.

25. Write an exponential function to model the population y of bacteria after x days.

26. How many bacteria are there after 6 days?

27. EDUCATION A college with a graduating class of 4000 students in the year 2002 predicts that it will have a graduating class of 4862 in 4 years. Write an exponential function to model the number of students y in the graduating class t years after 2002.

NAME _____ DATE _____ PERIOD _____

10-1 Reading to Learn Mathematics

Exponential Functions

Pre-Activity **How does an exponential function describe tournament play?**

Read the introduction to Lesson 10-1 at the top of page 523 in your textbook.

How many rounds of play would be needed for a tournament with 100 players?

Reading the Lesson

1. Indicate whether each of the following statements about the exponential function $y = 10^x$ is *true* or *false*.

a. The domain is the set of all positive real numbers.

b. The y-intercept is 1.

c. The function is one-to-one.

d. The y-axis is an asymptote of the graph.

e. The range is the set of all real numbers.

2. Determine whether each function represents exponential *growth* or *decay*.

a. $y = 0.2(3)^x$. **b.** $y = 3\left(\frac{2}{5}\right)^x$. **c.** $y = 0.4(1.01)^x$.

3. Supply the reason for each step in the following solution of an exponential equation.

$9^{2x-1} = 27x$	Original equation
$(3^2)^{2x-1} = (3^3)^x$	_____
$3^{2(2x-1)} = 3^{3x}$	_____
$2(2x-1) = 3x$	_____
$4x - 2 = 3x$	_____
$x - 2 = 0$	_____
$x = 2$	_____

Helping You Remember

4. One way to remember that polynomial functions and exponential functions are different is to contrast the polynomial function $y = x^2$ and the exponential function $y = 2^x$. Tell at least three ways they are different.

Lesson 10-1

NAME _____ DATE _____ PERIOD _____

10-1 Enrichment

Finding Solutions of $x^y = y^x$

Perhaps you have noticed that if x and y are interchanged in equations such as $x = y$ and $xy = 1$, the resulting equation is equivalent to the original equation. The same is true of the equation $x^y = y^x$. However, finding solutions of $x^y = y^x$ and drawing its graph is not a simple process.

Solve each problem. Assume that x and y are positive real numbers.

1. If $a > 0$, will (a, a) be a solution of $x^y = y^x$? Justify your answer.

2. If $c > 0$, $d > 0$, and (c, d) is a solution of $x^y = y^x$, will (d, c) also be a solution? Justify your answer.

3. Use 2 as a value for y in $x^y = y^x$. The equation becomes $x^2 = 2^x$.

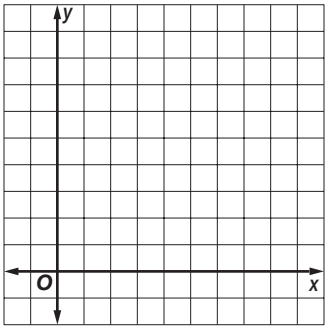
a. Find equations for two functions, $f(x)$ and $g(x)$ that you could graph to find the solutions of $x^2 = 2^x$. Then graph the functions on a separate sheet of graph paper.

b. Use the graph you drew for part **a** to state two solutions for $x^2 = 2^x$. Then use these solutions to state two solutions for $x^y = y^x$.

4. In this exercise, a graphing calculator will be very helpful. Use the technique of Exercise 3 to complete the tables below. Then graph $x^y = y^x$ for positive values of x and y. If there are asymptotes, show them in your diagram using dotted lines. Note that in the table, some values of y call for one value of x, others call for two.

x	y
	$\frac{1}{2}$
	$\frac{3}{4}$
	1
	2
	2
	3
	3

x	y
	4
	4
	5
	5
	8
	8



10-2 Study Guide and Intervention

Logarithms and Logarithmic Functions

Logarithmic Functions and Expressions

Definition of Logarithm with Base *b*	Let b and x be positive numbers, $b \neq 1$. The logarithm of x with base b is denoted $\log_b x$ and is defined as the exponent y that makes the equation $b^y = x$ true.

The inverse of the exponential function $y = b^x$ is the **logarithmic function** $x = b^y$. This function is usually written as $y = \log_b x$.

Properties of Logarithmic Functions	**1.** The function is continuous and one-to-one. **2.** The domain is the set of all positive real numbers. **3.** The y-axis is an asymptote of the graph. **4.** The range is the set of all real numbers. **5.** The graph contains the point (0, 1).

Example 1 **Write an exponential equation equivalent to $\log_3 243 = 5$.**

$3^5 = 243$

Example 2 **Write a logarithmic equation equivalent to $6^{-3} = \frac{1}{216}$.**

$\log_6 \frac{1}{216} = -3$

Example 3 **Evaluate $\log_8 16$.**

$8^{\frac{4}{3}} = 16$, so $\log_8 16 = \frac{4}{3}$.

Exercises

Write each equation in logarithmic form.

1. $2^7 = 128$ **2.** $3^{-4} = \frac{1}{81}$ **3.** $\left(\frac{1}{7}\right)^3 = \frac{1}{343}$

Write each equation in exponential form.

4. $\log_{15} 225 = 2$ **5.** $\log_3 \frac{1}{27} = -3$ **6.** $\log_4 32 = \frac{5}{2}$

Evaluate each expression.

7. $\log_4 64$ **8.** $\log_2 64$ **9.** $\log_{100} 100{,}000$

10. $\log_5 625$ **11.** $\log_{27} 81$ **12.** $\log_{25} 5$

13. $\log_2 \frac{1}{128}$ **14.** $\log_{10} 0.00001$ **15.** $\log_4 \frac{1}{32}$

Lesson 10-2

10-2 Study Guide and Intervention *(continued)*

Logarithms and Logarithmic Functions

Solve Logarithmic Equations and Inequalities

Logarithmic to Exponential Inequality	If $b > 1$, $x > 0$, and $\log_b x > y$, then $x > b^y$. If $b > 1$, $x > 0$, and $\log_b x < y$, then $0 < x < b^y$.
Property of Equality for Logarithmic Functions	If b is a positive number other than 1, then $\log_b x = \log_b y$ if and only if $x = y$.
Property of Inequality for Logarithmic Functions	If $b > 1$, then $\log_b x > \log_b y$ if and only if $x > y$, and $\log_b x < \log_b y$ if and only if $x < y$.

Example 1 **Solve $\log_2 2x = 3$.**

$\log_2 2x = 3$ Original equation

$2x = 2^3$ Definition of logarithm

$2x = 8$ Simplify.

$x = 4$ Simplify.

The solution is $x = 4$.

Example 2 **Solve $\log_5 (4x - 3) < 3$.**

$\log_5 (4x - 3) < 3$ Original equation

$0 < 4x - 3 < 5^3$ Logarithmic to exponential inequality

$3 < 4x < 125 + 3$ Addition Property of Inequalities

$\frac{3}{4} < x < 32$ Simplify.

The solution set is $\left\{x \middle| \frac{3}{4} < x < 32\right\}$.

Exercises

Solve each equation or inequality.

1. $\log_2 32 = 3x$

2. $\log_3 2c = -2$

3. $\log_{2x} 16 = -2$

4. $\log_{25} \left(\frac{x}{2}\right) = \frac{1}{2}$

5. $\log_4 (5x + 1) = 2$

6. $\log_8 (x - 5) = \frac{2}{3}$

7. $\log_4 (3x - 1) = \log_4 (2x + 3)$

8. $\log_2 (x^2 - 6) = \log_2 (2x + 2)$

9. $\log_{x+4} 27 = 3$

10. $\log_2 (x + 3) = 4$

11. $\log_x 1000 = 3$

12. $\log_8 (4x + 4) = 2$

13. $\log_2 2x > 2$

14. $\log_5 x > 2$

15. $\log_2 (3x + 1) < 4$

16. $\log_4 (2x) > -\frac{1}{2}$

17. $\log_3 (x + 3) < 3$

18. $\log_{27} 6x > \frac{2}{3}$

10-2 Skills Practice

Logarithms and Logarithmic Functions

Write each equation in logarithmic form.

1. $2^3 = 8$

2. $3^2 = 9$

3. $8^{-2} = \frac{1}{64}$

4. $\left(\frac{1}{3}\right)^2 = \frac{1}{9}$

Write each equation in exponential form.

5. $\log_3 243 = 5$

6. $\log_4 64 = 3$

7. $\log_9 3 = \frac{1}{2}$

8. $\log_5 \frac{1}{25} = -2$

Evaluate each expression.

9. $\log_5 25$

10. $\log_9 3$

11. $\log_{10} 1000$

12. $\log_{125} 5$

13. $\log_4 \frac{1}{64}$

14. $\log_5 \frac{1}{625}$

15. $\log_8 8^3$

16. $\log_{27} \frac{1}{3}$

Solve each equation or inequality. Check your solutions.

17. $\log_3 x = 5$

18. $\log_2 x = 3$

19. $\log_4 y < 0$

20. $\log_{\frac{1}{4}} x = 3$

21. $\log_2 n > -2$

22. $\log_b 3 = \frac{1}{2}$

23. $\log_6 (4x + 12) = 2$

24. $\log_2 (4x - 4) > 5$

25. $\log_3 (x + 2) = \log_3 (3x)$

26. $\log_6 (3y - 5) \geq \log_6 (2y + 3)$

Lesson 10-2

NAME _____ DATE _____ PERIOD _____

10-2 Practice

Logarithms and Logarithmic Functions

Write each equation in logarithmic form.

1. $5^3 = 125$

2. $7^0 = 1$

3. $3^4 = 81$

4. $3^{-4} = \frac{1}{81}$

5. $\left(\frac{1}{4}\right)^3 = \frac{1}{64}$

6. $7776^{\frac{1}{5}} = 6$

Write each equation in exponential form.

7. $\log_6 216 = 3$

8. $\log_2 64 = 6$

9. $\log_3 \frac{1}{81} = -4$

10. $\log_{10} 0.00001 = -5$

11. $\log_{25} 5 = \frac{1}{2}$

12. $\log_{32} 8 = \frac{3}{5}$

Evaluate each expression.

13. $\log_3 81$

14. $\log_{10} 0.0001$

15. $\log_2 \frac{1}{16}$

16. $\log_{\frac{1}{3}} 27$

17. $\log_9 1$

18. $\log_8 4$

19. $\log_7 \frac{1}{49}$

20. $\log_6 6^4$

21. $\log_3 \frac{1}{3}$

22. $\log_4 \frac{1}{256}$

23. $\log_9 9^{(n+1)}$

24. $2^{\log_2 32}$

Solve each equation or inequality. Check your solutions.

25. $\log_{10} n = -3$

26. $\log_4 x > 3$

27. $\log_4 x = \frac{3}{2}$

28. $\log_{\frac{1}{5}} x = -3$

29. $\log_7 q < 0$

30. $\log_6 (2y + 8) \geq 2$

31. $\log_y 16 = -4$

32. $\log_n \frac{1}{8} = -3$

33. $\log_b 1024 = 5$

34. $\log_8 (3x + 7) < \log_8 (7x + 4)$

35. $\log_7 (8x + 20) = \log_7 (x + 6)$

36. $\log_3 (x^2 - 2) = \log_3 x$

37. SOUND Sounds that reach levels of 130 decibels or more are painful to humans. What is the relative intensity of 130 decibels?

38. INVESTING Maria invests \$1000 in a savings account that pays 8% interest compounded annually. The value of the account A at the end of five years can be determined from the equation $\log A = \log[1000(1 + 0.08)^5]$. Find the value of A to the nearest dollar.

NAME _____ DATE _____ PERIOD _____

10-2 Reading to Learn Mathematics

Logarithms and Logarithmic Functions

Pre-Activity **Why is a logarithmic scale used to measure sound?**

Read the introduction to Lesson 10-2 at the top of page 531 in your textbook.

How many times louder than a whisper is normal conversation?

Reading the Lesson

1. a. Write an exponential equation that is equivalent to $\log_3 81 = 4$.

b. Write a logarithmic equation that is equivalent to $25^{-\frac{1}{2}} = \frac{1}{5}$.

c. Write an exponential equation that is equivalent to $\log_4 1 = 0$.

d. Write a logarithmic equation that is equivalent to $10^{-3} = 0.001$.

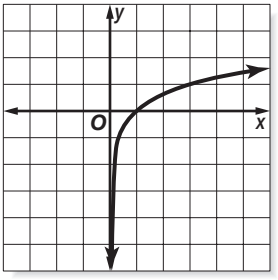
e. What is the inverse of the function $y = 5^x$?

f. What is the inverse of the function $y = \log_{10} x$?

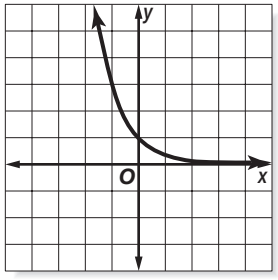
2. Match each function with its graph.

a. $y = 3^x$ **b.** $y = \log_3 x$ **c.** $y = \left(\frac{1}{3}\right)^x$

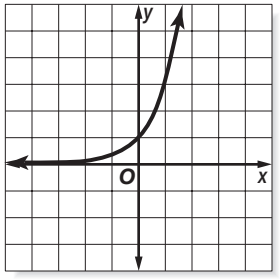
I.



II.



III.



3. Indicate whether each of the following statements about the exponential function $y = \log_5 x$ is *true* or *false*.

a. The y-axis is an asymptote of the graph.

b. The domain is the set of all real numbers.

c. The graph contains the point (5, 0).

d. The range is the set of all real numbers.

e. The y-intercept is 1.

Helping You Remember

4. An important skill needed for working with logarithms is changing an equation between logarithmic and exponential forms. Using the words *base*, *exponent*, and *logarithm*, describe an easy way to remember and apply the part of the definition of logarithm that says, "$\log_b x = y$ if and only if $b^y = x$."

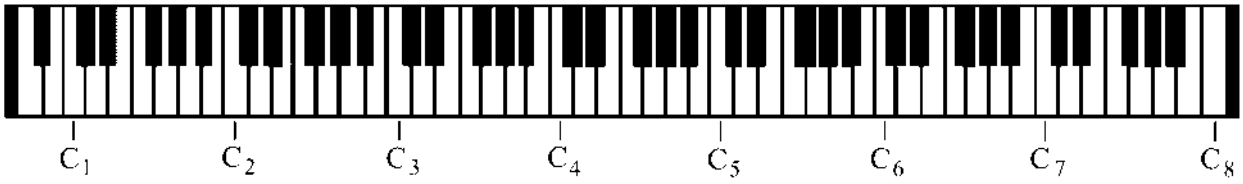
Lesson 10-2

NAME _____ DATE_____ PERIOD _____

10-2 Enrichment

Musical Relationships

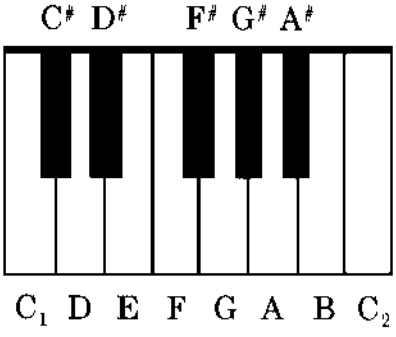
The frequencies of notes in a musical scale that are one octave apart are related by an exponential equation. For the eight C notes on a piano, the equation is $C_n = C_1 2^{n-1}$, where C_n represents the frequency of note C_n.



1. Find the relationship between C_1 and C_2.

2. Find the relationship between C_1 and C_4.

The frequencies of consecutive notes are related by a common ratio r. The general equation is $f_n = f_1 r^{n-1}$.

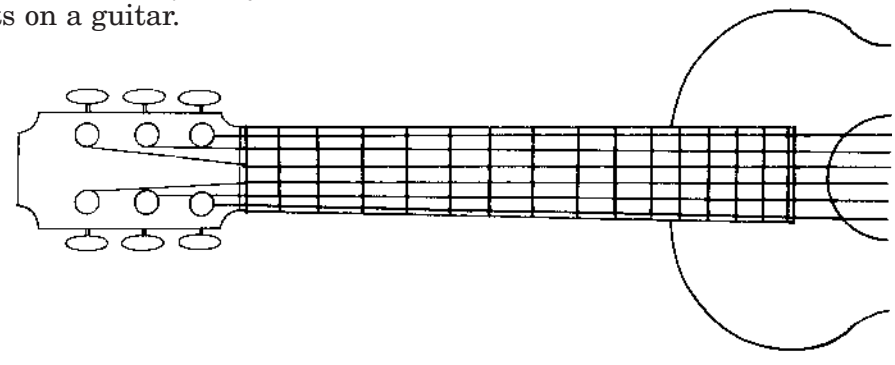


3. If the frequency of middle C is 261.6 cycles per second and the frequency of the next higher C is 523.2 cycles per second, find the common ratio r. (*Hint:* The two C's are 12 notes apart.) Write the answer as a radical expression.

4. Substitute decimal values for r and f_1 to find a specific equation for f_n.

5. Find the frequency of $F^{\#}$ above middle C.

6. Frets are a series of ridges placed across the fingerboard of a guitar. They are spaced so that the sound made by pressing a string against one fret has about 1.0595 times the wavelength of the sound made by using the next fret. The general equation is $w_n = w_0(1.0595)^n$. Describe the arrangement of the frets on a guitar.



10-3 Study Guide and Intervention

Properties of Logarithms

Properties of Logarithms Properties of exponents can be used to develop the following properties of logarithms.

Product Property of Logarithms	For all positive numbers m, n, and b, where $b \neq 1$, $\log_b mn = \log_b m + \log_b n$.
Quotient Property of Logarithms	For all positive numbers m, n, and b, where $b \neq 1$, $\log_b \frac{m}{n} = \log_b m - \log_b n$.
Power Property of Logarithms	For any real number p and positive numbers m and b, where $b \neq 1$, $\log_b m^p = p \log_b m$.

Example **Use $\log_3 28 \approx 3.0331$ and $\log_3 4 \approx 1.2619$ to approximate the value of each expression.**

a. $\log_3 36$

$$\begin{aligned}\log_3 36 &= \log_3 (3^2 \cdot 4)\\ &= \log_3 3^2 + \log_3 4\\ &= 2 + \log_3 4\\ &\approx 2 + 1.2619\\ &\approx 3.2619\end{aligned}$$

b. $\log_3 7$

$$\begin{aligned}\log_3 7 &= \log_3 \left(\frac{28}{4}\right)\\ &= \log_3 28 - \log_3 4\\ &\approx 3.0331 - 1.2619\\ &\approx 1.7712\end{aligned}$$

c. $\log_3 256$

$$\begin{aligned}\log_3 256 &= \log_3 (4^4)\\ &= 4 \cdot \log_3 4\\ &\approx 4(1.2619)\\ &\approx 5.0476\end{aligned}$$

Exercises

Use $\log_{12} 3 \approx 0.4421$ and $\log_{12} 7 \approx 0.7831$ to evaluate each expression.

1. $\log_{12} 21$

2. $\log_{12} \frac{7}{3}$

3. $\log_{12} 49$

4. $\log_{12} 36$

5. $\log_{12} 63$

6. $\log_{12} \frac{27}{49}$

7. $\log_{12} \frac{81}{49}$

8. $\log_{12} 16{,}807$

9. $\log_{12} 441$

Use $\log_5 3 \approx 0.6826$ and $\log_5 4 \approx 0.8614$ to evaluate each expression.

10. $\log_5 12$

11. $\log_5 100$

12. $\log_5 0.75$

13. $\log_5 144$

14. $\log_5 \frac{27}{16}$

15. $\log_5 375$

16. $\log_5 1.\overline{3}$

17. $\log_5 \frac{9}{16}$

18. $\log_5 \frac{81}{5}$

Lesson 10-3

10-3 Study Guide and Intervention *(continued)*

Properties of Logarithms

Solve Logarithmic Equations You can use the properties of logarithms to solve equations involving logarithms.

Example **Solve each equation.**

a. $\mathbf{2 \log_3 x - \log_3 4 = \log_3 25}$

$2 \log_3 x - \log_3 4 = \log_3 25$	Original equation
$\log_3 x^2 - \log_3 4 = \log_3 25$	Power Property
$\log_3 \frac{x^2}{4} = \log_3 25$	Quotient Property
$\frac{x^2}{4} = 25$	Property of Equality for Logarithmic Functions
$x^2 = 100$	Multiply each side by 4.
$x = \pm 10$	Take the square root of each side.

Since logarithms are undefined for $x < 0$, -10 is an extraneous solution. The only solution is 10.

b. $\mathbf{\log_2 x + \log_2 (x + 2) = 3}$

$\log_2 x + \log_2 (x + 2) = 3$	Original equation
$\log_2 x(x + 2) = 3$	Product Property
$x(x + 2) = 2^3$	Definition of logarithm
$x^2 + 2x = 8$	Distributive Property
$x^2 + 2x - 8 = 0$	Subtract 8 from each side.
$(x + 4)(x - 2) = 0$	Factor.
$x = 2$ or $x = -4$	Zero Product Property

Since logarithms are undefined for $x < 0$, -4 is an extraneous solution. The only solution is 2.

Exercises

Solve each equation. Check your solutions.

1. $\log_5 4 + \log_5 2x = \log_5 24$

2. $3 \log_4 6 - \log_4 8 = \log_4 x$

3. $\frac{1}{2} \log_6 25 + \log_6 x = \log_6 20$

4. $\log_2 4 - \log_2 (x + 3) = \log_2 8$

5. $\log_6 2x - \log_6 3 = \log_6 (x - 1)$

6. $2 \log_4 (x + 1) = \log_4 (11 - x)$

7. $\log_2 x - 3 \log_2 5 = 2 \log_2 10$

8. $3 \log_2 x - 2 \log_2 5x = 2$

9. $\log_3 (c + 3) - \log_3 (4c - 1) = \log_3 5$

10. $\log_5 (x + 3) - \log_5 (2x - 1) = 2$

NAME _____ DATE_____ PERIOD _____

10-3 Skills Practice

Properties of Logarithms

Use $\log_2 3 \approx 1.5850$ and $\log_2 5 \approx 2.3219$ to approximate the value of each expression.

1. $\log_2 25$

2. $\log_2 27$

3. $\log_2 \frac{3}{5}$

4. $\log_2 \frac{5}{3}$

5. $\log_2 15$

6. $\log_2 45$

7. $\log_2 75$

8. $\log_2 0.6$

9. $\log_2 \frac{1}{3}$

10. $\log_2 \frac{9}{5}$

Solve each equation. Check your solutions.

11. $\log_{10} 27 = 3 \log_{10} x$

12. $3 \log_7 4 = 2 \log_7 b$

13. $\log_4 5 + \log_4 x = \log_4 60$

14. $\log_6 2c + \log_6 8 = \log_6 80$

15. $\log_5 y - \log_5 8 = \log_5 1$

16. $\log_2 q - \log_2 3 = \log_2 7$

17. $\log_9 4 + 2 \log_9 5 = \log_9 w$

18. $3 \log_8 2 - \log_8 4 = \log_8 b$

19. $\log_{10} x + \log_{10} (3x - 5) = \log_{10} 2$

20. $\log_4 x + \log_4 (2x - 3) = \log_4 2$

21. $\log_3 d + \log_3 3 = 3$

22. $\log_{10} y - \log_{10} (2 - y) = 0$

23. $\log_2 s + 2 \log_2 5 = 0$

24. $\log_2 (x + 4) - \log_2 (x - 3) = 3$

25. $\log_4 (n + 1) - \log_4 (n - 2) = 1$

26. $\log_5 10 + \log_5 12 = 3 \log_5 2 + \log_5 a$

Lesson 10-3

NAME _____ DATE_____ PERIOD _____

10-3 Practice

Properties of Logarithms

Use $\log_{10} 5 \approx 0.6990$ and $\log_{10} 7 \approx 0.8451$ to approximate the value of each expression.

1. $\log_{10} 35$

2. $\log_{10} 25$

3. $\log_{10} \frac{7}{5}$

4. $\log_{10} \frac{5}{7}$

5. $\log_{10} 245$

6. $\log_{10} 175$

7. $\log_{10} 0.2$

8. $\log_{10} \frac{25}{7}$

Solve each equation. Check your solutions.

9. $\log_7 n = \frac{2}{3} \log_7 8$

10. $\log_{10} u = \frac{3}{2} \log_{10} 4$

11. $\log_6 x + \log_6 9 = \log_6 54$

12. $\log_8 48 - \log_8 w = \log_8 4$

13. $\log_9 (3u + 14) - \log_9 5 = \log_9 2u$

14. $4 \log_2 x + \log_2 5 = \log_2 405$

15. $\log_3 y = -\log_3 16 + \frac{1}{3} \log_3 64$

16. $\log_2 d = 5 \log_2 2 - \log_2 8$

17. $\log_{10} (3m - 5) + \log_{10} m = \log_{10} 2$

18. $\log_{10} (b + 3) + \log_{10} b = \log_{10} 4$

19. $\log_8 (t + 10) - \log_8 (t - 1) = \log_8 12$

20. $\log_3 (a + 3) + \log_3 (a + 2) = \log_3 6$

21. $\log_{10} (r + 4) - \log_{10} r = \log_{10} (r + 1)$

22. $\log_4 (x^2 - 4) - \log_4 (x + 2) = \log_4 1$

23. $\log_{10} 4 + \log_{10} w = 2$

24. $\log_8 (n - 3) + \log_8 (n + 4) = 1$

25. $3 \log_5 (x^2 + 9) - 6 = 0$

26. $\log_{16} (9x + 5) - \log_{16} (x^2 - 1) = \frac{1}{2}$

27. $\log_6 (2x - 5) + 1 = \log_6 (7x + 10)$

28. $\log_2 (5y + 2) - 1 = \log_2 (1 - 2y)$

29. $\log_{10} (c^2 - 1) - 2 = \log_{10} (c + 1)$

30. $\log_7 x + 2 \log_7 x - \log_7 3 = \log_7 72$

31. SOUND The loudness L of a sound in decibels is given by $L = 10 \log_{10} R$, where R is the sound's relative intensity. If the intensity of a certain sound is tripled, by how many decibels does the sound increase?

32. EARTHQUAKES An earthquake rated at 3.5 on the Richter scale is felt by many people, and an earthquake rated at 4.5 may cause local damage. The Richter scale magnitude reading m is given by $m = \log_{10} x$, where x represents the amplitude of the seismic wave causing ground motion. How many times greater is the amplitude of an earthquake that measures 4.5 on the Richter scale than one that measures 3.5?

NAME _____ DATE_____ PERIOD _____

10-3 Reading to Learn Mathematics

Properties of Logarithms

Pre-Activity **How are the properties of exponents and logarithms related?**

Read the introduction to Lesson 10-3 at the top of page 541 in your textbook.

Find the value of $\log_5 125$. Find the value of $\log_5 5$.

Find the value of $\log_5 (125 \div 5)$.

Which of the following statements is true?

A. $\log_5 (125 \div 5) = (\log_5 125) \div (\log_5 5)$

B. $\log_5 (125 \div 5) = \log_5 125 - \log_5 5$

Reading the Lesson

1. Each of the properties of logarithms can be stated in words or in symbols. Complete the statements of these properties in words.

a. The logarithm of a quotient is the _____ of the logarithms of the _____ and the _____.

b. The logarithm of a power is the _____ of the logarithm of the base and the _____.

c. The logarithm of a product is the _____ of the logarithms of its _____.

2. State whether each of the following equations is *true* or *false*. If the statement is true, name the property of logarithms that is illustrated.

a. $\log_3 10 = \log_3 30 - \log_3 3$

b. $\log_4 12 = \log_4 4 + \log_4 8$

c. $\log_2 81 = 2 \log_2 9$

d. $\log_8 30 = \log_8 5 \cdot \log_8 6$

3. The algebraic process of solving the equation $\log_2 x + \log_2 (x + 2) = 3$ leads to "$x = -4$ or $x = 2$." Does this mean that both -4 and 2 are solutions of the logarithmic equation? Explain your reasoning.

Helping You Remember

4. A good way to remember something is to relate it something you already know. Use words to explain how the Product Property for exponents can help you remember the product property for logarithms.

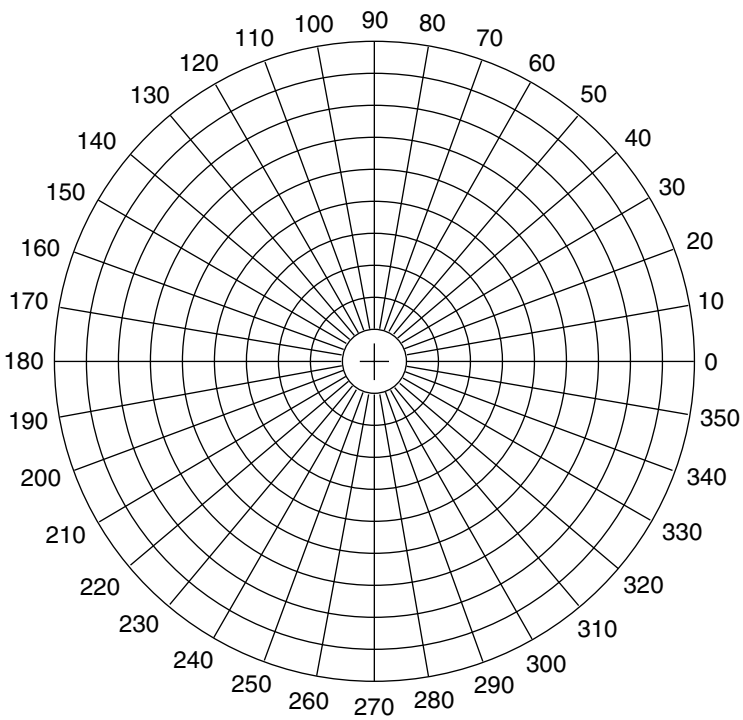
Lesson 10-3

NAME _____ DATE _____ PERIOD _____

10-3 Enrichment

Spirals

Consider an angle in standard position with its vertex at a point O called the pole. Its initial side is on a coordinatized axis called the *polar axis.* A point P on the terminal side of the angle is named by the *polar coordinates* (r, θ), where r is the directed distance of the point from O and θ is the measure of the angle. Graphs in this system may be drawn on polar coordinate paper such as the kind shown below.



1. Use a calculator to complete the table for $\log_2 r = \frac{\theta}{120}$.

(*Hint:* To find θ on a calculator, press 120 [×] [LOG] r [)] [÷] [LOG] 2 [)].)

r	1	2	3	4	5	6	7	8

2. Plot the points found in Exercise 1 on the grid above and connect to form a smooth curve.

This type of spiral is called a logarithmic spiral because the angle measures are proportional to the logarithms of the radii.

10-4 Study Guide and Intervention

Common Logarithms

Common Logarithms Base 10 logarithms are called **common logarithms**. The expression $\log_{10} x$ is usually written without the subscript as $\log x$. Use the LOG key on your calculator to evaluate common logarithms.

The relation between exponents and logarithms gives the following identity.

Inverse Property of Logarithms and Exponents	$10^{\log x} = x$

Example 1 **Evaluate log 50 to four decimal places.**

Use the LOG key on your calculator. To four decimal places, $\log 50 = 1.6990$.

Example 2 **Solve $3^{2x+1} = 12$.**

$3^{2x+1} = 12$	Original equation
$\log 3^{2x+1} = \log 12$	Property of Equality for Logarithms
$(2x + 1) \log 3 = \log 12$	Power Property of Logarithms
$2x + 1 = \dfrac{\log 12}{\log 3}$	Divide each side by log 3.
$2x = \dfrac{\log 12}{\log 3} - 1$	Subtract 1 from each side.
$x = \dfrac{1}{2}\left(\dfrac{\log 12}{\log 3} - 1\right)$	Multiply each side by $\frac{1}{2}$.
$x \approx 0.6309$	

Exercises

Use a calculator to evaluate each expression to four decimal places.

1. log 18 **2.** log 39 **3.** log 120

4. log 5.8 **5.** log 42.3 **6.** log 0.003

Solve each equation or inequality. Round to four decimal places.

7. $4^{3x} = 12$ **8.** $6^{x+2} = 18$

9. $5^{4x-2} = 120$ **10.** $7^{3x-1} \geq 21$

11. $2.4^{x+4} = 30$ **12.** $6.5^{2x} \geq 200$

13. $3.6^{4x-1} = 85.4$ **14.** $2^{x+5} = 3^{x-2}$

15. $9^{3x} = 4^{5x+2}$ **16.** $6^{x-5} = 2^{7x+3}$

Lesson 10-4

10-4 Study Guide and Intervention *(continued)*

Common Logarithms

Change of Base Formula The following formula is used to change expressions with different logarithmic bases to common logarithm expressions.

Change of Base Formula	For all positive numbers a, b, and n, where $a \neq 1$ and $b \neq 1$, $\log_a n = \dfrac{\log_b n}{\log_b a}$

Example **Express $\log_8 15$ in terms of common logarithms. Then approximate its value to four decimal places.**

$\log_8 15 = \dfrac{\log_{10} 15}{\log_{10} 8}$ Change of Base Formula

≈ 1.3023 Simplify.

The value of $\log_8 15$ is approximately 1.3023.

Exercises

Express each logarithm in terms of common logarithms. Then approximate its value to four decimal places.

1. $\log_3 16$

2. $\log_2 40$

3. $\log_5 35$

4. $\log_4 22$

5. $\log_{12} 200$

6. $\log_2 50$

7. $\log_5 0.4$

8. $\log_3 2$

9. $\log_4 28.5$

10. $\log_3 (20)^2$

11. $\log_6 (5)^4$

12. $\log_8 (4)^5$

13. $\log_5 (8)^3$

14. $\log_2 (3.6)^6$

15. $\log_{12} (10.5)^4$

16. $\log_3 \sqrt{150}$

17. $\log_4 \sqrt[3]{39}$

18. $\log_5 \sqrt[4]{1600}$

NAME _____ DATE_____ PERIOD _____

10-4 Skills Practice

Common Logarithms

Use a calculator to evaluate each expression to four decimal places.

1. log 6

2. log 15

3. log 1.1

4. log 0.3

Use the formula pH = −log[*H*+] to find the pH of each substance given its concentration of hydrogen ions.

5. gastric juices: $[H+] = 1.0 \times 10^{-1}$ mole per liter

6. tomato juice: $[H+] = 7.94 \times 10^{-5}$ mole per liter

7. blood: $[H+] = 3.98 \times 10^{-8}$ mole per liter

8. toothpaste: $[H+] = 1.26 \times 10^{-10}$ mole per liter

Solve each equation or inequality. Round to four decimal places.

9. $3^x > 243$

10. $16^v \leq \frac{1}{4}$

11. $8^p = 50$

12. $7^y = 15$

13. $5^{3b} = 106$

14. $4^{5k} = 37$

15. $12^{7p} = 120$

16. $9^{2m} = 27$

17. $3^{r-5} = 4.1$

18. $8^{y+4} > 15$

19. $7.6^{d+3} = 57.2$

20. $0.5^{t-8} = 16.3$

21. $42^{x^2} = 84$

22. $5^{x^2+1} = 10$

Express each logarithm in terms of common logarithms. Then approximate its value to four decimal places.

23. $\log_3 7$

24. $\log_5 66$

25. $\log_2 35$

26. $\log_6 10$

Lesson 10-4

NAME _____ DATE_____ PERIOD _____

10-4 Practice

Common Logarithms

Use a calculator to evaluate each expression to four decimal places.

1. log 101 **2.** log 2.2 **3.** log 0.05

Use the formula pH = −log[H+] to find the pH of each substance given its concentration of hydrogen ions.

4. milk: $[H+] = 2.51 \times 10^{-7}$ mole per liter

5. acid rain: $[H+] = 2.51 \times 10^{-6}$ mole per liter

6. black coffee: $[H+] = 1.0 \times 10^{-5}$ mole per liter

7. milk of magnesia: $[H+] = 3.16 \times 10^{-11}$ mole per liter

Solve each equation or inequality. Round to four decimal places.

8. $2^x < 25$ **9.** $5^a = 120$ **10.** $6^z = 45.6$

11. $9^m \geq 100$ **12.** $3.5^x = 47.9$ **13.** $8.2^y = 64.5$

14. $2^{b+1} \leq 7.31$ **15.** $4^{2x} = 27$ **16.** $2^{a-4} = 82.1$

17. $9^{z-2} > 38$ **18.** $5^{w+3} = 17$ **19.** $30^{x^2} = 50$

20. $5^{x^2-3} = 72$ **21.** $4^{2x} = 9^{x+1}$ **22.** $2^{n+1} = 5^{2n-1}$

Express each logarithm in terms of common logarithms. Then approximate its value to four decimal places.

23. $\log_5 12$ **24.** $\log_8 32$ **25.** $\log_{11} 9$

26. $\log_2 18$ **27.** $\log_9 6$ **28.** $\log_7 \sqrt{8}$

29. HORTICULTURE Siberian irises flourish when the concentration of hydrogen ions [H+] in the soil is not less than 1.58×10^{-8} mole per liter. What is the pH of the soil in which these irises will flourish?

30. ACIDITY The pH of vinegar is 2.9 and the pH of milk is 6.6. How many times greater is the hydrogen ion concentration of vinegar than of milk?

31. BIOLOGY There are initially 1000 bacteria in a culture. The number of bacteria doubles each hour. The number of bacteria N present after t hours is $N = 1000(2)^t$. How long will it take the culture to increase to 50,000 bacteria?

32. SOUND An equation for loudness L in decibels is given by $L = 10 \log R$, where R is the sound's relative intensity. An air-raid siren can reach 150 decibels and jet engine noise can reach 120 decibels. How many times greater is the relative intensity of the air-raid siren than that of the jet engine noise?

NAME _____ DATE _____ PERIOD _____

10-4 Reading to Learn Mathematics

Common Logarithms

Pre-Activity **Why is a logarithmic scale used to measure acidity?**

Read the introduction to Lesson 10-4 at the top of page 547 in your textbook.

Which substance is more acidic, milk or tomatoes?

Reading the Lesson

1. Rhonda used the following keystrokes to enter an expression on her graphing calculator:

[LOG] 17 [)] [ENTER]

The calculator returned the result 1.230448921.
Which of the following conclusions are correct?

a. The base 10 logarithm of 17 is about 1.2304.

b. The base 17 logarithm of 10 is about 1.2304.

c. The common logarithm of 17 is about 1.230449.

d. $10^{1.230448921}$ is very close to 17.

e. The common logarithm of 17 is exactly 1.230448921.

2. Match each expression from the first column with an expression from the second column that has the same value.

a. $\log_2 2$	**i.** $\log_4 1$
b. $\log 12$	**ii.** $\log_2 8$
c. $\log_3 1$	**iii.** $\log_{10} 12$
d. $\log_5 \frac{1}{5}$	**iv.** $\log_5 5$
e. $\log 1000$	**v.** $\log 0.1$

3. Calculators do not have keys for finding base 8 logarithms directly. However, you can use a calculator to find $\log_8 20$ if you apply the _____ formula.
Which of the following expressions are equal to $\log_8 20$?

A. $\log_{20} 8$ **B.** $\frac{\log_{10} 20}{\log_{10} 8}$ **C.** $\frac{\log 20}{\log 8}$ **D.** $\frac{\log 8}{\log 20}$

Helping You Remember

4. Sometimes it is easier to remember a formula if you can state it in words. State the change of base formula in words.

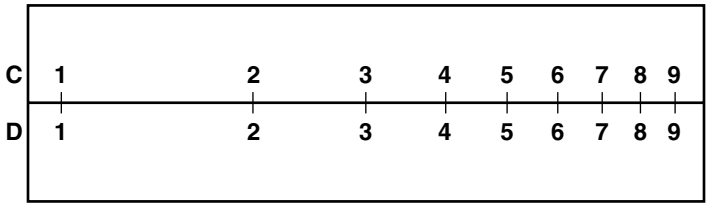
Lesson 10-4

NAME _____ DATE_____ PERIOD _____

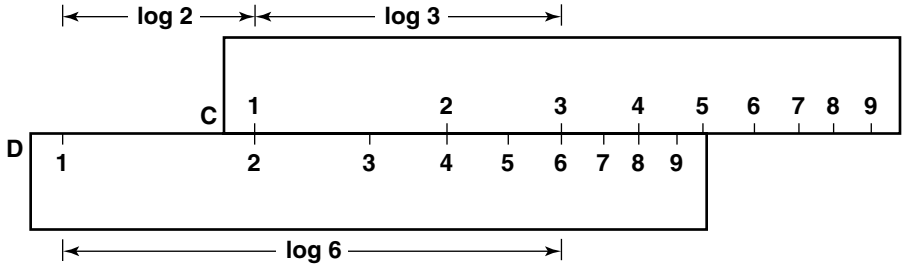
10-4 Enrichment

The Slide Rule

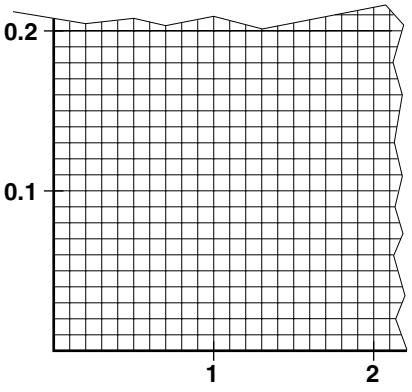
Before the invention of electronic calculators, computations were often performed on a slide rule. A slide rule is based on the idea of logarithms. It has two movable rods labeled with C and D scales. Each of the scales is logarithmic.



To multiply 2×3 on a slide rule, move the C rod to the right as shown below. You can find 2×3 by adding log 2 to log 3, and the slide rule adds the lengths for you. The distance you get is 0.778, or the logarithm of 6.

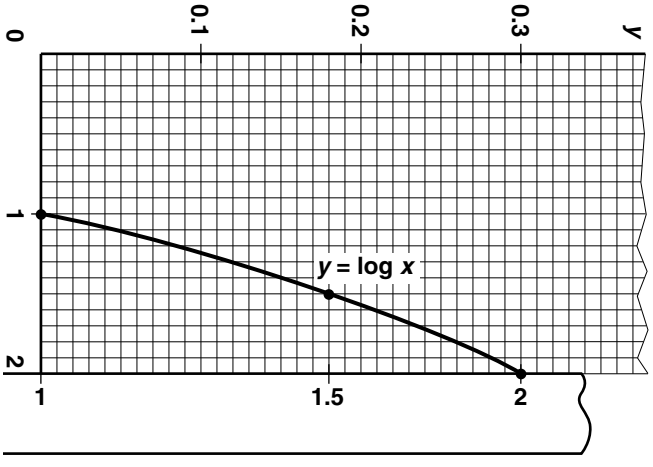


Follow the steps to make a slide rule.



1. Use graph paper that has small squares, such as 10 squares to the inch. Using the scales shown at the right, plot the curve $y = \log x$ for $x = 1, 1.5$, and the whole numbers from 2 through 10. Make an obvious heavy dot for each point plotted.

2. You will need two strips of cardboard. A 5-by-7 index card, cut in half the long way, will work fine. Turn the graph you made in Exercise 1 sideways and use it to mark a logarithmic scale on each of the two strips. The figure shows the mark for 2 being drawn.



3. Explain how to use a slide rule to divide 8 by 2.

10-5 Study Guide and Intervention

Base e and Natural Logarithms

Base *e* and Natural Logarithms The irrational number $e \approx 2.71828\ldots$ often occurs as the base for exponential and logarithmic functions that describe real-world phenomena.

Natural Base *e*	As n increases, $\left(1 + \frac{1}{n}\right)^n$ approaches $e \approx 2.71828\ldots$. $\ln x = \log_e x$

The functions $y = e^x$ and $y = \ln x$ are inverse functions.

Inverse Property of Base *e* and Natural Logarithms	$e^{\ln x} = x \qquad \ln e^x = x$

Natural base expressions can be evaluated using the e^x and ln keys on your calculator.

Example 1 **Evaluate ln 1685.**

Use a calculator.

$\ln 1685 \approx 7.4295$

Example 2 **Write a logarithmic equation equivalent to $e^{2x} = 7$.**

$e^{2x} = 7 \rightarrow \log_e 7 = 2x$ or $2x = \ln 7$

Example 3 **Evaluate $\ln e^{18}$.**

Use the Inverse Property of Base e and Natural Logarithms.

$\ln e^{18} = 18$

Exercises

Use a calculator to evaluate each expression to four decimal places.

1. ln 732 **2.** ln 84,350 **3.** ln 0.735 **4.** ln 100

5. ln 0.0824 **6.** ln 2.388 **7.** ln 128,245 **8.** ln 0.00614

Write an equivalent exponential or logarithmic equation.

9. $e^{15} = x$ **10.** $e^{3x} = 45$ **11.** $\ln 20 = x$ **12.** $\ln x = 8$

13. $e^{-5x} = 0.2$ **14.** $\ln (4x) = 9.6$ **15.** $e^{8.2} = 10x$ **16.** $\ln 0.0002 = x$

Evaluate each expression.

17. $\ln e^3$ **18.** $e^{\ln 42}$ **19.** $e^{\ln 0.5}$ **20.** $\ln e^{16.2}$

Lesson 10-5

10-5 Study Guide and Intervention *(continued)*

Base e and Natural Logarithms

Equations and Inequalities with *e* and ln All properties of logarithms from earlier lessons can be used to solve equations and inequalities with natural logarithms.

Example **Solve each equation or inequality.**

a. $\mathbf{3e^{2x} + 2 = 10}$

$3e^{2x} + 2 = 10$	Original equation
$3e^{2x} = 8$	Subtract 2 from each side.
$e^{2x} = \frac{8}{3}$	Divide each side by 3.
$\ln e^{2x} = \ln \frac{8}{3}$	Property of Equality for Logarithms
$2x = \ln \frac{8}{3}$	Inverse Property of Exponents and Logarithms
$x = \frac{1}{2} \ln \frac{8}{3}$	Multiply each side by $\frac{1}{2}$.
$x \approx 0.4904$	Use a calculator.

b. $\mathbf{\ln (4x - 1) < 2}$

$\ln (4x - 1) < 2$	Original inequality
$e^{\ln (4x - 1)} < e^2$	Write each side using exponents and base *e*.
$0 < 4x - 1 < e^2$	Inverse Property of Exponents and Logarithms
$1 < 4x < e^2 + 1$	Addition Property of Inequalities
$\frac{1}{4} < x < \frac{1}{4}(e^2 + 1)$	Multiplication Property of Inequalities
$0.25 < x < 2.0973$	Use a calculator.

Exercises

Solve each equation or inequality.

1. $e^{4x} = 120$

2. $e^x \leq 25$

3. $e^{x - 2} + 4 = 21$

4. $\ln 6x \geq 4$

5. $\ln (x + 3) - 5 = -2$

6. $e^{-8x} \leq 50$

7. $e^{4x - 1} - 3 = 12$

8. $\ln (5x + 3) = 3.6$

9. $2e^{3x} + 5 = 2$

10. $6 + 3e^{x + 1} = 21$

11. $\ln (2x - 5) = 8$

12. $\ln 5x + \ln 3x > 9$

NAME _____ DATE_____ PERIOD _____

10-5 Skills Practice

Base e and Natural Logarithms

Use a calculator to evaluate each expression to four decimal places.

1. e^3

2. e^{-2}

3. $\ln 2$

4. $\ln 0.09$

Write an equivalent exponential or logarithmic equation.

5. $e^x = 3$

6. $e^4 = 8x$

7. $\ln 15 = x$

8. $\ln x \approx 0.6931$

Evaluate each expression.

9. $e^{\ln 3}$

10. $e^{\ln 2x}$

11. $\ln e^{-2.5}$

12. $\ln e^y$

Solve each equation or inequality.

13. $e^x \geq 5$

14. $e^x < 3.2$

15. $2e^x - 1 = 11$

16. $5e^x + 3 = 18$

17. $e^{3x} = 30$

18. $e^{-4x} > 10$

19. $e^{5x} + 4 > 34$

20. $1 - 2e^{2x} = -19$

21. $\ln 3x = 2$

22. $\ln 8x = 3$

23. $\ln (x - 2) = 2$

24. $\ln (x + 3) = 1$

25. $\ln (x + 3) = 4$

26. $\ln x + \ln 2x = 2$

Lesson 10-5

NAME _____ DATE_____ PERIOD _____

10-5 Practice

Base e and Natural Logarithms

Use a calculator to evaluate each expression to four decimal places.

1. $e^{1.5}$ **2.** $\ln 8$ **3.** $\ln 3.2$ **4.** $e^{-0.6}$

5. $e^{4.2}$ **6.** $\ln 1$ **7.** $e^{-2.5}$ **8.** $\ln 0.037$

Write an equivalent exponential or logarithmic equation.

9. $\ln 50 = x$ **10.** $\ln 36 = 2x$ **11.** $\ln 6 \approx 1.7918$ **12.** $\ln 9.3 \approx 2.2300$

13. $e^x = 8$ **14.** $e^5 = 10x$ **15.** $e^{-x} = 4$ **16.** $e^2 = x + 1$

Evaluate each expression.

17. $e^{\ln 12}$ **18.** $e^{\ln 3x}$ **19.** $\ln e^{-1}$ **20.** $\ln e^{-2y}$

Solve each equation or inequality.

21. $e^x < 9$ **22.** $e^{-x} = 31$ **23.** $e^x = 1.1$ **24.** $e^x = 5.8$

25. $2e^x - 3 = 1$ **26.** $5e^x + 1 \geq 7$ **27.** $4 + e^x = 19$ **28.** $-3e^x + 10 < 8$

29. $e^{3x} = 8$ **30.** $e^{-4x} = 5$ **31.** $e^{0.5x} = 6$ **32.** $2e^{5x} = 24$

33. $e^{2x} + 1 = 55$ **34.** $e^{3x} - 5 = 32$ **35.** $9 + e^{2x} = 10$ **36.** $e^{-3x} + 7 \geq 15$

37. $\ln 4x = 3$ **38.** $\ln(-2x) = 7$ **39.** $\ln 2.5x = 10$ **40.** $\ln(x - 6) = 1$

41. $\ln(x + 2) = 3$ **42.** $\ln(x + 3) = 5$ **43.** $\ln 3x + \ln 2x = 9$ **44.** $\ln 5x + \ln x = 7$

INVESTING For Exercises 45 and 46, use the formula for continuously compounded interest, $A = Pe^{rt}$, where P is the principal, r is the annual interest rate, and t is the time in years.

45. If Sarita deposits \$1000 in an account paying 3.4% annual interest compounded continuously, what is the balance in the account after 5 years?

46. How long will it take the balance in Sarita's account to reach \$2000?

47. RADIOACTIVE DECAY The amount of a radioactive substance y that remains after t years is given by the equation $y = ae^{kt}$, where a is the initial amount present and k is the decay constant for the radioactive substance. If $a = 100$, $y = 50$, and $k = -0.035$, find t.

NAME _____ DATE _____ PERIOD _____

10-5 Reading to Learn Mathematics

Base e and Natural Logarithms

Pre-Activity

How is the natural base *e* used in banking?

Read the introduction to Lesson 10-5 at the top of page 554 in your textbook.

Suppose that you deposit $675 in a savings account that pays an annual interest rate of 5%. In each case listed below, indicate which method of compounding would result in more money in your account at the end of one year.

a. annual compounding or monthly compounding

b. quarterly compounding or daily compounding

c. daily compounding or continuous compounding

Reading the Lesson

1. Jagdish entered the following keystrokes in his calculator:

[LN] 5 [ )] [ENTER]

The calculator returned the result 1.609437912. Which of the following conclusions are correct?

a. The common logarithm of 5 is about 1.6094.

b. The natural logarithm of 5 is exactly 1.609437912.

c. The base 5 logarithm of e is about 1.6094.

d. The natural logarithm of 5 is about 1.609438.

e. $10^{1.609437912}$ is very close to 5.

f. $e^{1.609437912}$ is very close to 5.

2. Match each expression from the first column with its value in the second column. Some choices may be used more than once or not at all.

a. $e^{\ln 5}$	**I.** 1
b. $\ln 1$	**II.** 10
c. $e^{\ln e}$	**III.** -1
d. $\ln e^5$	**IV.** 5
e. $\ln e$	**V.** 0
f. $\ln\left(\frac{1}{e}\right)$	**VI.** e

Helping You Remember

3. A good way to remember something is to explain it to someone else. Suppose that you are studying with a classmate who is puzzled when asked to evaluate $\ln e^3$. How would you explain to him an easy way to figure this out?

Lesson 10-5

NAME _____ DATE_____ PERIOD _____

10-5 Enrichment

Approximations for π and e

The following expression can be used to approximate e. If greater and greater values of n are used, the value of the expression approximates e more and more closely.

$$\left(1 + \frac{1}{n}\right)^n$$

Another way to approximate e is to use this infinite sum. The greater the value of n, the closer the approximation.

$$e = 1 + 1 + \frac{1}{2} + \frac{1}{2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \ldots + \frac{1}{2 \cdot 3 \cdot 4 \cdot \ldots \cdot n} + \ldots$$

In a similar manner, π can be approximated using an infinite product discovered by the English mathematician John Wallis (1616–1703).

$$\frac{\pi}{2} = \frac{2}{1} \cdot \frac{2}{3} \cdot \frac{4}{3} \cdot \frac{4}{5} \cdot \frac{6}{5} \cdot \frac{6}{7} \cdot \ldots \cdot \frac{2n}{2n - 1} \cdot \frac{2n}{2n + 1} \ldots$$

Solve each problem.

1. Use a calculator with an e^x key to find e to 7 decimal places.

2. Use the expression $\left(1 + \frac{1}{n}\right)^n$ to approximate e to 3 decimal places. Use 5, 100, 500, and 7000 as values of n.

3. Use the infinite sum to approximate e to 3 decimal places. Use the whole numbers from 3 through 6 as values of n.

4. Which approximation method approaches the value of e more quickly?

5. Use a calculator with a π key to find π to 7 decimal places.

6. Use the infinite product to approximate π to 3 decimal places. Use the whole numbers from 3 through 6 as values of n.

7. Does the infinite product give good approximations for π quickly?

8. Show that $\pi^4 + \pi^5$ is equal to e^6 to 4 decimal places.

9. Which is larger, e^π or π^e?

10. The expression x reaches a maximum value at $x = e$. Use this fact to prove the inequality you found in Exercise 9.

10-6 Study Guide and Intervention

Exponential Growth and Decay

Exponential Decay Depreciation of value and radioactive decay are examples of **exponential decay**. When a quantity decreases by a fixed percent each time period, the amount of the quantity after t time periods is given by $y = a(1 - r)^t$, where a is the initial amount and r is the percent decrease expressed as a decimal.

Another exponential decay model often used by scientists is $y = ae^{-kt}$, where k is a constant.

Example **CONSUMER PRICES** **As technology advances, the price of many technological devices such as scientific calculators and camcorders goes down. One brand of hand-held organizer sells for $89.**

a. If its price decreases by 6% per year, how much will it cost after 5 years?

Use the exponential decay model with initial amount $89, percent decrease 0.06, and time 5 years.

$y = a(1 - r)^t$ Exponential decay formula

$y = 89(1 - 0.06)^5$ $a = 89$, $r = 0.06$, $t = 5$

$y = \$65.32$

After 5 years the price will be $65.32.

b. After how many years will its price be $50?

To find when the price will be $50, again use the exponential decay formula and solve for t.

$y = a(1 - r)^t$ Exponential decay formula

$50 = 89(1 - 0.06)^t$ $y = 50$, $a = 89$, $r = 0.06$

$\frac{50}{89} = (0.94)^t$ Divide each side by 89.

$\log\left(\frac{50}{89}\right) = \log (0.94)^t$ Property of Equality for Logarithms

$\log\left(\frac{50}{89}\right) = t \log 0.94$ Power Property

$t = \frac{\log\left(\frac{50}{89}\right)}{\log 0.94}$ Divide each side by log 0.94.

$t \approx 9.3$

The price will be $50 after about 9.3 years.

Exercises

1. **BUSINESS** A furniture store is closing out its business. Each week the owner lowers prices by 25%. After how many weeks will the sale price of a $500 item drop below $100?

CARBON DATING **Use the formula $y = ae^{-0.00012t}$, where a is the initial amount of Carbon-14, t is the number of years ago the animal lived, and y is the remaining amount after t years.**

2. How old is a fossil remain that has lost 95% of its Carbon-14?

3. How old is a skeleton that has 95% of its Carbon-14 remaining?

10-6 Study Guide and Intervention *(continued)*

Exponential Growth and Decay

Exponential Growth Population increase and growth of bacteria colonies are examples of **exponential growth**. When a quantity increases by a fixed percent each time period, the amount of that quantity after t time periods is given by $y = a(1 + r)^t$, where a is the initial amount and r is the percent increase (or rate of growth) expressed as a decimal.

Another exponential growth model often used by scientists is $y = ae^{kt}$, where k is a constant.

Example **A computer engineer is hired for a salary of $28,000. If she gets a 5% raise each year, after how many years will she be making $50,000 or more?**

Use the exponential growth model with $a = 28{,}000$, $y = 50{,}000$, and $r = 0.05$ and solve for t.

$y = a(1 + r)^t$	Exponential growth formula
$50{,}000 = 28{,}000(1 + 0.05)^t$	$y = 50{,}000$, $a = 28{,}000$, $r = 0.05$
$\frac{50}{28} = (1.05)^t$	Divide each side by 28,000.
$\log\left(\frac{50}{28}\right) = \log(1.05)^t$	Property of Equality of Logarithms
$\log\left(\frac{50}{28}\right) = t \log 1.05$	Power Property
$t = \frac{\log\left(\frac{50}{28}\right)}{\log 1.05}$	Divide each side by log 1.05.
$t \approx 11.9$ years	Use a calculator.

If raises are given annually, she will be making over $50,000 in 12 years.

Exercises

1. **BACTERIA GROWTH** A certain strain of bacteria grows from 40 to 326 in 120 minutes. Find k for the growth formula $y = ae^{kt}$, where t is in minutes.

2. **INVESTMENT** Carl plans to invest $500 at 8.25% interest, compounded continuously. How long will it take for his money to triple?

3. **SCHOOL POPULATION** There are currently 850 students at the high school, which represents full capacity. The town plans an addition to house 400 more students. If the school population grows at 7.8% per year, in how many years will the new addition be full?

4. **EXERCISE** Hugo begins a walking program by walking $\frac{1}{2}$ mile per day for one week. Each week thereafter he increases his mileage by 10%. After how many weeks is he walking more than 5 miles per day?

5. **VOCABULARY GROWTH** When Emily was 18 months old, she had a 10-word vocabulary. By the time she was 5 years old (60 months), her vocabulary was 2500 words. If her vocabulary increased at a constant percent per month, what was that increase?

10-6 Skills Practice

Exponential Growth and Decay

Solve each problem.

1. **FISHING** In an over-fished area, the catch of a certain fish is decreasing at an average rate of 8% per year. If this decline persists, how long will it take for the catch to reach half of the amount before the decline?

2. **INVESTING** Alex invests $2000 in an account that has a 6% annual rate of growth. To the nearest year, when will the investment be worth $3600?

3. **POPULATION** A current census shows that the population of a city is 3.5 million. Using the formula $P = ae^{rt}$, find the expected population of the city in 30 years if the growth rate r of the population is 1.5% per year, a represents the current population in millions, and t represents the time in years.

4. **POPULATION** The population P in thousands of a city can be modeled by the equation $P = 80e^{0.015t}$, where t is the time in years. In how many years will the population of the city be 120,000?

5. **BACTERIA** How many days will it take a culture of bacteria to increase from 2000 to 50,000 if the growth rate per day is 93.2%?

6. **NUCLEAR POWER** The element plutonium-239 is highly radioactive. Nuclear reactors can produce and also use this element. The heat that plutonium-239 emits has helped to power equipment on the moon. If the half-life of plutonium-239 is 24,360 years, what is the value of k for this element?

7. **DEPRECIATION** A Global Positioning Satellite (GPS) system uses satellite information to locate ground position. Abu's surveying firm bought a GPS system for $12,500. The GPS depreciated by a fixed rate of 6% and is now worth $8600. How long ago did Abu buy the GPS system?

8. **BIOLOGY** In a laboratory, an organism grows from 100 to 250 in 8 hours. What is the hourly growth rate in the growth formula $y = a(1 + r)^t$?

NAME _____ DATE_____ PERIOD _____

10-6 Practice

Exponential Growth and Decay

Solve each problem.

1. **INVESTING** The formula $A = P\left(1 + \frac{r}{2}\right)^{2t}$ gives the value of an investment after t years in an account that earns an annual interest rate r compounded twice a year. Suppose \$500 is invested at 6% annual interest compounded twice a year. In how many years will the investment be worth \$1000?

2. **BACTERIA** How many hours will it take a culture of bacteria to increase from 20 to 2000 if the growth rate per hour is 85%?

3. **RADIOACTIVE DECAY** A radioactive substance has a half-life of 32 years. Find the constant k in the decay formula for the substance.

4. **DEPRECIATION** A piece of machinery valued at \$250,000 depreciates at a fixed rate of 12% per year. After how many years will the value have depreciated to \$100,000?

5. **INFLATION** For Dave to buy a new car comparably equipped to the one he bought 8 years ago would cost \$12,500. Since Dave bought the car, the inflation rate for cars like his has been at an average annual rate of 5.1%. If Dave originally paid \$8400 for the car, how long ago did he buy it?

6. **RADIOACTIVE DECAY** Cobalt, an element used to make alloys, has several isotopes. One of these, cobalt-60, is radioactive and has a half-life of 5.7 years. Cobalt-60 is used to trace the path of nonradioactive substances in a system. What is the value of k for Cobalt-60?

7. **WHALES** Modern whales appeared 5−10 million years ago. The vertebrae of a whale discovered by paleontologists contain roughly 0.25% as much carbon-14 as they would have contained when the whale was alive. How long ago did the whale die? Use $k = 0.00012$.

8. **POPULATION** The population of rabbits in an area is modeled by the growth equation $P(t) = 8e^{0.26t}$, where P is in thousands and t is in years. How long will it take for the population to reach 25,000?

9. **DEPRECIATION** A computer system depreciates at an average rate of 4% per month. If the value of the computer system was originally \$12,000, in how many months is it worth \$7350?

10. **BIOLOGY** In a laboratory, a culture increases from 30 to 195 organisms in 5 hours. What is the hourly growth rate in the growth formula $y = a(1 + r)^t$?

NAME _____ DATE_____ PERIOD _____

10-6 Reading to Learn Mathematics

Exponential Growth and Decay

Pre-Activity

How can you determine the current value of your car?

Read the introduction to Lesson 10-6 at the top of page 560 in your textbook.

- Between which two years shown in the table did the car depreciate by the greatest amount?

- Describe two ways to calculate the value of the car 6 years after it was purchased. (Do not actually calculate the value.)

Reading the Lesson

1. State whether each situation is an example of exponential *growth* or *decay*.

a. A city had 42,000 residents in 1980 and 128,000 residents in 2000.

b. Raul compared the value of his car when he bought it new to the value when he traded ';lpit in six years later.

c. A paleontologist compared the amount of carbon-14 in the skeleton of an animal when it died to the amount 300 years later.

d. Maria deposited $750 in a savings account paying 4.5% annual interest compounded quarterly. She did not make any withdrawals or further deposits. She compared the balance in her passbook immediately after she opened the account to the balance 3 years later.

2. State whether each equation represents exponential growth or decay.

a. $y = 5e^{0.15t}$

b. $y = 1000(1 - 0.05)^t$

c. $y = 0.3e^{-1200t}$

d. $y = 2(1 + 0.0001)^t$

Helping You Remember

3. Visualizing their graphs is often a good way to remember the difference between mathematical equations. How can your knowledge of the graphs of exponential equations from Lesson 10-1 help you to remember that equations of the form $y = a(1 + r)^t$ represent exponential growth, while equations of the form $y = a(1 - r)^t$ represent exponential decay?

10-6 Enrichment

Effective Annual Yield

When interest is compounded more than once per year, the effective annual yield is higher than the annual interest rate. The effective annual yield, E, is the interest rate that would give the same amount of interest if the interest were compounded once per year. If P dollars are invested for one year, the value of the investment at the end of the year is $A = P(1 + E)$. If P dollars are invested for one year at a nominal rate r compounded n times per year, the value of the investment at the end of the year is $A = P\left(1 + \frac{r}{n}\right)^n$. Setting the amounts equal and solving for E will produce a formula for the effective annual yield.

$$P(1 + E) = P\left(1 + \frac{r}{n}\right)^n$$
$$1 + E = \left(1 + \frac{r}{n}\right)^n$$
$$E = \left(1 + \frac{r}{n}\right)^n - 1$$

If compounding is continuous, the value of the investment at the end of one year is $A = Pe^r$. Again set the amounts equal and solve for E. A formula for the effective annual yield under continuous compounding is obtained.

$$P(1 + E) = Pe^r$$
$$1 + E = e^r$$
$$E = e^r - 1$$

Example 1 **Find the effective annual yield of an investment made at 7.5% compounded monthly.**

$r = 0.075$

$n = 12$

$E = \left(1 + \frac{0.075}{12}\right)^{12} - 1 \approx 7.76\%$

Example 2 **Find the effective annual yield of an investment made at 6.25% compounded continuously.**

$r = 0.0625$

$E = e^{0.0625} - 1 \approx 6.45\%$

Find the effective annual yield for each investment.

1. 10% compounded quarterly

2. 8.5% compounded monthly

3. 9.25% compounded continuously

4. 7.75% compounded continuously

5. 6.5% compounded daily (assume a 365-day year)

6. Which investment yields more interest—9% compounded continuously or 9.2% compounded quarterly?

NAME _____ DATE _____ PERIOD _____

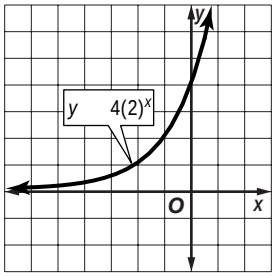
10

Chapter 10 Test, Form 1

SCORE _____

Write the letter for the correct answer in the blank at the right of each question.

1. Find the domain and range of the function whose graph is shown.



A. D = $\{x \mid x > 0\}$; R = $\{y \mid y > 0\}$
B. D = $\{x \mid x$ is any real number.$\}$; R = $\{y \mid y > 0\}$
C. D = $\{x \mid x > 0\}$; R = $\{y \mid y$ is any real number.$\}$
D. D = $\{x \mid x$ is any real number.$\}$; R = $\{y \mid y < 0\}$ **1.** _____

2. Which function represents exponential growth?

A. $y = 9\left(\frac{1}{3}\right)^x$ **B.** $y = 4x^4$ **C.** $y = 12\left(\frac{1}{5}\right)^x$ **D.** $y = 10(2)^x$ **2.** _____

3. The graph of which exponential function passes through the points (0, 4) and (1, 24)?

A. $y = 4(6)^x$ **B.** $y = 3(8)^x$ **C.** $y = 2(2)^x$ **D.** $y = 10(3)^x$ **3.** _____

4. Simplify $\left(x^{\sqrt{7}}\right)^{\sqrt{3}}$.

A. $x^{\sqrt{21}}$ **B.** $x^{\sqrt{7}+\sqrt{3}}$ **C.** $x^{\sqrt{10}}$ **D.** $x^{\sqrt{\frac{7}{3}}}$ **4.** _____

5. Solve $2^{3m-4} > 4$.

A. $m < 0$ **B.** $m > 0$ **C.** $m > 2$ **D.** $m > \frac{5}{3}$ **5.** _____

6. Write the equation $4^3 = 64$ in logarithmic form.

A. $\log_4 3 = 64$ **B.** $\log_3 4 = 64$ **C.** $\log_{64} 4 = 3$ **D.** $\log_4 64 = 3$ **6.** _____

7. Write the equation $\log_{12} 144 = 2$ in exponential form.

A. $144^2 = 12$ **B.** $12^2 = 144$ **C.** $2^{12} = 144$ **D.** $144^{12} = 2$ **7.** _____

8. Evaluate $\log_2 8$.

A. 3 **B.** 4 **C.** 16 **D.** 64 **8.** _____

9. Solve $\log_3 n = 2$.

A. 6 **B.** 5 **C.** 8 **D.** 9 **9.** _____

10. Solve $\log_2 2m > \log_2 (m + 5)$.

A. $m > \frac{5}{3}$ **B.** $m < 5$ **C.** $m > 5$ **D.** $m > -5$ **10.** _____

Assessment

10 Chapter 10 Test, Form 1 *(continued)*

11. Use $\log_5 2 \approx 0.4307$ to approximate the value of $\log_5 4$.
A. 0.8614 **B.** 0.8980 **C.** 1.3652 **D.** 0.1855 **11.** _____

12. Solve $\log_6 10 + \log_6 x = \log_6 40$.
A. 180 **B.** 4 **C.** 5 **D.** 30 **12.** _____

13. Solve $4^x = 20$. Round to four decimal places.
A. 0.4628 **B.** 1.5214 **C.** 0.6990 **D.** 2.1610 **13.** _____

14. Solve $3^x \geq 21$. Round to four decimal places.
A. $x \geq 0.8451$ **B.** $x \geq 2.7712$ **C.** $x \geq 0.3608$ **D.** $x \geq 7.0000$ **14.** _____

15. Express $\log_9 22$ in terms of common logarithms.
A. $\log \frac{22}{9}$ **B.** $\log 198$ **C.** $\frac{\log 22}{\log 9}$ **D.** $\frac{\log 9}{\log 22}$ **15.** _____

16. Evaluate $e^{\ln 4}$.
A. e^4 **B.** 4^e **C.** $\ln 4$ **D.** 4 **16.** _____

17. Solve $e^x > 2.7$.
A. $x < 0.9933$ **B.** $x > 0.9933$ **C.** $x > 1.0668$ **D.** $x < 1.0668$ **17.** _____

18. Solve $\ln 3x = 1$.
A. 20.0855 **B.** 0.3333 **C.** 0.9061 **D.** 8.1548 **18.** _____

19. **AUTOMOBILES** Lydia bought a car for \$20,000. It is expected to depreciate at a rate of 10% per year. What will be the value of the car in 2 years? Use $y = a(1 - r)^t$ and round to the nearest dollar.
A. \$16,200 **B.** \$16,000 **C.** \$19,980 **D.** \$18,050 **19.** _____

20. **ART** Martin bought a painting for \$5,000. It is expected to appreciate at 4% per year. How much will the painting be worth in 6 years? Use $y = a(1 + r)^t$ and round to the nearest cent.
A. \$6200.00 **B.** \$5360.38 **C.** \$37,647.68 **D.** \$6326.60 **20.** _____

Bonus Evaluate $3 \log_2 64 + e^{\ln 5} + \log_{\frac{1}{3}} 9$. **B:** _____

NAME _____ DATE _____ PERIOD _____

10 Chapter 10 Test, Form 2A

SCORE _____

Write the letter for the correct answer in the blank at the right of each question.

1. Find the domain and range of the function $y = 3\left(\frac{1}{5}\right)^x$.

A. D = $\{x \mid x$ is any real number.$\}$
R = $\{y \mid y < 0\}$

B. D = $\{x \mid x$ is any real number.$\}$
R = $\{y \mid y > 0\}$

C. D = $\{x \mid x > 0\}$
R = $\{y \mid y > 0\}$

D. D = $\{x \mid x > 0\}$
R = $\{y \mid y$ is any real number.$\}$

1. _____

2. Which function represents exponential *decay*?

A. $y = \frac{1}{100}(6)^x$ **B.** $y = (4x)^{\frac{1}{2}}$ **C.** $y = 2\left(\frac{4}{3}\right)^x$ **D.** $y = 12\left(\frac{1}{8}\right)^x$ **2.** _____

3. Use the equation of the exponential function whose graph passes through the points $(0, -3)$ and $(2, -48)$ to find the value of y when $x = -2$.

A. $-\frac{3}{4}$ **B.** $-\frac{3}{8}$ **C.** $-\frac{3}{16}$ **D.** 48 **3.** _____

4. Simplify $m^{9\sqrt{5}} \div m^{\sqrt{5}}$.

A. m^{45} **B.** m^9 **C.** $m^{8\sqrt{5}}$ **D.** $m^{10\sqrt{5}}$ **4.** _____

5. Solve $\left(\frac{1}{36}\right)^n = 216^{n+5}$.

A. 10 **B.** 3 **C.** -3 **D.** -10 **5.** _____

6. Solve $81^y < 27^{y+3}$

A. $y < -9$ **B.** $y > 9$ **C.** $y > -9$ **D.** $y < 9$ **6.** _____

7. Write the equation $6561^{\frac{1}{4}} = 9$ in logarithmic form.

A. $\log_{\frac{1}{4}} 9 = 6561$

B. $\log_{6561} 9 = \frac{1}{4}$

C. $\log_9 6561 = \frac{1}{4}$

D. $\log_{\frac{1}{4}} 6561 = 9$ **7.** _____

8. Evaluate $5^{\log_5 63}$.

A. 58 **B.** 315 **C.** $\log_5 63$ **D.** 63 **8.** _____

9. Solve $\log_{\frac{1}{5}} x = -1$.

A. $\frac{1}{25}$ **B.** -5 **C.** 5 **D.** $-\frac{1}{5}$ **9.** _____

10. Solve $\log_3 (5x + 1) \geq \log_3 (3x + 7)$

A. $x \geq 3$ **B.** $x \geq 4$ **C.** $x \leq 6$ **D.** $x \geq 27$ **10.** _____

Assessment

10 Chapter 10 Test, Form 2A *(continued)*

11. Use $\log_5 2 \approx 0.4307$ and $\log_5 3 \approx 0.6826$ to approximate the value of $\log_5 54$.
A. 0.1370 **B.** 2.4785 **C.** 0.8820 **D.** 0.7488 **11.** _____

12. Solve $\log_4 (m - 3) + \log_4 (m + 3) = 2$.
A. $\sqrt{11}$ **B.** 5 **C.** 1 **D.** -5.5 **12.** _____

13. Solve $6^{3n} = 43^{5n - 4}$. Round to four decimal places.
A. 1.1202 **B.** -1.9005 **C.** -0.2800 **D.** 2.1418 **13.** _____

14. Solve $5^{2x + 1} \geq 50$. Round to four decimal places.
A. $x \geq 4.5000$ **B.** $x \geq 0.7153$ **C.** $x \geq 0$ **D.** $x \geq 2.4307$ **14.** _____

15. Use common logarithms to approximate $\log_9 207$ to four decimal places.
A. 0.4120 **B.** 1.3617 **C.** 3.2702 **D.** 2.4270 **15.** _____

16. Evaluate $\ln e^{-9x}$.
A. $-9 \ln x$ **B.** $9 \ln x$ **C.** $-9x$ **D.** $9x$ **16.** _____

17. Solve $4 + 3e^{5x} = 27$.
A. 0.4074 **B.** 0.4394 **C.** 2.0369 **D.** 0.1769 **17.** _____

18. Solve $\ln (x + 5) \geq 2$.
A. $x \geq 2.3891$ **B.** $x \leq 2.3891$ **C.** $x \geq 12.3891$ **D.** $x \leq 12.3891$ **18.** _____

19. **CHEMISTRY** A particular compound decays according to the equation $y = ae^{-0.0974t}$, where t is in days. Find the half-life of this compound.
A. about 5.1 days **B.** about 7.4 days
C. about 7.1 days **D.** about 9.7 days **19.** _____

20. **TOURISM** At a town with a large convention center, the cost of a hotel room has increased 5.1% annually. If the average hotel room cost \$48.00 in 1980 and this growth continues, what will an average hotel room cost in 2012? Use $y = a(1 + r)^t$ and round to the nearest cent.
A. \$143.38 **B.** \$235.79 **C.** \$126.34 **D.** \$87.19 **20.** _____

Bonus Solve $5^{\log_5 2x - \log_5 (x - 3)} = \ln e^{x + 4}$. **B:** _____

NAME _____ DATE _____ PERIOD _____

10

Chapter 10 Test, Form 2B

SCORE _____

Write the letter for the correct answer in the blank at the right of each question.

1. Find the domain and range of the function $y = \frac{1}{2}(2)^x$.

A. D = $\{x \mid x$ is any real number.$\}$
R = $\{y \mid y > 0\}$

B. D = $\{x \mid x$ is any real number.$\}$
R = $\{y \mid y < 0\}$

C. D = $\{x \mid x > 0\}$
R = $\{y \mid y > 0\}$

D. D = $\{x \mid x > 0\}$
R = $\{y \mid y$ is any real number.$\}$

1. _____

2. Which function represents exponential *growth*?

A. $y = \frac{1}{20}\left(\frac{5}{2}\right)^x$ **B.** $y = 16(0.4)^x$ **C.** $y = \frac{1}{2}\left(\frac{1}{8}\right)^x$ **D.** $y = 8x^3$ **2.** _____

3. Use the equation of the exponential function whose graph passes through the points $(0, -2)$ and $(2, -50)$ to find the value of y when $x = -2$.

A. $-\frac{1}{100}$ **B.** 50 **C.** $-\frac{2}{25}$ **D.** $-\frac{1}{50}$ **3.** _____

4. Simplify $s^{7\sqrt{11}} \div s^{\sqrt{11}}$.

A. s^{77} **B.** $s^{6\sqrt{11}}$ **C.** $s^{8\sqrt{11}}$ **D.** s^7 **4.** _____

5. Solve $\left(\frac{1}{81}\right)^t = 243^{t-2}$.

A. $\frac{9}{2}$ **B.** $\frac{10}{9}$ **C.** $\frac{2}{9}$ **D.** $\frac{9}{10}$ **5.** _____

6. Solve $64^x < 32^{x+2}$.

A. $x > -10$ **B.** $x < -10$ **C.** $x > 10$ **D.** $x < 10$ **6.** _____

7. Write the equation $\log_{243} 81 = \frac{4}{5}$ in exponential form.

A. $81^{\frac{4}{5}} = 243$ **B.** $243^{\frac{4}{5}} = 81$ **C.** $\left(\frac{4}{5}\right)^{81} = 243$ **D.** $\left(\frac{4}{5}\right)^{243} = 81$ **7.** _____

8. Evaluate $9^{\log_9 54}$.

A. $\log_9 54$ **B.** 54 **C.** 6 **D.** 486 **8.** _____

9. Solve $\log_{\frac{1}{8}} x = -1$.

A. 8 **B.** -8 **C.** 0 **D.** $-\frac{1}{8}$ **9.** _____

10. Solve $\log_2 (7x - 3) \geq \log_2 (x + 12)$.

A. $x \leq \frac{5}{2}$ **B.** $x \leq -\frac{5}{2}$ **C.** $x \geq \frac{3}{2}$ **D.** $x \geq \frac{5}{2}$ **10.** _____

Assessment

10 Chapter 10 Test, Form 2B *(continued)*

11. Use $\log_5 2 \approx 0.4307$ and $\log_5 3 \approx 0.6826$ to approximate the value of $\log_5 12$.
A. 0.8681 **B.** 0.1266 **C.** 1.5440 **D.** 0.5880 **11.** _____

12. Solve $\log_3 a + \log_3 (a - 8) = 2$.
A. 8 **B.** 5 **C.** 9 **D.** −1, 9 **12.** _____

13. Solve $9^{2n} = 40^{4n - 7}$. Round to four decimal places.
A. 2.4922 **B.** 0.4012 **C.** −0.3560 **D.** −4.7209 **13.** _____

14. Solve $3^{5x - 1} \leq 30$. Round to four decimal places.
A. $x \leq 0.4000$ **B.** $x \leq 0.8192$ **C.** $x \leq 1.8000$ **D.** $x \leq 3.0959$ **14.** _____

15. Use common logarithms to approximate $\log_7 448$ to four decimal places.
A. 0.3188 **B.** 1.8062 **C.** 3.1372 **D.** −1.8062 **15.** _____

16. Evaluate $e^{\ln(-6x)}$.
A. $-6x$ **B.** $6 \ln x$ **C.** $6x$ **D.** $-6 \ln x$ **16.** _____

17. Solve $\ln (x + 2) = 3$.
A. 22.0855 **B.** 18.0855 **C.** 20.0855 **D.** −0.9014 **17.** _____

18. Solve $e^{-9x} \leq 6$.
A. $x \geq -1.8122$ **B.** $x \leq 1.7918$
C. $x \geq -0.08646$ **D.** $x \geq -0.1991$ **18.** _____

19. **CHEMISTRY** A particular compound decays according to the equation $y = ae^{-0.0736t}$, where t is in days. Find the half-life of the compound.
A. about 9.1 days **B.** about 9.4 days
C. about 6.8 days **D.** about 7.4 days **19.** _____

20. **FOOD PRICES** At a wholesale food distribution center, the price of sugar has increased 6.3% annually since 1980. Suppose sugar cost \$0.43 per pound in 1980 and this growth continues. What will a pound of sugar cost in 2017? Use $y = a(1 + r)^t$ and round to the nearest cent.
A. \$4.12 **B.** \$1.21 **C.** \$2.42 **D.** \$3.30 **20.** _____

Bonus Solve $2^{\log_2 5x - \log_2 (x + 1)} = \ln e^x$. **B:** _____

NAME _____ DATE _____ PERIOD _____

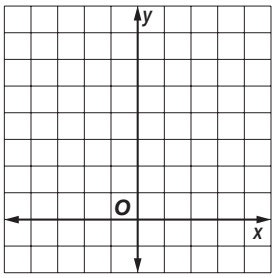
10

Chapter 10 Test, Form 2C

SCORE _____

1. Sketch the graph of $y = \frac{1}{2}(3)^x$. Then state the function's domain and range.

1.



_____

2. Determine whether the function $y = 0.8\left(\frac{2}{3}\right)^x$ represents exponential *growth* or *decay*. **2.** _____

3. Write an exponential function whose graph passes through the points $(0, -6)$ and $(-2, -54)$. **3.** _____

4. Simplify $z^{5\sqrt{7}} \cdot z^{\sqrt{7}}$. **4.** _____

5. Solve $\frac{1}{6} = 6^{n+4}$. **5.** _____

6. Solve $32^x < 16^{x+2}$. **6.** _____

7. Write the equation $\log_{81} \frac{1}{9} = -\frac{1}{2}$ in exponential form. **7.** _____

8. Evaluate $\log_9 9^7$. **8.** _____

9. Evaluate $\log_4 128$. **9.** _____

10. Solve $\log_{36} n = \frac{3}{2}$. **10.** _____

11. Solve $\log_5 (8x) > \log_5 (3x + 10)$. **11.** _____

Use $\log_5 2 \approx 0.4307$ and $\log_5 3 \approx 0.6826$ to approximate the value of each expression.

12. $\log_5 48$ **12.** _____

13. $\log_5 \frac{5}{3}$ **13.** _____

Assessment

10 Chapter 10 Test, Form 2C *(continued)*

For Questions 14–19, solve each equation or inequality. If necessary, round to four decimal places.

14. $\log_4 n = \frac{1}{4}\log_4 81 + \frac{1}{2}\log_4 25$ **14.** _____

15. $\log_2 (2x + 6) - \log_2 x = 3$ **15.** _____

16. $\log_3 (x + 3) + \log_3 (x - 2) = \log_3 14$ **16.** _____

17. $6^{n-2} = 50$ **17.** _____

18. $2^y = 5^{y-2}$ **18.** _____

19. $4^{3x+1} < 28$ **19.** _____

20. Express $\log_{12} 4$ in terms of common logarithms. Then approximate its value to four decimal places. **20.** _____

21. Evaluate $\ln e^{30}$. **21.** _____

22. Solve $\ln (x - 5) = 3$. **22.** _____

23. Solve $e^{-4x} \leq 9$. **23.** _____

24. **CHEMISTRY** After 12 hours, half of a 16-gram sample of a radio-active element remains. Find the constant k for this element for t hours, then write the equation for modeling its exponential decay. **24.** _____

25. **SAVINGS** A savings account deposit of \$150 is to earn 6.5% interest. After how many years will the investment be worth \$450? Use $y = a(1 + r)^t$ and round to the nearest tenth. **25.** _____

Bonus Evaluate $(\log_4 12^3)(\log_{12} 4^3)$. **B:** _____

NAME _____ DATE _____ PERIOD _____

10

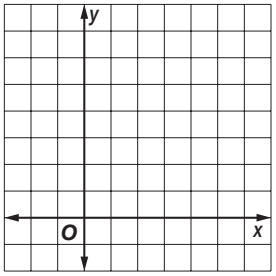
Chapter 10 Test, Form 2D

SCORE _____

Assessment

1. Sketch the graph of $y = 6\left(\frac{1}{2}\right)^x$. Then state the function's domain and range.

1.



_____

2. Determine whether the function $y = 0.3\left(\frac{8}{5}\right)^x$ represents exponential *growth* or *decay*. **2.** _____

3. Write an exponential function whose graph passes through the points $(0, -5)$ and $(-3, -40)$. **3.** _____

4. Simplify $m^{9\sqrt{6}} \div m^{\sqrt{6}}$. **4.** _____

5. Solve $\left(\frac{1}{4}\right)^{m-7} = 16$. **5.** _____

6. Solve $10^{x+3} > 100^{x-1}$. **6.** _____

7. Write the equation $5^{-4} = \frac{1}{625}$ in logarithmic form. **7.** _____

8. Evaluate $\log_6 6^8$. **8.** _____

9. Evaluate $\log_8 128$. **9.** _____

10. Solve $\log_{64} x = \frac{2}{3}$. **10.** _____

11. Solve $\log_4 (2x + 5) \leq \log_4 (3x - 2)$. **11.** _____

Use $\log_5 2 \approx 0.4307$ and $\log_5 3 \approx 0.6826$ to approximate the value of each expression.

12. $\log_5 18$ **12.** _____

13. $\log_5 \frac{5}{2}$ **13.** _____

10 Chapter 10 Test, Form 2D *(continued)*

For Questions 14–19, solve each equation or inequality. If necessary, round to four decimal places.

14. $\log_5 n = \frac{1}{3}\log_5 64 + \frac{1}{2}\log_5 49$ **14.** _____

15. $\log_6 (5 - 2a) - \log_6 (3a) = 1$ **15.** _____

16. $\log_3 (x - 3) + \log_3 (x + 2) = \log_3 6$ **16.** _____

17. $7^{n+3} = 80$ **17.** _____

18. $3^n = 6^{n-2}$ **18.** _____

19. $5^{4x-1} < 30$ **19.** _____

20. Express $\log_{15} 5$ in terms of common logarithms. Then approximate its value to four decimal places. **20.** _____

21. Evaluate $e^{\ln 22}$. **21.** _____

22. Solve $\ln (x + 4) = 4$. **22.** _____

23. Solve $e^{-3x} \leq 18$. **23.** _____

24. **CHEMISTRY** In 5 years, radioactivity reduces the mass of a 100-gram sample of an element to 75 grams. Find the constant k for this element for t in years, then write the equation for modeling this exponential decay. **24.** _____

25. **SAVINGS** A savings account deposit of \$300 is to earn 5.8% interest. After how many years will the investment be worth \$900? Use $y = a(1 + r)^t$ and round to the nearest tenth. **25.** _____

Bonus Evaluate $(\log_5 20^4)(\log_{20} 5^4)$. **B:** _____

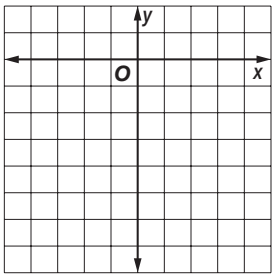
NAME _____ DATE _____ PERIOD _____

10 Chapter 10 Test, Form 3

SCORE _____

1. Sketch the graph of $y = -1.5(4)^x$. Then state the function's domain and range.

1.



_____

2. Determine whether the function $y = 0.4(3.8)^{-x}$ represents exponential *growth* or *decay*. **2.** _____

3. Write an exponential function whose graph passes through $(0, -0.3)$ and $(2, -10.8)$ **3.** _____

4. Solve $2^{4x} \cdot 32^{1-x} = 8^{x+2}$. **4.** _____

5. Solve $\left(\frac{1}{81}\right)^{4m+1} \geq \left(\frac{1}{27}\right)^{5m}$. **5.** _____

6. Evaluate $2^{\log_2 (8x-1)}$. **6.** _____

7. Evaluate $\log_3 243^x$. **7.** _____

8. Solve $\log_x [\log_2 (\log_3 9)] = 2$. **8.** _____

9. Solve $\log_3 (a^2 - 12) > \log_3 a$. **9.** _____

For Questions 10 and 11, use $\log_5 2 \approx 0.4307$ and $\log_5 3 \approx 0.6826$ to approximate the value of each expression.

10. $\log_5 \frac{15}{4}$ **10.** _____

11. $\log_5 1.2$ **11.** _____

12. Solve $\log_4 0.25 + 3 \log_4 x = 5 \log_4 2 + \frac{1}{3} \log_4 64$. **12.** _____

13. Solve $\log_4 (4b + 14) - \log_4 (b^2 - 3b - 17) = \frac{1}{2}$. **13.** _____

Assessment

10 Chapter 10 Test, Form 3 *(continued)*

14. Solve $\log_a (3n) - 2 \log_a x = \log_a x$ for n. **14.** _____

15. Solve $4.5^{x^2 + 2} = 32.7$. Round to four decimal places. **15.** _____

16. Solve $3^n = \sqrt{5^{n-2}}$. Round to four decimal places. **16.** _____

17. Solve $\left(\frac{1}{2}\right)^{2t} > 5^{3-4t}$. Round to four decimal places. **17.** _____

18. Express $\log_5 (2.1)^3$ in terms of common logarithms. Then approximate its value to four decimal places. **18.** _____

19. Evaluate $e^{4 \ln x}$. **19.** _____

20. Solve $\ln (x + 3) + \ln x = \ln 4$. **20.** _____

21. Solve $\ln (x^2 + 10) > \ln x + \ln 7$. **21.** _____

POPULATIONS For Questions 22 and 23, use the following information.

The population of Suffolk County in Massachusetts decreased from 663,906 in 1990 to 641,695 in 1999.

22. Write an exponential decay equation of the form $y = ae^{kt}$ for Suffolk County, where t is the number of years after 1990. **22.** _____

23. Use your equation to predict the population of Suffolk County in 2020. **23.** _____

24. **HOME OWNERSHIP** The Richardson family bought a house 12 years ago for \$95,000. The house is now worth \$167,000. Assuming a steady rate of growth, what was the yearly rate of appreciation? **24.** _____

25. **SCHOOL ENROLLMENT** At a certain school, the number of children entering kindergarten increased by 6.7% annually for 5 years and then decreased by 4.2% annually in the next 5 years. If 110 children enrolled in kindergarten at the beginning of this period, how many were enrolled after 10 years? **25.** _____

Bonus Solve $\log x^2 = (\log x)^2$. **B:** _____

NAME _____ DATE _____ PERIOD _____

10 Chapter 10 Open-Ended Assessment

SCORE _____

Demonstrate your knowledge by giving a clear, concise solution to each problem. Be sure to include all relevant drawings and justify your answers. You may show your solutions in more than one way or investigate beyond the requirements of the problem.

1. For the equation $y = ab^x$, where $a > 0$, we know that if $b > 1$, the function represents exponential growth, while it represents exponential decay if $0 < b < 1$.

a. Choose a positive value for a and let $b = 1$. Complete the table for these values of a and b. Is $y = ab^x$ an exponential function? Explain your reasoning.

x	−3	−2	−1	0	1	2	3
$y = ab^x$							

b. Choose a positive value for a and a negative value for b. Complete the table for these values. Is $y = ab^x$ an exponential function? Explain your reasoning.

x	−3	−2	−1	0	1	2	3
$y = ab^x$							

2. a. Solve the exponential equation $3^{5x} = 9^{x+6}$ by rewriting the equation so that each side has the same base.

b. Solve the equation in part **a** using common logarithms.

c. Which method do you prefer? Explain your reasoning.

d. Write and solve an exponential equation that you would also prefer to solve using the method you chose in part **c**.

3. a. How are the three equations below alike? How are they different?
$\log_3 x = 2 \qquad \log x = 2 \qquad \ln x = 2$

b. Solve each equation in part **a** above. Then write and solve a fourth equation that shares the same similarities and differences as the three given equations.

4. Ruby solved the exponential inequality $2^{2z} \geq 12^{z+1}$ and stated that the solution set was $\{z \mid z \geq -2.2619\}$. When she checked her solution however, Ruby found that $z = 1$, which *is* in her solution set, does *not* make the original inequality true. When she checked $z = -3$, which is *not* in her solution set, the original inequality *is* true.

a. Show how Ruby arrived at her solution using common logarithms.

b. What do her checks of $z = 1$ and $z = -3$ indicate about Ruby's solution?

c. What change must be made to the solution and why must that change be made?

5. ECONOMICS The Jones Corporation found that its annual profit could be modeled by the exponential equation $y = 10(0.99)^t$, while the Davis Company's annual profit is modeled by $y = 8(1.01)^t$. For both equations, profit is given in millions of dollars, and t is the number of years since 1990.

a. Find each company's annual profit for the years between 1990 and 2000 to the nearest dollar.

b. In which company would you prefer to own stock? Explain your reasoning.

c. Indicate how a comparison of the two profit equations would support your decision.

Assessment

NAME _____ DATE _____ PERIOD _____

10

Chapter 10 Vocabulary Test/Review

SCORE _____

Change of Base Formula	exponential growth	natural base, e	natural logarithmic function
common logarithm	exponential inequality	natural base exponential function	rate of decay
exponential decay	logarithm	natural logarithm	rate of growth
exponential equation	logarithmic function		
exponential function			

Choose from the terms above to complete each sentence.

1. A logarithm with base e is called a(n) _____.

2. The function $y = 10^x$ is an example of a(n) _____.

3. The equation $y = e^{-0.2t}$ is a model for _____.

4. The inverse of the function $y = e^x$ is the _____

5. The equation $y = 100(1 + 0.1)^t$ is a model for _____.

6. The value of $\log_3 50$ can be found by using the _____ with a calculator.

7. $y = \log_2 x$ is an example of a _____.

8. $5^{x+1} = 125$ and $9^x = 27^{2x+1}$ are examples of _____.

9. A logarithm with base 10 is called a(n) _____.

10. In the equation $y = 20(1 + 0.02)^t$, 0.02 is called the _____.

In your own words—
Define each term.

11. logarithm

12. natural base, e

NAME _____ DATE _____ PERIOD _____

10

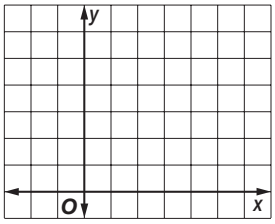
Chapter 10 Quiz

(Lessons 10–1 and 10–2)

SCORE _____

1. Sketch the graph of $y = 3\left(\frac{1}{2}\right)^x$. Then state the function's domain and range.

1.



_____

2. Write an exponential function whose graph passes through the points $(0, -5)$ and $(-2, -20)$. Then determine whether the function represents exponential *growth* or *decay*.

3. Simplify $3^{\sqrt{5}} \cdot 3^{2\sqrt{5}}$. **3.** _____

4. Solve $\left(\frac{1}{3}\right)^m = 27^{m+2}$. **4.** _____

5. Solve $25^{4t+1} \geq 125^{2t}$. **5.** _____

6. Write the equation $81^{\frac{1}{2}} = 9$ in logarithmic form. **6.** _____

7. Write the equation $\log_{216} 36 = \frac{2}{3}$ in exponential form. **7.** _____

8. Evaluate $\log_{16} 64$. **8.** _____

9. Solve $\log_{16} n = -\frac{1}{2}$. **9.** _____

10. Solve $\log_5 (4x - 1) \geq \log_5 (x + 2)$. **10.** _____

Assessment

NAME _____ DATE _____ PERIOD _____

10

Chapter 10 Quiz

(Lesson 10–3)

SCORE _____

Use $\log_5 2 \approx 0.4307$ and $\log_5 3 \approx 0.6826$ to approximate the value of each expression.

1. $\log_5 \frac{10}{3}$ **2.** $\log_5 24$

1. _____ **2.** _____

Solve each equation.

3. $\log_7 36 - \log_7 (2x) = \log_7 4$ **3.** _____

4. $\log_3 x = \frac{1}{2} \log_3 25 - 5 \log_3 2$ **4.** _____

5. $\log_2 (x + 1) + \log_2 (x - 5) = 4$ **5.** _____

NAME _____ DATE _____ PERIOD _____

10

Chapter 10 Quiz

SCORE _____

(Lessons 10–4 and 10–5)

Use a calculator to evaluate each expression to four decimal places.

1. log 1.5 **2.** ln 4.1

For Questions 3–7, solve each equation or inequality. Round to four decimal places.

3. $4^{2m} = 130$ **4.** $5^{x+4} = 2^{3x}$

5. $7^{t-5} < 21.5$ **6.** $\ln(x + 5) = 3$

7. $4 + 2e^{5x} \geq 28$

8. Express $\log_3 25$ in terms of common logarithms. Then approximate its value to four decimal places.

9. Write an equivalent logarithmic equation for $e^3 = 2x$.

10. Evaluate $e^{\ln 0.3}$.

1. _____ **2.** _____ **3.** _____ **4.** _____ **5.** _____ **6.** _____ **7.** _____ **8.** _____ **9.** _____ **10.** _____

NAME _____ DATE _____ PERIOD _____

10

Chapter 10 Quiz

SCORE _____

(Lesson 10–6)

1. A substance decays according to the equation $y = ae^{-0.0025t}$, where t is in minutes. Find the half-life of the substance. Round to the nearest tenth.

2. A-1 Electric has a piece of machinery valued at $55,000. It depreciates at a rate of 12.5% per year. After how many years will the value have depreciated to $38,000? Round to the nearest tenth.

3. **Standardized Test Practice** In 1925, the population of a city was 90,000. Since then, the population has increased by 2.1% per year. If it continues to grow at this rate, what will the population be in 2020?

A. 4,073,333 **B.** 136,382 **C.** 648,169 **D.** 6.6×10^{12}

4. The Morgans bought a house worth $125,000. Assuming that the house will appreciate 8% per year, what will the house be worth in eight years? Round to the nearest dollar.

5. A type of bacteria doubles in number every 25 minutes. Find the constant k for this type of bacteria, then write the equation for modeling this exponential growth.

1. _____ **2.** _____ **3.** _____ **4.** _____ **5.** _____

NAME _____ DATE _____ PERIOD _____

10

Chapter 10 Mid-Chapter Test

SCORE _____

(Lessons 10–1 through 10–4)

Part I ***Write the letter for the correct answer in the blank at the right of each question.***

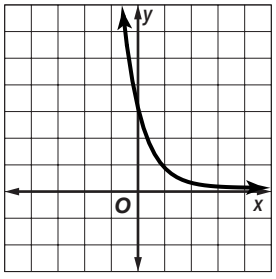
1. Find the domain and range of the function shown.

A. D = $\{x \mid x > 0\}$, R = $\{y \mid y$ is any real number.$\}$

B. D = $\{x \mid x$ is any real number.$\}$, R = $\{y \mid y < 0\}$

C. D = $\{x \mid x$ is any real number.$\}$, R = $\{y \mid y > 0\}$

D. D = $\{x \mid x > 0\}$, R = $\{y \mid y > 0\}$



1. _____

2. Simplify the expression $y^{5\sqrt{7}} \div y^{\sqrt{7}}$.

A. y^{35} **B.** y^{5} **C.** $y^{6\sqrt{7}}$ **D.** $y^{4\sqrt{7}}$ **2.** _____

3. Write the equation $4^{-3} = \frac{1}{64}$ in logarithmic form.

A. $\log_{-3} 4 = \frac{1}{64}$ **B.** $\log_{4} \frac{1}{64} = -3$

C. $\log_{\frac{1}{64}} (-3) = 4$ **D.** $\log_{4} (-3) = \frac{1}{64}$ **3.** _____

4. Evaluate $\log_4 32$.

A. $\frac{5}{2}$ **B.** 8 **C.** 3 **D.** $\frac{2}{5}$ **4.** _____

5. Solve $\log_3 (7x - 3) \geq \log_3 (5x)$.

A. $x \geq \frac{3}{2}$ **B.** $x > \frac{3}{7}$ **C.** $x \geq 0$ **D.** $x \geq \frac{2}{3}$ **5.** _____

6. Use $\log_3 5 \approx 1.4650$ and $\log_3 7 \approx 1.7712$ to approximate the value of $\log_3 \frac{21}{5}$.

A. 3.6270 **B.** 3.8486 **C.** 1.8916 **D.** 1.3062 **6.** _____

Part II

7. Write an exponential function whose graph passes through the points (0, −3) and (4, −48). **7.** _____

For Questions 9–13, solve each equation or inequality. **8.** _____

8. $\left(\frac{1}{8}\right)^{x} \geq 4^{x-5}$ **9.** $\log_{\frac{1}{5}} m = -2$ **9.** _____

10. $\log_7 (x + 3) - \log_7 (x - 3) = 1$ **10.** _____

11. $\log_3 (y - 8) + \log_3 (y + 4) = \log_3 13$ **11.** _____

12. Use $\log_2 3 \approx 1.5850$ and $\log_2 7 \approx 2.8074$ to approximate the value of $\log_2 84$. **12.** _____

Assessment

NAME _____ DATE _____ PERIOD _____

10

Chapter 10 Cumulative Review

(Chapters 1–10)

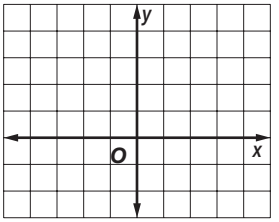
1. Name the sets of numbers to which $-\sqrt{16}$ belongs. (Lesson 1–2) **1.** _____

For Questions 2 and 3, use the following information.

A new camp site will contain t tent sites, with 25 square meters of land each, and r recreational vehicle (RV) sites with 40 square meters of land each. No more than 90 camp sites can be built on the 3000 square meters of land available.

2. Write a system of inequalities to represent the number of sites built. Then list the coordinates of the vertices of the feasible region. (Lesson 3–4) **2.** _____ **3.** _____

3. The site owner will charge \$14 per tent site and \$20 per RV site per day. Write a function for the total profit per day. Then determine the number of each type of site needed to earn a maximum profit, and find the maximum profit per day. (Lesson 3–4) **4.** _____ **5.** _____



4. Simplify $\sqrt[3]{27x^3y}$. (Lesson 5–6)

5. Solve $2x^2 = x + 1$ by graphing. If exact roots cannot be found, state the consecutive integers between which the roots are located. (Lesson 6–2)

6. Find all the zeros of the function $h(x) = x^3 - 5x^2 + 8x - 6$. (Lesson 7–5) **6.** _____

7. Find $(f + g)(x)$, $(f - g)(x)$, $(f \cdot g)(x)$, and $\left(\frac{f}{g}\right)(x)$ for $f(x) = x^2 + 2x - 15$ and $g(x) = 2x - 1$. (Lesson 7–7) **7.** _____

8. Write an equation for the hyperbola with vertices (0, 5) and (0, −5) and a conjugate axis at length 6 units. (Lesson 8–5) **8.** _____

9. Find the exact solution(s) of the system of equations
$x^2 + y^2 = 13$
$x^2 - 8y^2 = 4$. (Lesson 8–7) **9.** _____

For Questions 10 and 11, simplify.

10. $\frac{10mn^4}{s^2} \div \left(\frac{5mn}{s^3}\right)^2$ (Lesson 9–1) **10.** _____

11. $\frac{4}{x - 5} - \frac{3}{5 - x}$ (Lesson 9–2) **11.** _____

12. Identify the function represented by the equation $y = -2[\![x]\!]$. (Lesson 9–5) **12.** _____

13. Write an exponential function whose graph passes through the points (0, 4) and (−2, 100). (Lesson 10–1) **13.** _____

14. Solve $\log_{\frac{1}{9}} x = -2$. (Lesson 10–2) **14.** _____

15. Solve $6^{5n} > 54^{2n+3}$. Round to four decimal places. (Lesson 10–4) **15.** _____

NAME _____ DATE _____ PERIOD _____

10 Standardized Test Practice

(Chapters 1–10)

Part 1: Multiple Choice

Instructions: Fill in the appropriate oval for the best answer.

1. If $\frac{x}{8} > x$, which could be a value for x?

A. -1 **B.** 0 **C.** 2 **D.** $\frac{1}{4}$ **1.** Ⓐ Ⓑ Ⓒ Ⓓ

2. If $0 < a < 1$, which of the following increases as a decreases?

E. $a - 1$ **F.** $a^2 - 1$ **G.** $\frac{1}{a}$ **H.** a^2 **2.** Ⓔ Ⓕ Ⓖ Ⓗ

3. If $3x - 2$ is an odd integer, what is the next consecutive odd integer?

A. $3x - 1$ **B.** $3x - 3$ **C.** $3x + 1$ **D.** $3x$ **3.** Ⓐ Ⓑ Ⓒ Ⓓ

4. Jody sold 4 more than twice the number of cars that Laura sold. If Laura sold c cars, how many more did Jody sell than Laura?

E. 4 **F.** $c + 4$ **G.** $3c + 4$ **H.** $2c + 4$ **4.** Ⓔ Ⓕ Ⓖ Ⓗ

5. If $8 - 3z = 16 + 5z$, then what is the value of $4z$?

A. -16 **B.** -4 **C.** 1 **D.** 12 **5.** Ⓐ Ⓑ Ⓒ Ⓓ

6. The radius of a wheel is 6 inches. How many revolutions will it make if it is rolled a distance of 288π inches?

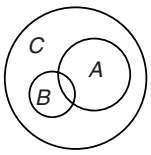
E. 8 **F.** 8π **G.** 24 **H.** 24π **6.** Ⓔ Ⓕ Ⓖ Ⓗ

7. What is the 8th term in the sequence 3, 2, 0, −4, −12, …?

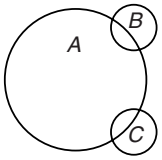
A. -124 **B.** -60 **C.** -36 **D.** -144 **7.** Ⓐ Ⓑ Ⓒ Ⓓ

8. Which Venn diagram models the relationships among the sets $A = \{1, 2, 3\}$, $B = \{-4, 0\}$, and $C = \{\text{positive integers}\}$?

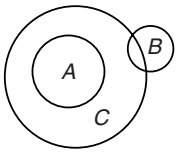
E.



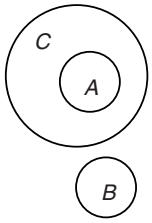
F.



G.



H.



8. Ⓔ Ⓕ Ⓖ Ⓗ

9. A total of \$270 is to be divided among four children. Each will receive an amount that is proportional to his or her age. If the children are 5, 10, 14, and 16 years old, how much money does the youngest child receive?

A. \$96 **B.** \$6 **C.** \$30 **D.** \$54 **9.** Ⓐ Ⓑ Ⓒ Ⓓ

Assessment

10 Standardized Test Practice *(continued)*

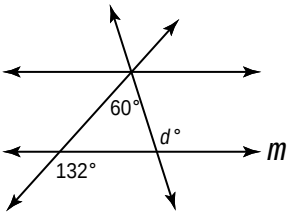
Part 2: Grid In

Instructions: Enter your answer by writing each digit of the answer in a column box and then shading in the appropriate oval that corresponds to that entry.

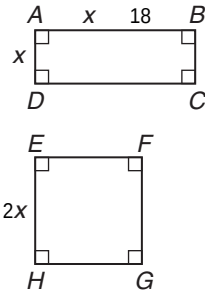
10. If $m^2 + n^2 = 140$ and $mn = 49$, what is the value of $(m - n)^2$?

11. What is the slope of a line that is perpendicular to the graph of $5x + 4y = 7$?

12. If $\ell \parallel m$ in the figure shown, what is the value of d?



13. Find the perimeter of square *EFGH* if the areas of rectangle *ABCD* and square *EFGH* are equal.



10.

	/	/	
.	.	.	.
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

11.

	/	/	
.	.	.	.
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

12.

	/	/	
.	.	.	.
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

13.

	/	/	
.	.	.	.
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Part 3: Quantitative Comparison

Instructions: Compare the quantities in columns A and B. Shade in
(A) if the quantity in column A is greater;
(B) if the quantity in column B is greater;
(C) if the quantities are equal; or
(D) if the relationship cannot be determined from the information given.

	Column A	Column B	
14.	23% of 1400	14% of 2300	**14.** (A) (B) (C) (D)
15.	A right triangle has sides 6, 6, and c.		**15.** (A) (B) (C) (D)
	8	c	
16.	$\frac{x^2}{3} = y$		**16.** (A) (B) (C) (D)
	x	y	
17.	$(-3)^{104}$	$(-3)^{333}$	**17.** (A) (B) (C) (D)
18.	$S = \{19, 22, 11, 17, 35\}$		**18.** (A) (B) (C) (D)
	The median of set S	The mean of set S	

NAME _____ DATE _____ PERIOD _____

Unit 3 Test

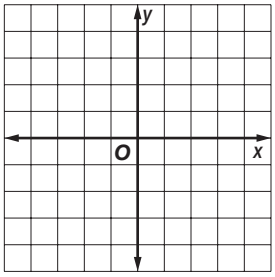
(Chapters 8–10)

SCORE _____

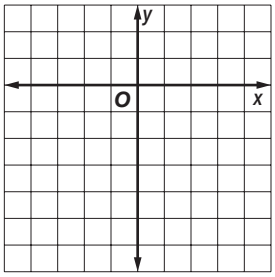
1. Find the midpoint of the line segment with end points at $(-6, 2)$ and $(3, 8)$. **1.** _____

2. Write an equation for the parabola with focus $(4, 0)$ and directrix $y = 2$. **2.** _____

3. Graph $16x^2 + 9y^2 = 144$. **3.**



4. Graph $\frac{x^2}{9} - \frac{(y + 2)^2}{9} = 1$. **4.**



5. Write an equation for the circle with center $(-4, \sqrt{7})$ and radius 5 units. **5.** _____

6. Write an equation for the ellipse with end points of the major axis at $(7, 1)$ and $(-7, 1)$ and end points of the minor axis at $(0, 5)$ and $(0, -3)$. **6.** _____

7. Write $36x^2 - 360x - 25y^2 - 100y = 100$ in standard form. Then state whether the graph of the equation is a *parabola*, *circle*, *ellipse*, or *hyperbola*. **7.**

8. Find the exact solutions of the system of equations $x^2 + 2y^2 = 18$ and $x = 2y$. **8.** _____

9. Simplify $\frac{4xy^3}{z^2} \div \left(\frac{8x^2y}{z^3}\right)^2$. **9.** _____

10. Simplify $\frac{d}{d^2 - 9} + \frac{5}{2d + 6}$. **10.** _____

11. Find the LCM of $m^2 - 4m - 5$ and $m^2 + 8m + 7$. **11.** _____

12. Determine the equations of any vertical asymptotes and the value of x for any holes in the graph of $f(x) = \frac{x^2 - 11x + 18}{x - 2}$. **12.** _____

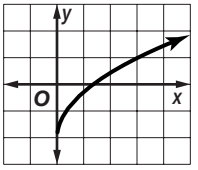
Assessment

Unit 3 Test *(continued)*

(Chapters 8–10)

13. If y varies jointly as x and z and $y = 100$ when $x = 10$ and $z = 5$, find y when $x = 12$ and $z = 6$. **13.** _____

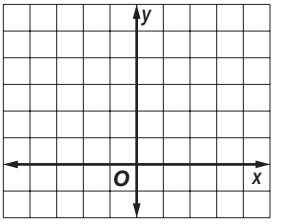
14. Identify the type of function represented by the graph. **14.** _____



15. Identify the function represented by $y = \frac{3}{x}$. **15.** _____

16. Solve $\frac{8}{t+5} = \frac{t-3}{t+5} + \frac{1}{3}$. **16.** _____

17. Sketch the graph of $y = 1.5(2)^x$. Then state the function's domain and range. **17.**



_____

18. Determine whether $y = 1.5\left(\frac{1}{6}\right)^x$ represents exponential *growth* or *decay*. **18.** _____

19. Simplify $x^5 \cdot x^{\pi}$. **19.** _____

For Questions 20–24, solve each equation or inequality. Round to four decimal places if necessary.

20. $\left(\frac{1}{5}\right)^{t-2} = 125$ **20.** _____

21. $\log_4 (x - 9) = 2$ **21.** _____

22. $\log_4 z + \log_4 (z - 3) = 1$ **22.** _____

23. $3.9^{m-4} = 10.21$ **23.** _____

24. $e^{3x} \geq 21$ **24.** _____

25. Evaluate $\log_6 6^9$. **25.** _____

26. Evaluate $\ln e^{-3x}$. **26.** _____

27. Use $\log_5 2 \approx 0.4307$ and $\log_5 3 \approx 0.6826$ to approximate the value of $\log_5 12$. **27.** _____

28. Express $\log_6 19$ in terms of common logarithms. Then approximate its value to four decimal places. **28.** _____

29. In a certain area, the sale price of new single-family homes has increased 4.1% per year since 1992. If a house was purchased in this area in 1992 for \$75,000 and the growth continues, what will the sale price be in 2006? Use $y = a(1 + r)^t$ and round to the nearest cent. **29.** _____

NAME _____ DATE _____ PERIOD _____

10 Standardized Test Practice

Student Record Sheet *(Use with pages 572–573 of the Student Edition.)*

Part 1 *Multiple Choice*

Select the best answer from the choices given and fill in the corresponding oval.

1 Ⓐ Ⓑ Ⓒ Ⓓ	**4** Ⓐ Ⓑ Ⓒ Ⓓ	**7** Ⓐ Ⓑ Ⓒ Ⓓ	**9** Ⓐ Ⓑ Ⓒ Ⓓ
2 Ⓐ Ⓑ Ⓒ Ⓓ	**5** Ⓐ Ⓑ Ⓒ Ⓓ	**8** Ⓐ Ⓑ Ⓒ Ⓓ	**10** Ⓐ Ⓑ Ⓒ Ⓓ
3 Ⓐ Ⓑ Ⓒ Ⓓ	**6** Ⓐ Ⓑ Ⓒ Ⓓ		

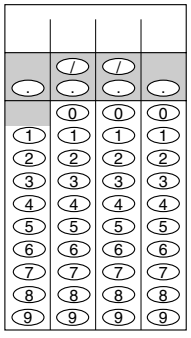
Part 2 *Short Response/Grid In*

Solve the problem and write your answer in the blank.

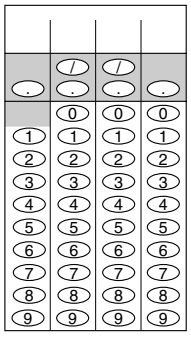
For Questions 12–18, also enter your answer by writing each number or symbol in a box. Then fill in the corresponding oval for that number or symbol.

11 _____

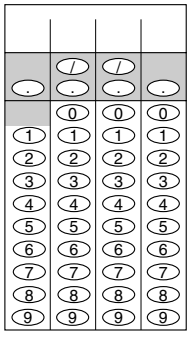
12 _____



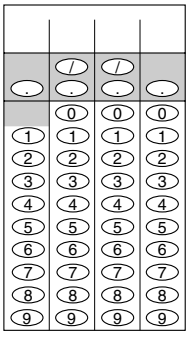
13 _____



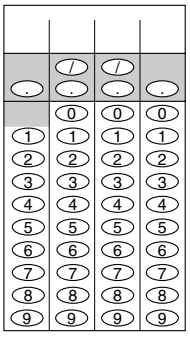
14 _____



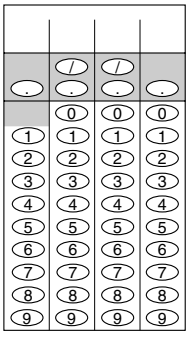
15 _____



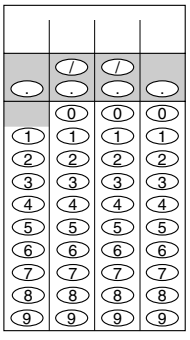
16 _____



17 _____



18 _____



Part 3 *Quantitative Comparison*

Select the best answer from the choices given and fill in the corresponding oval.

19 Ⓐ Ⓑ Ⓒ Ⓓ	**21** Ⓐ Ⓑ Ⓒ Ⓓ	**23** Ⓐ Ⓑ Ⓒ Ⓓ
20 Ⓐ Ⓑ Ⓒ Ⓓ	**22** Ⓐ Ⓑ Ⓒ Ⓓ	

Answers

NAME _____ DATE _____ PERIOD ____

10-1 Study Guide and Intervention

Exponential Functions

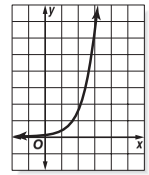
Exponential Functions An **exponential function** has the form $y = ab^x$, where $a \neq 0$, $b > 0$, and $b \neq 1$.

Properties of an Exponential Function	**1.** The function is continuous and one-to-one. **2.** The domain is the set of all real numbers. **3.** The x-axis is the asymptote of the graph. **4.** The range is the set of all positive numbers if $a > 0$ and all negative numbers if $a < 0$. **5.** The graph contains the point $(0, a)$.
Exponential Growth and Decay	If $a > 0$ and $b > 1$, the function $y = ab^x$ represents exponential growth. If $a > 0$ and $0 < b < 1$, the function $y = ab^x$ represents exponential decay.

Example 1 **Sketch the graph of $y = 0.1(4)^x$. Then state the function's domain and range.**

Make a table of values. Connect the points to form a smooth curve.

x	−1	0	1	2	3
y	0.025	0.1	0.4	1.6	6.4



The domain of the function is all real numbers, while the range is the set of all positive real numbers.

Example 2 **Determine whether each function represents exponential *growth* or *decay*.**

a. $y = 0.5(2)^x$
exponential growth, since the base, 2, is greater than 1

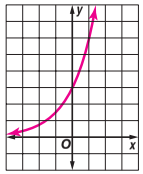
b. $y = -2.8(2)^x$
neither, since −2.8, the value of a is less than 0.

c. $y = 1.1(0.5)^x$
exponential decay, since the base, 0.5, is between 0 and 1

Exercises

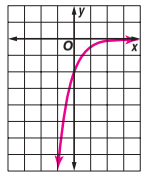
Sketch the graph of each function. Then state the function's domain and range.

1. $y = 3(2)^x$



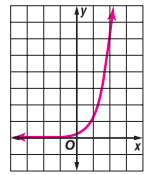
Domain: all real numbers; Range: all positive real numbers

2. $y = -2\left(\frac{1}{4}\right)^x$



Domain: all real numbers; Range: all negative real numbers

3. $y = 0.25(5)^x$



Domain: all real numbers; Range: all positive real numbers

Determine whether each function represents exponential growth or decay.

4. $y = 0.3(1.2)^x$ growth

5. $y = -5\left(\frac{4}{5}\right)^x$ neither

6. $y = 3(10)^{-x}$ decay

Lesson 10-1

NAME _____ DATE _____ PERIOD ____

10-1 Study Guide and Intervention *(continued)*

Exponential Functions

Exponential Equations and Inequalities All the properties of rational exponents that you know also apply to real exponents. Remember that $a^m \cdot a^n = a^{m+n}$, $(a^m)^n = a^{mn}$, and $a^m \div a^n = a^{m-n}$.

Property of Equality for Exponential Functions	If b is a positive number other than 1, then $b^x = b^y$ if and only if $x = y$.
Property of Inequality for Exponential Functions	If $b > 1$ then $b^x > b^y$ if and only if $x > y$ and $b^x < b^y$ if and only if $x < y$.

Example 1 **Solve $4^{x-1} = 2^{x+5}$.**

$4^{x-1} = 2^{x+5}$	Original equation
$(2^2)^{x-1} = 2^{x+5}$	Rewrite 4 as 2^2.
$2(x-1) = x+5$	Prop. of Inequality for Exponential Functions
$2x - 2 = x + 5$	Distributive Property
$x = 7$	Subtract x and add 2 to each side.

Example 2 **Solve $5^{2x-1} > \frac{1}{125}$.**

$5^{2x-1} > \frac{1}{125}$	Original inequality
$5^{2x-1} > 5^{-3}$	Rewrite $\frac{1}{125}$ as 5^{-3}.
$2x - 1 > -3$	Prop. of Inequality for Exponential Functions
$2x > -2$	Add 1 to each side.
$x > -1$	Divide each side by 2.

The solution set is $\{x | x > -1\}$.

Exercises

Simplify each expression.

1. $(3^{\sqrt{2}})^{\sqrt{2}}$ 9

2. $25^{\sqrt{2}} \cdot 125^{\sqrt{2}}$ $5^{5\sqrt{2}}$ or $3125^{\sqrt{2}}$

3. $(x^{\sqrt{2}}y^{3\sqrt{2}})^{\sqrt{2}}$ x^2y^6

4. $(x^{\sqrt{6}})(x^{\sqrt{5}})$ $x^{\sqrt{6}+\sqrt{5}}$

5. $(x^{\sqrt{6}})^{\sqrt{5}}$ $x^{\sqrt{30}}$

6. $(2x^{\pi})(5x^{3\pi})$ $10x^{4\pi}$

Solve each equation or inequality. Check your solution.

7. $3^{2x-1} = 3^{x+2}$ 3

8. $2^{3x} = 4^{x+2}$ 4

9. $3^{2x-1} = \frac{1}{9}$ $-\frac{1}{2}$

10. $4^{x+1} = 8^{2x+3}$ $-\frac{7}{4}$

11. $8^{x-2} = \frac{1}{16}$ $\frac{2}{3}$

12. $25^{2x} = 125^{x+2}$ 6

13. $4^{\sqrt{x}} = 16^{\sqrt{5}}$ 20

14. $x^{\sqrt{3}} = 36^{\sqrt[3]{4}}$ 6

15. $x^{\sqrt{2}} = 81^{\frac{1}{\sqrt{8}}}$ 3

16. $3^{x-4} < \frac{1}{27}$ $x < 1$

17. $4^{2x-2} > 2^{x+1}$ $x > \frac{5}{3}$

18. $5^{2x} < 125^{x-5}$ $x > 15$

19. $10^{4x+1} > 100^{x-2}$ $x > -\frac{5}{2}$

20. $7^{3x} < 49^{x^2}$ $x > \frac{3}{2}$ or $x < 0$

21. $8^{2x-5} < 4^{x+8}$ $x < \frac{31}{4}$

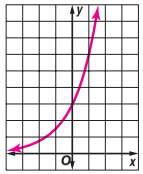
NAME _____ DATE _____ PERIOD ____

10-1 Skills Practice

Exponential Functions

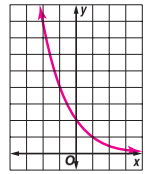
Sketch the graph of each function. Then state the function's domain and range.

1. $y = 3(2)^x$



domain: all real numbers; range: all positive numbers

2. $y = 2\left(\frac{1}{2}\right)^x$



domain: all real numbers; range: all positive numbers

Determine whether each function represents exponential *growth* or *decay*.

3. $y = 3(6)^x$ growth

4. $y = 2\left(\frac{9}{10}\right)^x$ decay

5. $y = 10^{-x}$ decay

6. $y = 2(2.5)^x$ growth

Write an exponential function whose graph passes through the given points.

7. (0, 1) and (−1, 3) $y = \left(\frac{1}{3}\right)^x$

8. (0, 4) and (1, 12) $y = 4(3)^x$

9. (0, 3) and (−1, 6) $y = 3\left(\frac{1}{2}\right)^x$

10. (0, 5) and (1, 15) $y = 5(3)^x$

11. (0, 0.1) and (1, 0.5) $y = 0.1(5)^x$

12. (0, 0.2) and (1, 1.6) $y = 0.2(8)^x$

Simplify each expression.

13. $(3^{\sqrt{3}})^{\sqrt{3}}$ 27

14. $(x^{\sqrt{2}})^{\sqrt{7}}$ $x^{\sqrt{14}}$

15. $5^{2\sqrt{3}} \cdot 5^{4\sqrt{3}}$ $5^{6\sqrt{3}}$

16. $x^{3\pi} \div x^{\pi}$ $x^{2\pi}$

Solve each equation or inequality. Check your solution.

17. $3^x > 9$ $x > 2$

18. $2^{2x+3} = 32$ 1

19. $49^x \le \frac{1}{7}$ $x \le -\frac{1}{2}$

20. $4^{3x-2} = 16$ $\frac{4}{3}$

21. $3^{2x+5} = 27^x$ 5

22. $27^x = 3^{2x+3}$ 3

Lesson 10-1

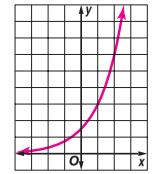
NAME _____ DATE _____ PERIOD ____

10-1 Practice (Average)

Exponential Functions

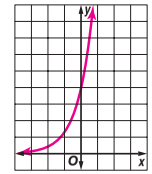
Sketch the graph of each function. Then state the function's domain and range.

1. $y = 1.5(2)^x$



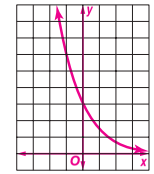
domain: all real numbers; range: all positive numbers

2. $y = 4(3)^x$



domain: all real numbers; range: all positive numbers

3. $y = 3(0.5)^x$



domain: all real numbers; range: all positive numbers

Determine whether each function represents exponential *growth* or *decay*.

4. $y = 5(0.6)^x$ decay

5. $y = 0.1(2)^x$ growth

6. $y = 5 \cdot 4^{-x}$ decay

Write an exponential function whose graph passes through the given points.

7. (0, 1) and (−1, 4) $y = \left(\frac{1}{4}\right)^x$

8. (0, 2) and (1, 10) $y = 2(5)^x$

9. (0, −3) and (1, −1.5) $y = -3(0.5)^x$

10. (0, 0.8) and (1, 1.6) $y = 0.8(2)^x$

11. (0, −0.4) and (2, −10) $y = -0.4(5)^x$

12. (0, π) and (3, 8π) $y = \pi(2)^x$

Simplify each expression.

13. $(2^{\sqrt{2}})^{\sqrt{8}}$ 16

14. $(n^{\sqrt{3}})^{\sqrt{75}}$ n^{15}

15. $y^{\sqrt{6}} \cdot y^{5\sqrt{6}}$ $y^{6\sqrt{6}}$

16. $13^{\sqrt{6}} \cdot 13^{\sqrt{24}}$ $13^{3\sqrt{6}}$

17. $n^3 \div n^{\pi}$ $n^{3-\pi}$

18. $125^{\sqrt{11}} \div 5^{\sqrt{11}}$ $5^{2\sqrt{11}}$

Solve each equation or inequality. Check your solution.

19. $3^{3x-5} > 81$ $x > 3$

20. $7^{6x} = 7^{2x-20}$ −5

21. $3^{6n-5} < 9^{4n-3}$ $n > \frac{1}{2}$

22. $9^{2x-1} = 27^{x+4}$ 14

23. $2^{3n-1} \ge \left(\frac{1}{8}\right)^n$ $n \ge \frac{1}{6}$

24. $16^{4n-1} = 128^{2n+1}$ $\frac{11}{2}$

BIOLOGY For Exercises 25 and 26, use the following information.

The initial number of bacteria in a culture is 12,000. The number after 3 days is 96,000.

25. Write an exponential function to model the population y of bacteria after x days. $y = 12{,}000(2)^x$

26. How many bacteria are there after 6 days? 768,000

27. EDUCATION A college with a graduating class of 4000 students in the year 2002 predicts that it will have a graduating class of 4862 in 4 years. Write an exponential function to model the number of students y in the graduating class t years after 2002. $y = 4000(1.05)^t$

NAME _____ DATE _____ PERIOD ____

10-1 Reading to Learn Mathematics

Exponential Functions

Pre-Activity **How does an exponential function describe tournament play?**

Read the introduction to Lesson 10-1 at the top of page 523 in your textbook.

How many rounds of play would be needed for a tournament with 100 players? **7**

Reading the Lesson

1. Indicate whether each of the following statements about the exponential function $y = 10^x$ is *true* or *false*.

a. The domain is the set of all positive real numbers. **false**

b. The y-intercept is 1. **true**

c. The function is one-to-one. **true**

d. The y-axis is an asymptote of the graph. **false**

e. The range is the set of all real numbers. **false**

2. Determine whether each function represents exponential *growth* or *decay*.

a. $y = 0.2(3)^x$. **growth** **b.** $y = 3\left(\frac{2}{5}\right)^x$. **decay** **c.** $y = 0.4(1.01)^x$. **growth**

3. Supply the reason for each step in the following solution of an exponential equation.

$9^{2x-1} = 27x$	Original equation
$(3^2)^{2x-1} = (3^3)^x$	**Rewrite each side with a base of 3.**
$3^{2(2x-1)} = 3^{3x}$	**Power of a Power**
$2(2x-1) = 3x$	**Property of Equality for Exponential Functions**
$4x - 2 = 3x$	**Distributive Property**
$x - 2 = 0$	**Subtract 3*x* from each side.**
$x = 2$	**Add 2 to each side.**

Helping You Remember

4. One way to remember that polynomial functions and exponential functions are different is to contrast the polynomial function $y = x^2$ and the exponential function $y = 2^x$. Tell at least three ways they are different.

Sample answer: In $y = x^2$, the variable x is a base, but in $y = 2^x$, the variable x is an exponent. The graph of $y = x^2$ is symmetric with respect to the y-axis, but the graph of $y = 2^x$ is not. The graph of $y = x^2$ touches the x-axis at (0, 0), but the graph of $y = 2^x$ has the x-axis as an asymptote. You can compute the value of $y = x^2$ mentally for $x = 100$, but you cannot compute the value of $y = 2^x$ mentally for $x = 100$.

Lesson 10-1

NAME _____ DATE _____ PERIOD ____

10-1 Enrichment

Finding Solutions of $x^y = y^x$

Perhaps you have noticed that if x and y are interchanged in equations such as $x = y$ and $xy = 1$, the resulting equation is equivalent to the original equation. The same is true of the equation $x^y = y^x$. However, finding solutions of $x^y = y^x$ and drawing its graph is not a simple process.

Solve each problem. Assume that x and y are positive real numbers.

1. If $a > 0$, will (a, a) be a solution of $x^y = y^x$? Justify your answer.

Yes, since $a^a = a^a$ must be true (Reflexive Prop. of Equality).

2. If $c > 0$, $d > 0$, and (c, d) is a solution of $x^y = y^x$, will (d, c) also be a solution? Justify your answer.

Yes; replacing x with d, y with c gives $d^c = c^d$; but if (c, d) is a solution, $c^d = d^c$. So, by the Symmetric Property of Equality, $d^c = c^d$ is true.

3. Use 2 as a value for y in $x^y = y^x$. The equation becomes $x^2 = 2^x$.

a. Find equations for two functions, $f(x)$ and $g(x)$ that you could graph to find the solutions of $x^2 = 2^x$. Then graph the functions on a separate sheet of graph paper. **$f(x) = x^2$, $g(x) = 2^x$**

See students' graphs.

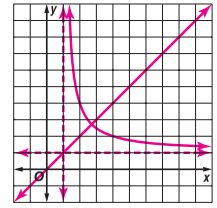
b. Use the graph you drew for part **a** to state two solutions for $x^2 = 2^x$. Then use these solutions to state two solutions for $x^y = y^x$.

2, 4; (2, 2), (4, 2)

4. In this exercise, a graphing calculator will be very helpful. Use the technique of Exercise 3 to complete the tables below. Then graph $x^y = y^x$ for positive values of x and y. If there are asymptotes, show them in your diagram using dotted lines. Note that in the table, some values of y call for one value of x, others call for two.

x	y
$\frac{1}{2}$	$\frac{1}{2}$
$\frac{3}{4}$	$\frac{3}{4}$
1	1
2	2
4	2
3	3
2.5	3

x	y
4	4
2	4
5	5
1.8	5
8	8
1.5	8



NAME _____ DATE _____ PERIOD ____

10-2 Study Guide and Intervention

Logarithms and Logarithmic Functions

Logarithmic Functions and Expressions

Definition of Logarithm with Base *b*	Let b and x be positive numbers, $b \neq 1$. The logarithm of x with base b is denoted $\log_b x$ and is defined as the exponent y that makes the equation $b^y = x$ true.

The inverse of the exponential function $y = b^x$ is the **logarithmic function** $x = b^y$. This function is usually written as $y = \log_b x$.

Properties of Logarithmic Functions	**1.** The function is continuous and one-to-one. **2.** The domain is the set of all positive real numbers. **3.** The y-axis is an asymptote of the graph. **4.** The range is the set of all real numbers. **5.** The graph contains the point (0, 1).

Example 1 **Write an exponential equation equivalent to $\log_3 243 = 5$.**

$3^5 = 243$

Example 2 **Write a logarithmic equation equivalent to $6^{-3} = \frac{1}{216}$.**

$\log_6 \frac{1}{216} = -3$

Example 3 **Evaluate $\log_8 16$.**

$8^{\frac{4}{3}} = 16$, so $\log_8 16 = \frac{4}{3}$.

Exercises

Write each equation in logarithmic form.

1. $2^7 = 128$ $\log_2 128 = 7$

2. $3^{-4} = \frac{1}{81}$ $\log_3 \frac{1}{81} = -4$

3. $\left(\frac{1}{7}\right)^3 = \frac{1}{343}$ $\log_{\frac{1}{7}} \frac{1}{343} = 3$

Write each equation in exponential form.

4. $\log_{15} 225 = 2$ $15^2 = 225$

5. $\log_3 \frac{1}{27} = -3$ $3^{-3} = \frac{1}{27}$

6. $\log_4 32 = \frac{5}{2}$ $4^{\frac{5}{2}} = 32$

Evaluate each expression.

7. $\log_4 64$ **3**

8. $\log_2 64$ **6**

9. $\log_{100} 100{,}000$ **2.5**

10. $\log_5 625$ **4**

11. $\log_{27} 81$ $\frac{4}{3}$

12. $\log_{25} 5$ $\frac{1}{2}$

13. $\log_2 \frac{1}{128}$ **−7**

14. $\log_{10} 0.00001$ **−5**

15. $\log_4 \frac{1}{32}$ **−2.5**

Lesson 10-2

NAME _____ DATE _____ PERIOD ____

10-2 Study Guide and Intervention *(continued)*

Logarithms and Logarithmic Functions

Solve Logarithmic Equations and Inequalities

Logarithmic to Exponential Inequality	If $b > 1$, $x > 0$, and $\log_b x > y$, then $x > b^y$. If $b > 1$, $x > 0$, and $\log_b x < y$, then $0 < x < b^y$.
Property of Equality for Logarithmic Functions	If b is a positive number other than 1, then $\log_b x = \log_b y$ if and only if $x = y$.
Property of Inequality for Logarithmic Functions	If $b > 1$, then $\log_b x > \log_b y$ if and only if $x > y$, and $\log_b x < \log_b y$ if and only if $x < y$.

Example 1 **Solve $\log_2 2x = 3$.**

$\log_2 2x = 3$	Original equation
$2x = 2^3$	Definition of logarithm
$2x = 8$	Simplify.
$x = 4$	Simplify.

The solution is $x = 4$.

Example 2 **Solve $\log_5 (4x - 3) < 3$.**

$\log_5 (4x - 3) < 3$	Original equation
$0 < 4x - 3 < 5^3$	Logarithmic to exponential inequality
$3 < 4x < 125 + 3$	Addition Property of Inequalities
$\frac{3}{4} < x < 32$	Simplify.

The solution set is $\left\{x \middle| \frac{3}{4} < x < 32\right\}$.

Exercises

Solve each equation or inequality.

1. $\log_2 32 = 3x$ $\frac{5}{3}$

2. $\log_3 2c = -2$ $\frac{1}{18}$

3. $\log_{2x} 16 = -2$ $\frac{1}{8}$

4. $\log_{25} \left(\frac{x}{2}\right) = \frac{1}{2}$ **10**

5. $\log_4 (5x + 1) = 2$ **3**

6. $\log_8 (x - 5) = \frac{2}{3}$ **9**

7. $\log_4 (3x - 1) = \log_4 (2x + 3)$ **4**

8. $\log_2 (x^2 - 6) = \log_2 (2x + 2)$ **4**

9. $\log_{x+4} 27 = 3$ **−1**

10. $\log_2 (x + 3) = 4$ **13**

11. $\log_x 1000 = 3$ **10**

12. $\log_8 (4x + 4) = 2$ **15**

13. $\log_2 2x > 2$ $x > 2$

14. $\log_5 x > 2$ $x > 25$

15. $\log_2 (3x + 1) < 4$ $-\frac{1}{3} < x < 5$

16. $\log_4 (2x) > -\frac{1}{2}$ $x > \frac{1}{4}$

17. $\log_3 (x + 3) < 3$ $-3 < x < 24$

18. $\log_{27} 6x > \frac{2}{3}$ $x > \frac{3}{2}$

NAME _____ DATE _____ PERIOD ____

10-2 Skills Practice

Logarithms and Logarithmic Functions

Write each equation in logarithmic form.

1. $2^3 = 8$ $\log_2 8 = 3$
2. $3^2 = 9$ $\log_3 9 = 2$
3. $8^{-2} = \frac{1}{64}$ $\log_8 \frac{1}{64} = -2$
4. $\left(\frac{1}{3}\right)^2 = \frac{1}{9}$ $\log_{\frac{1}{3}} \frac{1}{9} = 2$

Write each equation in exponential form.

5. $\log_3 243 = 5$ $3^5 = 243$
6. $\log_4 64 = 3$ $4^3 = 64$
7. $\log_9 3 = \frac{1}{2}$ $9^{\frac{1}{2}} = 3$
8. $\log_5 \frac{1}{25} = -2$ $5^{-2} = \frac{1}{25}$

Evaluate each expression.

9. $\log_5 25$ 2
10. $\log_9 3$ $\frac{1}{2}$
11. $\log_{10} 1000$ 3
12. $\log_{125} 5$ $\frac{1}{3}$
13. $\log_4 \frac{1}{64}$ -3
14. $\log_5 \frac{1}{625}$ -4
15. $\log_8 8^3$ 3
16. $\log_{27} \frac{1}{3}$ $-\frac{1}{3}$

Solve each equation or inequality. Check your solutions.

17. $\log_3 x = 5$ 243
18. $\log_2 x = 3$ 8
19. $\log_4 y < 0$ $0 < y < 1$
20. $\log_{\frac{1}{4}} x = 3$ $\frac{1}{64}$
21. $\log_2 n > -2$ $n > \frac{1}{4}$
22. $\log_b 3 = \frac{1}{2}$ 9
23. $\log_6 (4x + 12) = 2$ 6
24. $\log_2 (4x - 4) > 5$ $x > 9$
25. $\log_3 (x + 2) = \log_3 (3x)$ 1
26. $\log_6 (3y - 5) \geq \log_6 (2y + 3)$ $y \geq 8$

Lesson 10-2

NAME _____ DATE _____ PERIOD ____

10-2 Practice (Average)

Logarithms and Logarithmic Functions

Write each equation in logarithmic form.

1. $5^3 = 125$ $\log_5 125 = 3$
2. $7^0 = 1$ $\log_7 1 = 0$
3. $3^4 = 81$ $\log_3 81 = 4$
4. $3^{-4} = \frac{1}{81}$ $\log_3 \frac{1}{81} = -4$
5. $\left(\frac{1}{4}\right)^3 = \frac{1}{64}$ $\log_{\frac{1}{4}} \frac{1}{64} = 3$
6. $7776^{\frac{1}{5}} = 6$ $\log_{7776} 6 = \frac{1}{5}$

Write each equation in exponential form.

7. $\log_6 216 = 3$ $6^3 = 216$
8. $\log_2 64 = 6$ $2^6 = 64$
9. $\log_3 \frac{1}{81} = -4$ $3^{-4} = \frac{1}{81}$
10. $\log_{10} 0.00001 = -5$ $10^{-5} = 0.00001$
11. $\log_{25} 5 = \frac{1}{2}$ $25^{\frac{1}{2}} = 5$
12. $\log_{32} 8 = \frac{3}{5}$ $32^{\frac{3}{5}} = 8$

Evaluate each expression.

13. $\log_3 81$ 4
14. $\log_{10} 0.0001$ -4
15. $\log_2 \frac{1}{16}$ -4
16. $\log_{\frac{1}{3}} 27$ -3
17. $\log_9 1$ 0
18. $\log_8 4$ $\frac{2}{3}$
19. $\log_7 \frac{1}{49}$ -2
20. $\log_6 6^4$ 4
21. $\log_3 \frac{1}{3}$ -1
22. $\log_4 \frac{1}{256}$ -4
23. $\log_9 9^{(n+1)}$ $n + 1$
24. $2^{\log_2 32}$ 32

Solve each equation or inequality. Check your solutions.

25. $\log_{10} n = -3$ $\frac{1}{1000}$
26. $\log_4 x > 3$ $x > 64$
27. $\log_4 x = \frac{3}{2}$ 8
28. $\log_{\frac{1}{5}} x = -3$ 125
29. $\log_7 q < 0$ $0 < q < 1$
30. $\log_6 (2y + 8) \geq 2$ $y \geq 14$
31. $\log_y 16 = -4$ $\frac{1}{2}$
32. $\log_n \frac{1}{8} = -3$ 2
33. $\log_b 1024 = 5$ 4
34. $\log_8 (3x + 7) < \log_8 (7x + 4)$ $x > \frac{3}{4}$
35. $\log_7 (8x + 20) = \log_7 (x + 6)$ -2
36. $\log_3 (x^2 - 2) = \log_3 x$ 2

37. **SOUND** Sounds that reach levels of 130 decibels or more are painful to humans. What is the relative intensity of 130 decibels? 10^{13}

38. **INVESTING** Maria invests \$1000 in a savings account that pays 8% interest compounded annually. The value of the account A at the end of five years can be determined from the equation $\log A = \log[1000(1 + 0.08)^5]$. Find the value of A to the nearest dollar. \$1469

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10-2 Reading to Learn Mathematics

Logarithms and Logarithmic Functions

Pre-Activity **Why is a logarithmic scale used to measure sound?**

Read the introduction to Lesson 10-2 at the top of page 531 in your textbook.

How many times louder than a whisper is normal conversation?
10^4 or 10,000 times

Reading the Lesson

1. a. Write an exponential equation that is equivalent to $\log_3 81 = 4$. **$3^4 = 81$**

b. Write a logarithmic equation that is equivalent to $25^{-\frac{1}{2}} = \frac{1}{5}$. **$\log_{25} \frac{1}{5} = -\frac{1}{2}$**

c. Write an exponential equation that is equivalent to $\log_4 1 = 0$. **$4^0 = 1$**

d. Write a logarithmic equation that is equivalent to $10^{-3} = 0.001$. **$\log_{10} 0.001 = -3$**

e. What is the inverse of the function $y = 5^x$? **$y = \log_5 x$**

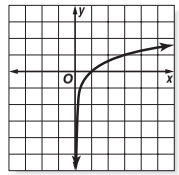
f. What is the inverse of the function $y = \log_{10} x$? **$y = 10^x$**

2. Match each function with its graph.

a. $y = 3^x$ **IV**

b. $y = \log_3 x$ **I**

c. $y = \left(\frac{1}{3}\right)^x$ **II**

I.


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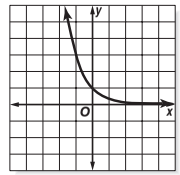

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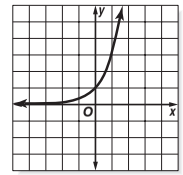


3. Indicate whether each of the following statements about the exponential function $y = \log_5 x$ is *true* or *false*.

a. The y-axis is an asymptote of the graph. **true**

b. The domain is the set of all real numbers. **false**

c. The graph contains the point (5, 0). **false**

d. The range is the set of all real numbers. **true**

e. The y-intercept is 1. **false**

Helping You Remember

4. An important skill needed for working with logarithms is changing an equation between logarithmic and exponential forms. Using the words *base*, *exponent*, and *logarithm*, describe an easy way to remember and apply the part of the definition of logarithm that says, "$\log_b x = y$ if and only if $b^y = x$." **Sample answer: In these equations, *b* stands for *base*. In log form, *b* is the subscript, and in exponential form, *b* is the number that is raised to a power. A *logarithm* is an *exponent*, so *y*, which is the log in the first equation, becomes the exponent in the second equation.**

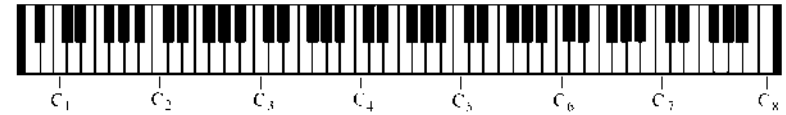
Lesson 10-2

NAME _____ DATE _____ PERIOD ____

10-2 Enrichment

Musical Relationships

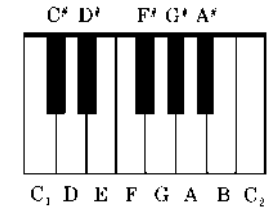
The frequencies of notes in a musical scale that are one octave apart are related by an exponential equation. For the eight C notes on a piano, the equation is $C_n = C_1 2^{n-1}$, where C_n represents the frequency of note C_n.



1. Find the relationship between C_1 and C_2. **$C_2 = 2C_1$**

2. Find the relationship between C_1 and C_4. **$C_4 = 8C_1$**

The frequencies of consecutive notes are related by a common ratio r. The general equation is $f_n = f_1 r^{n-1}$.



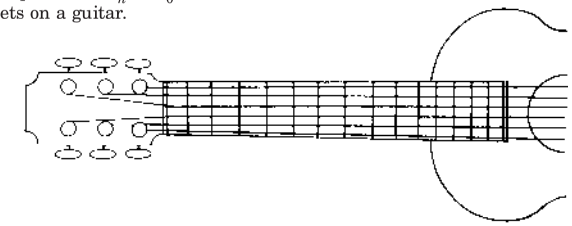
3. If the frequency of middle C is 261.6 cycles per second and the frequency of the next higher C is 523.2 cycles per second, find the common ratio r. (*Hint:* The two C's are 12 notes apart.) Write the answer as a radical expression.
$r = \sqrt[12]{2}$

4. Substitute decimal values for r and f_1 to find a specific equation for f_n.
$f_n = 261.1(1.05946)^{n-1}$

5. Find the frequency of F# above middle C.
$f_7 = 261.6(1.05946)^6 \approx 369.95$

6. Frets are a series of ridges placed across the fingerboard of a guitar. They are spaced so that the sound made by pressing a string against one fret has about 1.0595 times the wavelength of the sound made by using the next fret. The general equation is $w_n = w_0(1.0595)^n$. Describe the arrangement of the frets on a guitar.

The frets are spaced in a logarithmic scale.



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10-3 Study Guide and Intervention

Properties of Logarithms

Properties of Logarithms Properties of exponents can be used to develop the following properties of logarithms.

Product Property of Logarithms	For all positive numbers m, n, and b, where $b \neq 1$, $\log_b mn = \log_b m + \log_b n$.
Quotient Property of Logarithms	For all positive numbers m, n, and b, where $b \neq 1$, $\log_b \frac{m}{n} = \log_b m - \log_b n$.
Power Property of Logarithms	For any real number p and positive numbers m and b, where $b \neq 1$, $\log_b m^p = p \log_b m$.

Example **Use $\log_3 28 \approx 3.0331$ and $\log_3 4 \approx 1.2619$ to approximate the value of each expression.**

a. $\log_3 36$

$$\begin{aligned}\log_3 36 &= \log_3 (3^2 \cdot 4)\\ &= \log_3 3^2 + \log_3 4\\ &= 2 + \log_3 4\\ &\approx 2 + 1.2619\\ &\approx 3.2619\end{aligned}$$

b. $\log_3 7$

$$\begin{aligned}\log_3 7 &= \log_3 \left(\frac{28}{4}\right)\\ &= \log_3 28 - \log_3 4\\ &\approx 3.0331 - 1.2619\\ &\approx 1.7712\end{aligned}$$

c. $\log_3 256$

$$\begin{aligned}\log_3 256 &= \log_3 (4^4)\\ &= 4 \cdot \log_3 4\\ &\approx 4(1.2619)\\ &\approx 5.0476\end{aligned}$$

Exercises

Use $\log_{12} 3 \approx 0.4421$ and $\log_{12} 7 \approx 0.7831$ to evaluate each expression.

1. $\log_{12} 21$ **1.2252**

2. $\log_{12} \frac{7}{3}$ **0.3410**

3. $\log_{12} 49$ **1.5662**

4. $\log_{12} 36$ **1.4421**

5. $\log_{12} 63$ **1.6673**

6. $\log_{12} \frac{27}{49}$ **−0.2399**

7. $\log_{12} \frac{81}{49}$ **0.2022**

8. $\log_{12} 16{,}807$ **3.9155**

9. $\log_{12} 441$ **2.4504**

Use $\log_5 3 \approx 0.6826$ and $\log_5 4 \approx 0.8614$ to evaluate each expression.

10. $\log_5 12$ **1.5440**

11. $\log_5 100$ **2.8614**

12. $\log_5 0.75$ **−0.1788**

13. $\log_5 144$ **3.0880**

14. $\log_5 \frac{27}{16}$ **0.3250**

15. $\log_5 375$ **3.6826**

16. $\log_5 1.\overline{3}$ **0.1788**

17. $\log_5 \frac{9}{16}$ **−0.3576**

18. $\log_5 \frac{81}{5}$ **1.7304**

Lesson 10-3

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10-3 Study Guide and Intervention *(continued)*

Properties of Logarithms

Solve Logarithmic Equations You can use the properties of logarithms to solve equations involving logarithms.

Example **Solve each equation.**

a. $2 \log_3 x - \log_3 4 = \log_3 25$

$2 \log_3 x - \log_3 4 = \log_3 25$	Original equation
$\log_3 x^2 - \log_3 4 = \log_3 25$	Power Property
$\log_3 \frac{x^2}{4} = \log_3 25$	Quotient Property
$\frac{x^2}{4} = 25$	Property of Equality for Logarithmic Functions
$x^2 = 100$	Multiply each side by 4.
$x = \pm 10$	Take the square root of each side.

Since logarithms are undefined for $x < 0$, -10 is an extraneous solution. The only solution is 10.

b. $\log_2 x + \log_2 (x + 2) = 3$

$\log_2 x + \log_2 (x + 2) = 3$	Original equation
$\log_2 x(x + 2) = 3$	Product Property
$x(x + 2) = 2^3$	Definition of logarithm
$x^2 + 2x = 8$	Distributive Property
$x^2 + 2x - 8 = 0$	Subtract 8 from each side.
$(x + 4)(x - 2) = 0$	Factor.
$x = 2$ or $x = -4$	Zero Product Property

Since logarithms are undefined for $x < 0$, -4 is an extraneous solution. The only solution is 2.

Exercises

Solve each equation. Check your solutions.

1. $\log_5 4 + \log_5 2x = \log_5 24$ **3**

2. $3 \log_4 6 - \log_4 8 = \log_4 x$ **27**

3. $\frac{1}{2} \log_6 25 + \log_6 x = \log_6 20$ **4**

4. $\log_2 4 - \log_2 (x + 3) = \log_2 8$ **$-\frac{5}{2}$**

5. $\log_6 2x - \log_6 3 = \log_6 (x - 1)$ **3**

6. $2 \log_4 (x + 1) = \log_4 (11 - x)$ **2**

7. $\log_2 x - 3 \log_2 5 = 2 \log_2 10$ **12,500**

8. $3 \log_2 x - 2 \log_2 5x = 2$ **100**

9. $\log_3 (c + 3) - \log_3 (4c - 1) = \log_3 5$ **$\frac{8}{19}$**

10. $\log_5 (x + 3) - \log_5 (2x - 1) = 2$ **$\frac{4}{7}$**

NAME _____ DATE _____ PERIOD ____

10-3 Skills Practice

Properties of Logarithms

Use $\log_2 3 \approx 1.5850$ and $\log_2 5 \approx 2.3219$ to approximate the value of each expression.

1. $\log_2 25$ **4.6438**
2. $\log_2 27$ **4.755**
3. $\log_2 \frac{3}{5}$ **−0.7369**
4. $\log_2 \frac{5}{3}$ **0.7369**
5. $\log_2 15$ **3.9069**
6. $\log_2 45$ **5.4919**
7. $\log_2 75$ **6.2288**
8. $\log_2 0.6$ **−0.7369**
9. $\log_2 \frac{1}{3}$ **−1.5850**
10. $\log_2 \frac{9}{5}$ **0.8481**

Solve each equation. Check your solutions.

11. $\log_{10} 27 = 3 \log_{10} x$ **3**
12. $3 \log_7 4 = 2 \log_7 b$ **8**
13. $\log_4 5 + \log_4 x = \log_4 60$ **12**
14. $\log_6 2c + \log_6 8 = \log_6 80$ **5**
15. $\log_5 y - \log_5 8 = \log_5 1$ **8**
16. $\log_2 q - \log_2 3 = \log_2 7$ **21**
17. $\log_9 4 + 2 \log_9 5 = \log_9 w$ **100**
18. $3 \log_8 2 - \log_8 4 = \log_8 b$ **2**
19. $\log_{10} x + \log_{10} (3x - 5) = \log_{10} 2$ **2**
20. $\log_4 x + \log_4 (2x - 3) = \log_4 2$ **2**
21. $\log_3 d + \log_3 3 = 3$ **9**
22. $\log_{10} y - \log_{10} (2 - y) = 0$ **1**
23. $\log_2 s + 2 \log_2 5 = 0$ **$\frac{1}{25}$**
24. $\log_2 (x + 4) - \log_2 (x - 3) = 3$ **4**
25. $\log_4 (n + 1) - \log_4 (n - 2) = 1$ **3**
26. $\log_5 10 + \log_5 12 = 3 \log_5 2 + \log_5 a$ **15**

Lesson 10-3

NAME _____ DATE _____ PERIOD ____

10-3 Practice (Average)

Properties of Logarithms

Use $\log_{10} 5 \approx 0.6990$ and $\log_{10} 7 \approx 0.8451$ to approximate the value of each expression.

1. $\log_{10} 35$ **1.5441**
2. $\log_{10} 25$ **1.3980**
3. $\log_{10} \frac{7}{5}$ **0.1461**
4. $\log_{10} \frac{5}{7}$ **−0.1461**
5. $\log_{10} 245$ **2.3892**
6. $\log_{10} 175$ **2.2431**
7. $\log_{10} 0.2$ **−0.6990**
8. $\log_{10} \frac{25}{7}$ **0.5529**

Solve each equation. Check your solutions.

9. $\log_7 n = \frac{2}{3} \log_7 8$ **4**
10. $\log_{10} u = \frac{3}{2} \log_{10} 4$ **8**
11. $\log_6 x + \log_6 9 = \log_6 54$ **6**
12. $\log_8 48 - \log_8 w = \log_8 4$ **12**
13. $\log_9 (3u + 14) - \log_9 5 = \log_9 2u$ **2**
14. $4 \log_2 x + \log_2 5 = \log_2 405$ **3**
15. $\log_3 y = -\log_3 16 + \frac{1}{3} \log_3 64$ **$\frac{1}{4}$**
16. $\log_2 d = 5 \log_2 2 - \log_2 8$ **4**
17. $\log_{10} (3m - 5) + \log_{10} m = \log_{10} 2$ **2**
18. $\log_{10} (b + 3) + \log_{10} b = \log_{10} 4$ **1**
19. $\log_8 (t + 10) - \log_8 (t - 1) = \log_8 12$ **2**
20. $\log_3 (a + 3) + \log_3 (a + 2) = \log_3 6$ **0**
21. $\log_{10} (r + 4) - \log_{10} r = \log_{10} (r + 1)$ **2**
22. $\log_4 (x^2 - 4) - \log_4 (x + 2) = \log_4 1$ **3**
23. $\log_{10} 4 + \log_{10} w = 2$ **25**
24. $\log_8 (n - 3) + \log_8 (n + 4) = 1$ **4**
25. $3 \log_5 (x^2 + 9) - 6 = 0$ **±4**
26. $\log_{16} (9x + 5) - \log_{16} (x^2 - 1) = \frac{1}{2}$ **3**
27. $\log_6 (2x - 5) + 1 = \log_6 (7x + 10)$ **8**
28. $\log_2 (5y + 2) - 1 = \log_2 (1 - 2y)$ **0**
29. $\log_{10} (c^2 - 1) - 2 = \log_{10} (c + 1)$ **101**
30. $\log_7 x + 2 \log_7 x - \log_7 3 = \log_7 72$ **6**

31. **SOUND** The loudness L of a sound in decibels is given by $L = 10 \log_{10} R$, where R is the sound's relative intensity. If the intensity of a certain sound is tripled, by how many decibels does the sound increase? **about 4.8 db**

32. **EARTHQUAKES** An earthquake rated at 3.5 on the Richter scale is felt by many people, and an earthquake rated at 4.5 may cause local damage. The Richter scale magnitude reading m is given by $m = \log_{10} x$, where x represents the amplitude of the seismic wave causing ground motion. How many times greater is the amplitude of an earthquake that measures 4.5 on the Richter scale than one that measures 3.5? **10 times**

NAME _____ DATE _____ PERIOD ____

10-3 Reading to Learn Mathematics

Properties of Logarithms

Pre-Activity **How are the properties of exponents and logarithms related?**

Read the introduction to Lesson 10-3 at the top of page 541 in your textbook.

Find the value of $\log_5 125$. **3** Find the value of $\log_5 5$. **1**

Find the value of $\log_5 (125 \div 5)$. **2**

Which of the following statements is true? **B**

A. $\log_5 (125 \div 5) = (\log_5 125) \div (\log_5 5)$

B. $\log_5 (125 \div 5) = \log_5 125 - \log_5 5$

Reading the Lesson

1. Each of the properties of logarithms can be stated in words or in symbols. Complete the statements of these properties in words.

a. The logarithm of a quotient is the **difference** of the logarithms of the **numerator** and the **denominator**.

b. The logarithm of a power is the **product** of the logarithm of the base and the **exponent**.

c. The logarithm of a product is the **sum** of the logarithms of its **factors**.

2. State whether each of the following equations is *true* or *false*. If the statement is true, name the property of logarithms that is illustrated.

a. $\log_3 10 = \log_3 30 - \log_3 3$ **true; Quotient Property**

b. $\log_4 12 = \log_4 4 + \log_4 8$ **false**

c. $\log_2 81 = 2 \log_2 9$ **true; Power Property**

d. $\log_8 30 = \log_8 5 \cdot \log_8 6$ **false**

3. The algebraic process of solving the equation $\log_2 x + \log_2 (x + 2) = 3$ leads to "$x = -4$ or $x = 2$." Does this mean that both -4 and 2 are solutions of the logarithmic equation? Explain your reasoning. **Sample answer: No; 2 is a solution because it checks: $\log_2 2 + \log_2 (2 + 2) = \log_2 2 + \log_2 4 = 1 + 2 = 3$. However, because $\log_2 (-4)$ and $\log_2 (-2)$ are undefined, -4 is an extraneous solution and must be eliminated. The only solution is 2.**

Helping You Remember

4. A good way to remember something is to relate it something you already know. Use words to explain how the Product Property for exponents can help you remember the product property for logarithms. **Sample answer: When you *multiply* two numbers or expressions with the same base, you *add* the exponents and keep the *same base*. Logarithms are exponents, so to find the logarithm of a *product*, you *add* the logarithms of the factors, keeping the *same base*.**

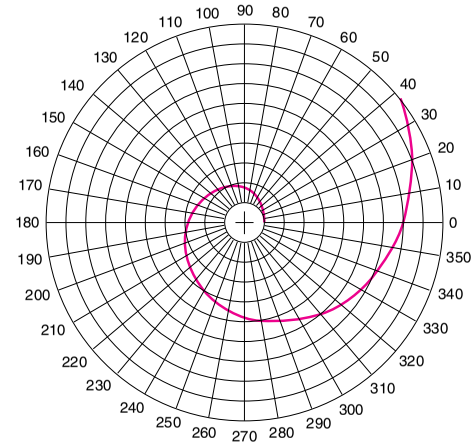
Lesson 10-3

NAME _____ DATE _____ PERIOD ____

10-3 Enrichment

Spirals

Consider an angle in standard position with its vertex at a point O called the pole. Its initial side is on a coordinatized axis called the *polar axis*. A point P on the terminal side of the angle is named by the *polar coordinates* (r, θ), where r is the directed distance of the point from O and θ is the measure of the angle. Graphs in this system may be drawn on polar coordinate paper such as the kind shown below.



1. Use a calculator to complete the table for $\log_2 r = \frac{\theta}{120}$.

(*Hint:* To find θ on a calculator, press 120 [×] [LOG] r [)] [÷] [LOG] 2 [)].)

r	1	2	3	4	5	6	7	8
θ	0°	120°	190°	240°	279°	310°	337°	360°

2. Plot the points found in Exercise 1 on the grid above and connect to form a smooth curve.

This type of spiral is called a logarithmic spiral because the angle measures are proportional to the logarithms of the radii.

NAME _____ DATE _____ PERIOD ____

10-4 Study Guide and Intervention

Common Logarithms

Common Logarithms Base 10 logarithms are called **common logarithms**. The expression $\log_{10} x$ is usually written without the subscript as $\log x$. Use the LOG key on your calculator to evaluate common logarithms.

The relation between exponents and logarithms gives the following identity.

Inverse Property of Logarithms and Exponents	$10^{\log x} = x$

Example 1 **Evaluate log 50 to four decimal places.**

Use the LOG key on your calculator. To four decimal places, $\log 50 = 1.6990$.

Example 2 **Solve $3^{2x+1} = 12$.**

$3^{2x+1} = 12$	Original equation
$\log 3^{2x+1} = \log 12$	Property of Equality for Logarithms
$(2x+1)\log 3 = \log 12$	Power Property of Logarithms
$2x + 1 = \frac{\log 12}{\log 3}$	Divide each side by log 3.
$2x = \frac{\log 12}{\log 3} - 1$	Subtract 1 from each side.
$x = \frac{1}{2}\left(\frac{\log 12}{\log 3} - 1\right)$	Multiply each side by $\frac{1}{2}$.
$x \approx 0.6309$	

Exercises

Use a calculator to evaluate each expression to four decimal places.

1. $\log 18$ **1.2553**
2. $\log 39$ **1.5911**
3. $\log 120$ **2.0792**
4. $\log 5.8$ **0.7634**
5. $\log 42.3$ **1.6263**
6. $\log 0.003$ **−2.5229**

Solve each equation or inequality. Round to four decimal places.

7. $4^{3x} = 12$ **0.5975**
8. $6^{x+2} = 18$ **−0.3869**
9. $5^{4x-2} = 120$ **1.2437**
10. $7^{3x-1} \geq 21$ **$\{x \mid x \geq 0.8549\}$**
11. $2.4^{x+4} = 30$ **−0.1150**
12. $6.5^{2x} \geq 200$ **$\{x \mid x \geq 1.4153\}$**
13. $3.6^{4x-1} = 85.4$ **1.1180**
14. $2^{x+5} = 3^{x-2}$ **13.9666**
15. $9^{3x} = 4^{5x+2}$ **−8.1595**
16. $6^{x-5} = 2^{7x+3}$ **−3.6069**

Lesson 10-4

NAME _____ DATE _____ PERIOD ____

10-4 Study Guide and Intervention *(continued)*

Common Logarithms

Change of Base Formula The following formula is used to change expressions with different logarithmic bases to common logarithm expressions.

Change of Base Formula	For all positive numbers a, b, and n, where $a \neq 1$ and $b \neq 1$, $\log_a n = \frac{\log_b n}{\log_b a}$

Example **Express $\log_8 15$ in terms of common logarithms. Then approximate its value to four decimal places.**

$\log_8 15 = \frac{\log_{10} 15}{\log_{10} 8}$	Change of Base Formula
≈ 1.3023	Simplify.

The value of $\log_8 15$ is approximately 1.3023.

Exercises

Express each logarithm in terms of common logarithms. Then approximate its value to four decimal places.

1. $\log_3 16$ **$\frac{\log 16}{\log 3}$, 2.5237**
2. $\log_2 40$ **$\frac{\log 40}{\log 2}$, 5.3219**
3. $\log_5 35$ **$\frac{\log 35}{\log 5}$, 2.2091**
4. $\log_4 22$ **$\frac{\log 22}{\log 4}$, 2.2297**
5. $\log_{12} 200$ **$\frac{\log 200}{\log 12}$, 2.1322**
6. $\log_2 50$ **$\frac{\log 50}{\log 2}$, 5.6439**
7. $\log_5 0.4$ **$\frac{\log 0.4}{\log 5}$, −0.5693**
8. $\log_3 2$ **$\frac{\log 2}{\log 3}$, 0.6309**
9. $\log_4 28.5$ **$\frac{\log 28.5}{\log 4}$, 2.4164**
10. $\log_3 (20)^2$ **$\frac{2 \log 20}{\log 3}$, 5.4537**
11. $\log_6 (5)^4$ **$\frac{4 \log 5}{\log 6}$, 3.5930**
12. $\log_8 (4)^5$ **$\frac{5 \log 4}{\log 8}$, 3.3333**
13. $\log_5 (8)^3$ **$\frac{3 \log 8}{\log 5}$, 3.8761**
14. $\log_2 (3.6)^6$ **$\frac{6 \log 3.6}{\log 2}$, 11.0880**
15. $\log_{12} (10.5)^4$ **$\frac{4 \log 10.5}{\log 12}$, 3.7851**
16. $\log_3 \sqrt{150}$ **$\frac{\log 150}{2 \log 3}$, 2.2804**
17. $\log_4 \sqrt[3]{39}$ **$\frac{\log 39}{3 \log 4}$, 0.8809**
18. $\log_5 \sqrt[4]{1600}$ **$\frac{\log 1600}{4 \log 5}$, 1.1460**

NAME _____ DATE _____ PERIOD ____

10-4 Skills Practice

Common Logarithms

Use a calculator to evaluate each expression to four decimal places.

1. log 6 **0.7782**

2. log 15 **1.1761**

3. log 1.1 **0.0414**

4. log 0.3 **−0.5229**

Use the formula pH = −log[*H*+] to find the pH of each substance given its concentration of hydrogen ions.

5. gastric juices: $[H+] = 1.0 \times 10^{-1}$ mole per liter **1.0**

6. tomato juice: $[H+] = 7.94 \times 10^{-5}$ mole per liter **4.1**

7. blood: $[H+] = 3.98 \times 10^{-8}$ mole per liter **7.4**

8. toothpaste: $[H+] = 1.26 \times 10^{-10}$ mole per liter **9.9**

Solve each equation or inequality. Round to four decimal places.

9. $3^x > 243$ $\{x|x > 5\}$

10. $16^v \le \frac{1}{4}$ $\left\{v \middle| v \le -\frac{1}{2}\right\}$

11. $8^p = 50$ **1.8813**

12. $7^y = 15$ **1.3917**

13. $5^{3b} = 106$ **0.9659**

14. $4^{5k} = 37$ **0.5209**

15. $12^{7p} = 120$ **0.2752**

16. $9^{2m} = 27$ **0.75**

17. $3^{r-5} = 4.1$ **6.2843**

18. $8^{y+4} > 15$ $\{y|y > -2.6977\}$

19. $7.6^{d+3} = 57.2$ **−1.0048**

20. $0.5^{t-8} = 16.3$ **3.9732**

21. $42^{x^2} = 84$ **±1.0888**

22. $5^{x^2+1} = 10$ **±0.6563**

Express each logarithm in terms of common logarithms. Then approximate its value to four decimal places.

23. $\log_3 7$ $\frac{\log_{10} 7}{\log_{10} 3}$; **1.7712**

24. $\log_5 66$ $\frac{\log_{10} 66}{\log_{10} 5}$; **2.6032**

25. $\log_2 35$ $\frac{\log_{10} 35}{\log_{10} 2}$; **5.1293**

26. $\log_6 10$ $\frac{\log_{10} 10}{\log_{10} 6}$; **1.2851**

Lesson 10-4

NAME _____ DATE _____ PERIOD ____

10-4 Practice (Average)

Common Logarithms

Use a calculator to evaluate each expression to four decimal places.

1. log 101 **2.0043**

2. log 2.2 **0.3424**

3. log 0.05 **−1.3010**

Use the formula pH = −log[*H*+] to find the pH of each substance given its concentration of hydrogen ions.

4. milk: $[H+] = 2.51 \times 10^{-7}$ mole per liter **6.6**

5. acid rain: $[H+] = 2.51 \times 10^{-6}$ mole per liter **5.6**

6. black coffee: $[H+] = 1.0 \times 10^{-5}$ mole per liter **5.0**

7. milk of magnesia: $[H+] = 3.16 \times 10^{-11}$ mole per liter **10.5**

Solve each equation or inequality. Round to four decimal places.

8. $2^x < 25$ $\{x|x < 4.6439\}$

9. $5^a = 120$ **2.9746**

10. $6^z = 45.6$ **2.1319**

11. $9^m \ge 100$ $\{m|m \ge 2.0959\}$

12. $3.5^x = 47.9$ **3.0885**

13. $8.2^y = 64.5$ **1.9802**

14. $2^{b+1} \le 7.31$ $\{b|b \le 1.8699\}$

15. $4^{2x} = 27$ **1.1887**

16. $2^{a-4} = 82.1$ **10.3593**

17. $9^{z-2} > 38$ $\{z|z > 3.6555\}$

18. $5^{w+3} = 17$ **−1.2396**

19. $30^{x^2} = 50$ **±1.0725**

20. $5^{x^2-3} = 72$ **±2.3785**

21. $4^{2x} = 9^{x+1}$ **3.8188**

22. $2^{n+1} = 5^{2n-1}$ **0.9117**

Express each logarithm in terms of common logarithms. Then approximate its value to four decimal places.

23. $\log_5 12$ $\frac{\log_{10} 12}{\log_{10} 5}$; **1.5440**

24. $\log_8 32$ $\frac{\log_{10} 32}{\log_{10} 8}$; **1.6667**

25. $\log_{11} 9$ $\frac{\log_{10} 9}{\log_{10} 11}$; **0.9163**

26. $\log_2 18$ $\frac{\log_{10} 18}{\log_{10} 2}$; **4.1699**

27. $\log_9 6$ $\frac{\log_{10} 6}{\log_{10} 9}$; **0.8155**

28. $\log_7 \sqrt{8}$ $\frac{\log_{10} 8}{2 \log_{10} 7}$; **0.5343**

29. HORTICULTURE Siberian irises flourish when the concentration of hydrogen ions $[H+]$ in the soil is not less than 1.58×10^{-8} mole per liter. What is the pH of the soil in which these irises will flourish? **7.8 or less**

30. ACIDITY The pH of vinegar is 2.9 and the pH of milk is 6.6. How many times greater is the hydrogen ion concentration of vinegar than of milk? **about 5000**

31. BIOLOGY There are initially 1000 bacteria in a culture. The number of bacteria doubles each hour. The number of bacteria N present after t hours is $N = 1000(2)^t$. How long will it take the culture to increase to 50,000 bacteria? **about 5.6 h**

32. SOUND An equation for loudness L in decibels is given by $L = 10 \log R$, where R is the sound's relative intensity. An air-raid siren can reach 150 decibels and jet engine noise can reach 120 decibels. How many times greater is the relative intensity of the air-raid siren than that of the jet engine noise? **1000**

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10-4 Reading to Learn Mathematics

Common Logarithms

Pre-Activity **Why is a logarithmic scale used to measure acidity?**

Read the introduction to Lesson 10-4 at the top of page 547 in your textbook.

Which substance is more acidic, milk or tomatoes?
tomatoes

Reading the Lesson

1. Rhonda used the following keystrokes to enter an expression on her graphing calculator:

[LOG] 17 [ )] [ENTER]

The calculator returned the result 1.230448921.
Which of the following conclusions are correct? **a, c, and d**

a. The base 10 logarithm of 17 is about 1.2304.

b. The base 17 logarithm of 10 is about 1.2304.

c. The common logarithm of 17 is about 1.230449.

d. $10^{1.230448921}$ is very close to 17.

e. The common logarithm of 17 is exactly 1.230448921.

2. Match each expression from the first column with an expression from the second column that has the same value.

a. $\log_2 2$ **iv**	**i.** $\log_4 1$
b. $\log 12$ **iii**	**ii.** $\log_2 8$
c. $\log_3 1$ **i**	**iii.** $\log_{10} 12$
d. $\log_5 \frac{1}{5}$ **v**	**iv.** $\log_5 5$
e. $\log 1000$ **ii**	**v.** $\log 0.1$

3. Calculators do not have keys for finding base 8 logarithms directly. However, you can use a calculator to find $\log_8 20$ if you apply the **change of base** formula.
Which of the following expressions are equal to $\log_8 20$? **B and C**

A. $\log_{20} 8$ **B.** $\frac{\log_{10} 20}{\log_{10} 8}$ **C.** $\frac{\log 20}{\log 8}$ **D.** $\frac{\log 8}{\log 20}$

Helping You Remember

4. Sometimes it is easier to remember a formula if you can state it in words. State the change of base formula in words. **Sample answer: To change the logarithm of a number from one base to another, divide the log of the original number in the old base by the log of the new base in the old base.**

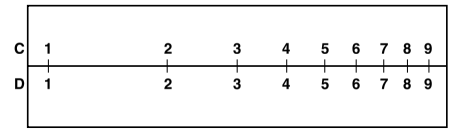
Lesson 10-4

NAME _____ DATE _____ PERIOD ____

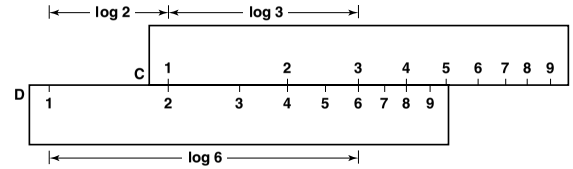
10-4 Enrichment

The Slide Rule

Before the invention of electronic calculators, computations were often performed on a slide rule. A slide rule is based on the idea of logarithms. It has two movable rods labeled with C and D scales. Each of the scales is logarithmic.

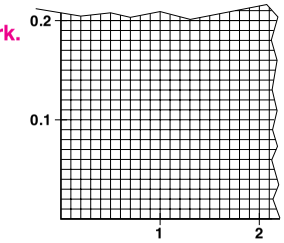


To multiply 2×3 on a slide rule, move the C rod to the right as shown below. You can find 2×3 by adding log 2 to log 3, and the slide rule adds the lengths for you. The distance you get is 0.778, or the logarithm of 6.

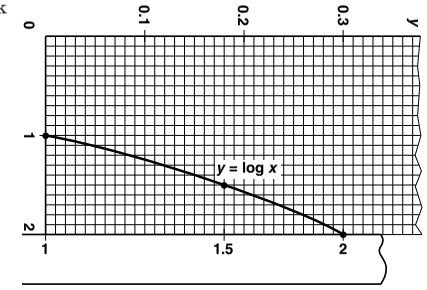


Follow the steps to make a slide rule. **1–2. See students' work.**

1. Use graph paper that has small squares, such as 10 squares to the inch. Using the scales shown at the right, plot the curve $y = \log x$ for $x = 1, 1.5$, and the whole numbers from 2 through 10. Make an obvious heavy dot for each point plotted.



2. You will need two strips of cardboard. A 5-by-7 index card, cut in half the long way, will work fine. Turn the graph you made in Exercise 1 sideways and use it to mark a logarithmic scale on each of the two strips. The figure shows the mark for 2 being drawn.



3. Explain how to use a slide rule to divide 8 by 2. **Line up the 2 on the C scale with the 8 on the D scale. The quotient is the number on the D scale below the 1 on the C scale.**

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10-5 Study Guide and Intervention

Base e and Natural Logarithms

Base *e* and Natural Logarithms The irrational number $e \approx 2.71828\ldots$ often occurs as the base for exponential and logarithmic functions that describe real-world phenomena.

Natural Base *e*	As n increases, $\left(1 + \frac{1}{n}\right)^n$ approaches $e \approx 2.71828\ldots$. $\ln x = \log_e x$

The functions $y = e^x$ and $y = \ln x$ are inverse functions.

Inverse Property of Base *e* and Natural Logarithms	$e^{\ln x} = x \qquad \ln e^x = x$

Natural base expressions can be evaluated using the e^x and ln keys on your calculator.

Example 1 **Evaluate ln 1685.**

Use a calculator.

$\ln 1685 \approx 7.4295$

Example 2 **Write a logarithmic equation equivalent to $e^{2x} = 7$.**

$e^{2x} = 7 \rightarrow \log_e 7 = 2x$ or $2x = \ln 7$

Example 3 **Evaluate $\ln e^{18}$.**

Use the Inverse Property of Base e and Natural Logarithms.

$\ln e^{18} = 18$

Exercises

Use a calculator to evaluate each expression to four decimal places.

1. $\ln 732$ **6.5958**

2. $\ln 84{,}350$ **11.3427**

3. $\ln 0.735$ **−0.3079**

4. $\ln 100$ **4.6052**

5. $\ln 0.0824$ **−2.4962**

6. $\ln 2.388$ **0.8705**

7. $\ln 128{,}245$ **11.7617**

8. $\ln 0.00614$ **−5.0929**

Write an equivalent exponential or logarithmic equation.

9. $e^{15} = x$ **$\ln x = 15$**

10. $e^{3x} = 45$ **$3x = \ln 45$**

11. $\ln 20 = x$ **$e^x = 20$**

12. $\ln x = 8$ **$x = e^8$**

13. $e^{-5x} = 0.2$ **$-5x = \ln 0.2$**

14. $\ln (4x) = 9.6$ **$4x = e^{9.6}$**

15. $e^{8.2} = 10x$ **$\ln 10x = 8.2$**

16. $\ln 0.0002 = x$ **$e^x = 0.0002$**

Evaluate each expression.

17. $\ln e^3$ **3**

18. $e^{\ln 42}$ **42**

19. $e^{\ln 0.5}$ **0.5**

20. $\ln e^{16.2}$ **16.2**

Lesson 10-5

NAME _____ DATE _____ PERIOD ____

10-5 Study Guide and Intervention *(continued)*

Base e and Natural Logarithms

Equations and Inequalities with *e* and ln All properties of logarithms from earlier lessons can be used to solve equations and inequalities with natural logarithms.

Example **Solve each equation or inequality.**

a. $3e^{2x} + 2 = 10$

$3e^{2x} + 2 = 10$	Original equation
$3e^{2x} = 8$	Subtract 2 from each side.
$e^{2x} = \frac{8}{3}$	Divide each side by 3.
$\ln e^{2x} = \ln \frac{8}{3}$	Property of Equality for Logarithms
$2x = \ln \frac{8}{3}$	Inverse Property of Exponents and Logarithms
$x = \frac{1}{2} \ln \frac{8}{3}$	Multiply each side by $\frac{1}{2}$.
$x \approx 0.4904$	Use a calculator.

b. $\ln (4x - 1) < 2$

$\ln (4x - 1) < 2$	Original inequality
$e^{\ln (4x - 1)} < e^2$	Write each side using exponents and base e.
$0 < 4x - 1 < e^2$	Inverse Property of Exponents and Logarithms
$1 < 4x < e^2 + 1$	Addition Property of Inequalities
$\frac{1}{4} < x < \frac{1}{4}(e^2 + 1)$	Multiplication Property of Inequalities
$0.25 < x < 2.0973$	Use a calculator.

Exercises

Solve each equation or inequality.

1. $e^{4x} = 120$ **1.1969**

2. $e^x \leq 25$ **$\{x | x \leq 3.2189\}$**

3. $e^{x-2} + 4 = 21$ **4.8332**

4. $\ln 6x \geq 4$ **$x \geq 9.0997$**

5. $\ln (x + 3) - 5 = -2$ **17.0855**

6. $e^{-8x} \leq 50$ **$\{x | x \geq -0.4890\}$**

7. $e^{4x-1} - 3 = 12$ **0.9270**

8. $\ln (5x + 3) = 3.6$ **6.7196**

9. $2e^{3x} + 5 = 2$ **no solution**

10. $6 + 3e^{x+1} = 21$ **0.6094**

11. $\ln (2x - 5) = 8$ **1492.9790**

12. $\ln 5x + \ln 3x > 9$ **$\{x | x > 23.2423\}$**

NAME _____ DATE _____ PERIOD ____

10-5 Skills Practice

Base e and Natural Logarithms

Use a calculator to evaluate each expression to four decimal places.

1. e^3 **20.0855**
2. e^{-2} **0.1353**
3. $\ln 2$ **0.6931**
4. $\ln 0.09$ **−2.4079**

Write an equivalent exponential or logarithmic equation.

5. $e^x = 3$ $x = \ln 3$
6. $e^4 = 8x$ $4 = \ln 8x$
7. $\ln 15 = x$ $e^x = 15$
8. $\ln x \approx 0.6931$ $x \approx e^{0.6931}$

Evaluate each expression.

9. $e^{\ln 3}$ **3**
10. $e^{\ln 2x}$ $2x$
11. $\ln e^{-2.5}$ **−2.5**
12. $\ln e^y$ y

Solve each equation or inequality.

13. $e^x \geq 5$ $\{x|x \geq 1.6094\}$
14. $e^x < 3.2$ $\{x|x < 1.1632\}$
15. $2e^x - 1 = 11$ **1.7918**
16. $5e^x + 3 = 18$ **1.0986**
17. $e^{3x} = 30$ **1.1337**
18. $e^{-4x} > 10$ $\{x|x < -0.5756\}$
19. $e^{5x} + 4 > 34$ $\{x|x > 0.6802\}$
20. $1 - 2e^{2x} = -19$ **1.1513**
21. $\ln 3x = 2$ **2.4630**
22. $\ln 8x = 3$ **2.5107**
23. $\ln (x - 2) = 2$ **9.3891**
24. $\ln (x + 3) = 1$ **−0.2817**
25. $\ln (x + 3) = 4$ **51.5982**
26. $\ln x + \ln 2x = 2$ **1.9221**

Lesson 10-5

NAME _____ DATE _____ PERIOD ____

10-5 Practice (Average)

Base e and Natural Logarithms

Use a calculator to evaluate each expression to four decimal places.

1. $e^{1.5}$ **4.4817**
2. $\ln 8$ **2.0794**
3. $\ln 3.2$ **1.1632**
4. $e^{-0.6}$ **0.5488**
5. $e^{4.2}$ **66.6863**
6. $\ln 1$ **0**
7. $e^{-2.5}$ **0.0821**
8. $\ln 0.037$ **−3.2968**

Write an equivalent exponential or logarithmic equation.

9. $\ln 50 = x$ $e^x = 50$
10. $\ln 36 = 2x$ $e^{2x} = 36$
11. $\ln 6 \approx 1.7918$ $e^{1.7918} \approx 6$
12. $\ln 9.3 \approx 2.2300$ $e^{2.2300} \approx 9.3$
13. $e^x = 8$ $x = \ln 8$
14. $e^5 = 10x$ $5 = \ln 10x$
15. $e^{-x} = 4$ $x = -\ln 4$
16. $e^2 = x + 1$ $2 = \ln (x + 1)$

Evaluate each expression.

17. $e^{\ln 12}$ **12**
18. $e^{\ln 3x}$ $3x$
19. $\ln e^{-1}$ **−1**
20. $\ln e^{-2y}$ $-2y$

Solve each equation or inequality.

21. $e^x < 9$ $\{x|x < 2.1972\}$
22. $e^{-x} = 31$ **−3.4340**
23. $e^x = 1.1$ **0.0953**
24. $e^x = 5.8$ **1.7579**
25. $2e^x - 3 = 1$ **0.6931**
26. $5e^x + 1 \geq 7$ $\{x|x \geq 0.1823\}$
27. $4 + e^x = 19$ **2.7081**
28. $-3e^x + 10 < 8$ $\{x|x > -0.4055\}$
29. $e^{3x} = 8$ **0.6931**
30. $e^{-4x} = 5$ **−0.4024**
31. $e^{0.5x} = 6$ **3.5835**
32. $2e^{5x} = 24$ **0.4970**
33. $e^{2x} + 1 = 55$ **1.9945**
34. $e^{3x} - 5 = 32$ **1.2036**
35. $9 + e^{2x} = 10$ **0**
36. $e^{-3x} + 7 \geq 15$ $\{x|x \leq -0.6931\}$
37. $\ln 4x = 3$ **5.0214**
38. $\ln (-2x) = 7$ **−548.3166**
39. $\ln 2.5x = 10$ **8810.5863**
40. $\ln (x - 6) = 1$ **8.7183**
41. $\ln (x + 2) = 3$ **18.0855**
42. $\ln (x + 3) = 5$ **145.4132**
43. $\ln 3x + \ln 2x = 9$ **36.7493**
44. $\ln 5x + \ln x = 7$ **14.8097**

INVESTING For Exercises 45 and 46, use the formula for continuously compounded interest, $A = Pe^{rt}$, where P is the principal, r is the annual interest rate, and t is the time in years.

45. If Sarita deposits $1000 in an account paying 3.4% annual interest compounded continuously, what is the balance in the account after 5 years? **$1185.30**
46. How long will it take the balance in Sarita's account to reach $2000? **about 20.4 yr**
47. **RADIOACTIVE DECAY** The amount of a radioactive substance y that remains after t years is given by the equation $y = ae^{kt}$, where a is the initial amount present and k is the decay constant for the radioactive substance. If $a = 100$, $y = 50$, and $k = -0.035$, find t. **about 19.8 yr**

NAME _____ DATE _____ PERIOD ____

10-5 Reading to Learn Mathematics

Base e and Natural Logarithms

Pre-Activity **How is the natural base *e* used in banking?**

Read the introduction to Lesson 10-5 at the top of page 554 in your textbook.

Suppose that you deposit $675 in a savings account that pays an annual interest rate of 5%. In each case listed below, indicate which method of compounding would result in more money in your account at the end of one year.

a. annual compounding or monthly compounding **monthly**

b. quarterly compounding or daily compounding **daily**

c. daily compounding or continuous compounding **continuous**

Reading the Lesson

1. Jagdish entered the following keystrokes in his calculator:

[LN] 5 [)] [ENTER]

The calculator returned the result 1.609437912. Which of the following conclusions are correct? **d and f**

a. The common logarithm of 5 is about 1.6094.

b. The natural logarithm of 5 is exactly 1.609437912.

c. The base 5 logarithm of e is about 1.6094.

d. The natural logarithm of 5 is about 1.609438.

e. $10^{1.609437912}$ is very close to 5.

f. $e^{1.609437912}$ is very close to 5.

2. Match each expression from the first column with its value in the second column. Some choices may be used more than once or not at all.

a. $e^{\ln 5}$ **IV**	**I.** 1
b. $\ln 1$ **V**	**II.** 10
c. $e^{\ln e}$ **VI**	**III.** -1
d. $\ln e^5$ **IV**	**IV.** 5
e. $\ln e$ **I**	**V.** 0
f. $\ln\left(\frac{1}{e}\right)$ **III**	**VI.** e

Helping You Remember

3. A good way to remember something is to explain it to someone else. Suppose that you are studying with a classmate who is puzzled when asked to evaluate $\ln e^3$. How would you explain to him an easy way to figure this out? **Sample answer: *ln* means *natural log*. The natural log of e^3 is the power to which you raise e to get e^3. This is obviously 3.**

Lesson 10-5

NAME _____ DATE _____ PERIOD ____

10-5 Enrichment

Approximations for π and e

The following expression can be used to approximate e. If greater and greater values of n are used, the value of the expression approximates e more and more closely.

$$\left(1 + \frac{1}{n}\right)^n$$

Another way to approximate e is to use this infinite sum. The greater the value of n, the closer the approximation.

$$e = 1 + 1 + \frac{1}{2} + \frac{1}{2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \ldots + \frac{1}{2 \cdot 3 \cdot 4 \cdot \ldots \cdot n} + \ldots$$

In a similar manner, π can be approximated using an infinite product discovered by the English mathematician John Wallis (1616–1703).

$$\frac{\pi}{2} = \frac{2}{1} \cdot \frac{2}{3} \cdot \frac{4}{3} \cdot \frac{4}{5} \cdot \frac{6}{5} \cdot \frac{6}{7} \cdot \ldots \cdot \frac{2n}{2n-1} \cdot \frac{2n}{2n+1} \ldots$$

Solve each problem.

1. Use a calculator with an e^x key to find e to 7 decimal places. **2.7182818**

2. Use the expression $\left(1 + \frac{1}{n}\right)^n$ to approximate e to 3 decimal places. Use 5, 100, 500, and 7000 as values of n. **2.488, 2.705, 2.716, 2.718**

3. Use the infinite sum to approximate e to 3 decimal places. Use the whole numbers from 3 through 6 as values of n. **2.667, 2.708, 2.717, 2.718**

4. Which approximation method approaches the value of e more quickly? **the infinite sum**

5. Use a calculator with a π key to find π to 7 decimal places. **3.1415927**

6. Use the infinite product to approximate π to 3 decimal places. Use the whole numbers from 3 through 6 as values of n. **2.926, 2.972, 3.002, 3.023**

7. Does the infinite product give good approximations for π quickly? **no**

8. Show that $\pi^4 + \pi^5$ is equal to e^6 to 4 decimal places. **To 4 decimal places, they both equal 403.4288.**

9. Which is larger, e^π or π^e? $e^\pi > \pi^e$

10. The expression x reaches a maximum value at $x = e$. Use this fact to prove the inequality you found in Exercise 9.

$e^{\frac{1}{e}} > \pi^{\frac{1}{\pi}}$; $\left(e^{\frac{1}{e}}\right)^{\pi e} > \left(\pi^{\frac{1}{\pi}}\right)^{\pi e}$; $e^\pi > \pi^e$

NAME _____ DATE _____ PERIOD ____

10-6 Study Guide and Intervention

Exponential Growth and Decay

Exponential Decay Depreciation of value and radioactive decay are examples of **exponential decay**. When a quantity decreases by a fixed percent each time period, the amount of the quantity after t time periods is given by $y = a(1 - r)^t$, where a is the initial amount and r is the percent decrease expressed as a decimal.

Another exponential decay model often used by scientists is $y = ae^{-kt}$, where k is a constant.

Example **CONSUMER PRICES As technology advances, the price of many technological devices such as scientific calculators and camcorders goes down. One brand of hand-held organizer sells for $89.**

a. If its price decreases by 6% per year, how much will it cost after 5 years?

Use the exponential decay model with initial amount $89, percent decrease 0.06, and time 5 years.

$y = a(1 - r)^t$ Exponential decay formula

$y = 89(1 - 0.06)^5$ $a = 89$, $r = 0.06$, $t = 5$

$y = \$65.32$

After 5 years the price will be $65.32.

b. After how many years will its price be $50?

To find when the price will be $50, again use the exponential decay formula and solve for t.

$y = a(1 - r)^t$ Exponential decay formula

$50 = 89(1 - 0.06)^t$ $y = 50$, $a = 89$, $r = 0.06$

$\frac{50}{89} = (0.94)^t$ Divide each side by 89.

$\log\left(\frac{50}{89}\right) = \log(0.94)^t$ Property of Equality for Logarithms

$\log\left(\frac{50}{89}\right) = t \log 0.94$ Power Property

$t = \frac{\log\left(\frac{50}{89}\right)}{\log 0.94}$ Divide each side by log 0.94.

$t \approx 9.3$

The price will be $50 after about 9.3 years.

Exercises

1. **BUSINESS** A furniture store is closing out its business. Each week the owner lowers prices by 25%. After how many weeks will the sale price of a $500 item drop below $100? **6 weeks**

CARBON DATING Use the formula $y = ae^{-0.00012t}$, where a is the initial amount of Carbon-14, t is the number of years ago the animal lived, and y is the remaining amount after t years.

2. How old is a fossil remain that has lost 95% of its Carbon-14? **about 25,000 years old**

3. How old is a skeleton that has 95% of its Carbon-14 remaining? **about 427.5 years old**

Lesson 10-6

NAME _____ DATE _____ PERIOD ____

10-6 Study Guide and Intervention *(continued)*

Exponential Growth and Decay

Exponential Growth Population increase and growth of bacteria colonies are examples of **exponential growth**. When a quantity increases by a fixed percent each time period, the amount of that quantity after t time periods is given by $y = a(1 + r)^t$, where a is the initial amount and r is the percent increase (or rate of growth) expressed as a decimal.

Another exponential growth model often used by scientists is $y = ae^{kt}$, where k is a constant.

Example **A computer engineer is hired for a salary of $28,000. If she gets a 5% raise each year, after how many years will she be making $50,000 or more?**

Use the exponential growth model with $a = 28{,}000$, $y = 50{,}000$, and $r = 0.05$ and solve for t.

$y = a(1 + r)^t$ Exponential growth formula

$50{,}000 = 28{,}000(1 + 0.05)^t$ $y = 50{,}000$, $a = 28{,}000$, $r = 0.05$

$\frac{50}{28} = (1.05)^t$ Divide each side by 28,000.

$\log\left(\frac{50}{28}\right) = \log(1.05)^t$ Property of Equality of Logarithms

$\log\left(\frac{50}{28}\right) = t \log 1.05$ Power Property

$t = \frac{\log\left(\frac{50}{28}\right)}{\log 1.05}$ Divide each side by log 1.05.

$t \approx 11.9$ years Use a calculator.

If raises are given annually, she will be making over $50,000 in 12 years.

Exercises

1. **BACTERIA GROWTH** A certain strain of bacteria grows from 40 to 326 in 120 minutes. Find k for the growth formula $y = ae^{kt}$, where t is in minutes. **about 0.0175**

2. **INVESTMENT** Carl plans to invest $500 at 8.25% interest, compounded continuously. How long will it take for his money to triple? **about 14 years**

3. **SCHOOL POPULATION** There are currently 850 students at the high school, which represents full capacity. The town plans an addition to house 400 more students. If the school population grows at 7.8% per year, in how many years will the new addition be full? **about 5 years**

4. **EXERCISE** Hugo begins a walking program by walking $\frac{1}{2}$ mile per day for one week. Each week thereafter he increases his mileage by 10%. After how many weeks is he walking more than 5 miles per day? **24 weeks**

5. **VOCABULARY GROWTH** When Emily was 18 months old, she had a 10-word vocabulary. By the time she was 5 years old (60 months), her vocabulary was 2500 words. If her vocabulary increased at a constant percent per month, what was that increase? **about 14%**

NAME _____ DATE _____ PERIOD ____

10-6 Skills Practice

Exponential Growth and Decay

Solve each problem.

1. **FISHING** In an over-fished area, the catch of a certain fish is decreasing at an average rate of 8% per year. If this decline persists, how long will it take for the catch to reach half of the amount before the decline? **about 8.3 yr**

2. **INVESTING** Alex invests \$2000 in an account that has a 6% annual rate of growth. To the nearest year, when will the investment be worth \$3600? **10 yr**

3. **POPULATION** A current census shows that the population of a city is 3.5 million. Using the formula $P = ae^{rt}$, find the expected population of the city in 30 years if the growth rate r of the population is 1.5% per year, a represents the current population in millions, and t represents the time in years. **about 5.5 million**

4. **POPULATION** The population P in thousands of a city can be modeled by the equation $P = 80e^{0.015t}$, where t is the time in years. In how many years will the population of the city be 120,000? **about 27 yr**

5. **BACTERIA** How many days will it take a culture of bacteria to increase from 2000 to 50,000 if the growth rate per day is 93.2%? **about 4.9 days**

6. **NUCLEAR POWER** The element plutonium-239 is highly radioactive. Nuclear reactors can produce and also use this element. The heat that plutonium-239 emits has helped to power equipment on the moon. If the half-life of plutonium-239 is 24,360 years, what is the value of k for this element? **about 0.00002845**

7. **DEPRECIATION** A Global Positioning Satellite (GPS) system uses satellite information to locate ground position. Abu's surveying firm bought a GPS system for \$12,500. The GPS depreciated by a fixed rate of 6% and is now worth \$8600. How long ago did Abu buy the GPS system? **about 6.0 yr**

8. **BIOLOGY** In a laboratory, an organism grows from 100 to 250 in 8 hours. What is the hourly growth rate in the growth formula $y = a(1 + r)^t$? **about 12.13%**

Lesson 10-6

NAME _____ DATE _____ PERIOD ____

10-6 Practice (Average)

Exponential Growth and Decay

Solve each problem.

1. **INVESTING** The formula $A = P\left(1 + \frac{r}{2}\right)^{2t}$ gives the value of an investment after t years in an account that earns an annual interest rate r compounded twice a year. Suppose \$500 is invested at 6% annual interest compounded twice a year. In how many years will the investment be worth \$1000? **about 11.7 yr**

2. **BACTERIA** How many hours will it take a culture of bacteria to increase from 20 to 2000 if the growth rate per hour is 85%? **about 7.5 h**

3. **RADIOACTIVE DECAY** A radioactive substance has a half-life of 32 years. Find the constant k in the decay formula for the substance. **about 0.02166**

4. **DEPRECIATION** A piece of machinery valued at \$250,000 depreciates at a fixed rate of 12% per year. After how many years will the value have depreciated to \$100,000? **about 7.2 yr**

5. **INFLATION** For Dave to buy a new car comparably equipped to the one he bought 8 years ago would cost \$12,500. Since Dave bought the car, the inflation rate for cars like his has been at an average annual rate of 5.1%. If Dave originally paid \$8400 for the car, how long ago did he buy it? **about 8 yr**

6. **RADIOACTIVE DECAY** Cobalt, an element used to make alloys, has several isotopes. One of these, cobalt-60, is radioactive and has a half-life of 5.7 years. Cobalt-60 is used to trace the path of nonradioactive substances in a system. What is the value of k for Cobalt-60? **about 0.1216**

7. **WHALES** Modern whales appeared 5–10 million years ago. The vertebrae of a whale discovered by paleontologists contain roughly 0.25% as much carbon-14 as they would have contained when the whale was alive. How long ago did the whale die? Use $k = 0.00012$. **about 50,000 yr**

8. **POPULATION** The population of rabbits in an area is modeled by the growth equation $P(t) = 8e^{0.26t}$, where P is in thousands and t is in years. How long will it take for the population to reach 25,000? **about 4.4 yr**

9. **DEPRECIATION** A computer system depreciates at an average rate of 4% per month. If the value of the computer system was originally \$12,000, in how many months is it worth \$7350? **about 12 mo**

10. **BIOLOGY** In a laboratory, a culture increases from 30 to 195 organisms in 5 hours. What is the hourly growth rate in the growth formula $y = a(1 + r)^t$? **about 45.4%**

NAME _____ DATE _____ PERIOD ____

10-6 Reading to Learn Mathematics

Exponential Growth and Decay

Pre-Activity **How can you determine the current value of your car?**

Read the introduction to Lesson 10-6 at the top of page 560 in your textbook.

- Between which two years shown in the table did the car depreciate by the greatest amount?
  between years 0 and 1
- Describe two ways to calculate the value of the car 6 years after it was purchased. (Do not actually calculate the value.)
  Sample answer: 1. Multiply \$9200.66 by 0.16 and subtract the result from \$9200.66. 2. Multiply \$9200.66 by 0.84.

Reading the Lesson

1. State whether each situation is an example of exponential *growth* or *decay*.

a. A city had 42,000 residents in 1980 and 128,000 residents in 2000. **growth**

b. Raul compared the value of his car when he bought it new to the value when he traded ';lpit in six years later. **decay**

c. A paleontologist compared the amount of carbon-14 in the skeleton of an animal when it died to the amount 300 years later. **decay**

d. Maria deposited \$750 in a savings account paying 4.5% annual interest compounded quarterly. She did not make any withdrawals or further deposits. She compared the balance in her passbook immediately after she opened the account to the balance 3 years later. **growth**

2. State whether each equation represents exponential growth or decay.

a. $y = 5e^{0.15t}$ **growth**

b. $y = 1000(1 - 0.05)^t$ **decay**

c. $y = 0.3e^{-1200t}$ **decay**

d. $y = 2(1 + 0.0001)^t$ **growth**

Helping You Remember

3. Visualizing their graphs is often a good way to remember the difference between mathematical equations. How can your knowledge of the graphs of exponential equations from Lesson 10-1 help you to remember that equations of the form $y = a(1 + r)^t$ represent exponential growth, while equations of the form $y = a(1 - r)^t$ represent exponential decay?
Sample answer: If $a > 0$, the graph of $y = ab^x$ is always increasing if $b > 1$ and is always decreasing if $0 < b < 1$. Since r is always a positive number, if $b = 1 + r$, the base will be greater than 1 and the function will be increasing (growth), while if $b = 1 - r$, the base will be less than 1 and the function will be decreasing (decay).

Lesson 10-6

NAME _____ DATE _____ PERIOD ____

10-6 Enrichment

Effective Annual Yield

When interest is compounded more than once per year, the effective annual yield is higher than the annual interest rate. The effective annual yield, E, is the interest rate that would give the same amount of interest if the interest were compounded once per year. If P dollars are invested for one year, the value of the investment at the end of the year is $A = P(1 + E)$. If P dollars are invested for one year at a nominal rate r compounded n times per year, the value of the investment at the end of the year is $A = P\left(1 + \frac{r}{n}\right)^n$. Setting the amounts equal and solving for E will produce a formula for the effective annual yield.

$$P(1 + E) = P\left(1 + \frac{r}{n}\right)^n$$
$$1 + E = \left(1 + \frac{r}{n}\right)^n$$
$$E = \left(1 + \frac{r}{n}\right)^n - 1$$

If compounding is continuous, the value of the investment at the end of one year is $A = Pe^r$. Again set the amounts equal and solve for E. A formula for the effective annual yield under continuous compounding is obtained.

$$P(1 + E) = Pe^r$$
$$1 + E = e^r$$
$$E = e^r - 1$$

Example 1 **Find the effective annual yield of an investment made at 7.5% compounded monthly.**

$r = 0.075$

$n = 12$

$E = \left(1 + \frac{0.075}{12}\right)^{12} - 1 \approx 7.76\%$

Example 2 **Find the effective annual yield of an investment made at 6.25% compounded continuously.**

$r = 0.0625$

$E = e^{0.0625} - 1 \approx 6.45\%$

Find the effective annual yield for each investment.

1. 10% compounded quarterly **10.38%**

2. 8.5% compounded monthly **8.84%**

3. 9.25% compounded continuously **9.69%**

4. 7.75% compounded continuously **8.06%**

5. 6.5% compounded daily (assume a 365-day year) **6.72%**

6. Which investment yields more interest—9% compounded continuously or 9.2% compounded quarterly? **9.2% quarterly**

Chapter 10 Assessment Answer Key

Form 1
Page 609

1. B
2. D
3. A
4. A
5. C
6. D
7. B
8. A
9. D
10. C

Page 610

11. A
12. B
13. D
14. B
15. C
16. D
17. B
18. C
19. A
20. D

B: 21

Form 2A
Page 611

1. B
2. D
3. C
4. C
5. C
6. D
7. B
8. D
9. C
10. A

(continued on the next page)

Chapter 10 Assessment Answer Key

Form 2A *(continued)*
Page 612

11. B
12. B
13. A
14. B
15. D
16. C
17. A
18. A
19. C
20. B

B: 4

Form 2B
Page 613

1. A
2. A
3. C
4. B
5. B
6. D
7. B
8. B
9. A
10. D

Page 614

11. C
12. C
13. A
14. B
15. C
16. A
17. B
18. D
19. B
20. A

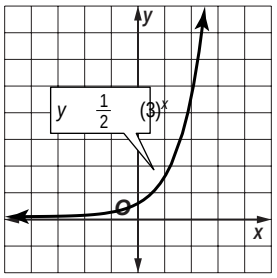
B: 4

Answers

Chapter 10 Assessment Answer Key

Form 2C

Page 615

1. 

D = {x | x is any real number.}; R = {y | $y > 0$}

2. decay

3. $y = -6\left(\frac{1}{3}\right)^x$

4. $z^{6\sqrt{7}}$

5. -5

6.

7. $81^{-\frac{1}{2}} = \frac{1}{9}$

8. 7

9. $\frac{7}{2}$

10. 216

11. $x > 2$

12. 2.4054

13. 0.3174

Page 616

14. 15

15. 1

16. 4

17. 4.1833

18. 3.5129

19.

20. $\frac{\log 4}{\log 12} \approx 0.5579$

21. 30

22. 25.0855

23.

24. $k \approx -0.0578$; $y = ae^{-0.0578t}$

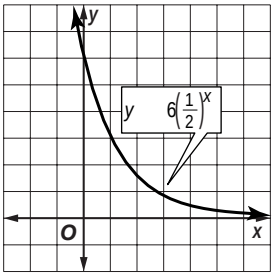
25. 17.4 yr

B: 9

Chapter 10 Assessment Answer Key

Form 2D

Page 617

1. 

D = {x | x is any real number.}; R = {y | $y > 0$}

2. growth

3. $y = -5\left(\frac{1}{2}\right)^x$

4. $m^{8\sqrt{6}}$

5. 5

6. $x < 5$

7. $\log_5 \frac{1}{625} = -4$

8. 8

9. $\frac{7}{3}$

10. 16

11. $x \geq 7$

12. 1.7959

13. 0.5693

Page 618

14. 28

15. $\frac{1}{4}$

16. 4

17. −0.7481

18. 5.1699

19. $x < 0.7783$

20. $\frac{\log 5}{\log 15} \approx 0.5943$

21. 22

22. 50.5982

23. $x \geq -0.9635$

24. $k \approx -0.0575$; $y = ae^{-0.0575t}$

25. 19.5 yr

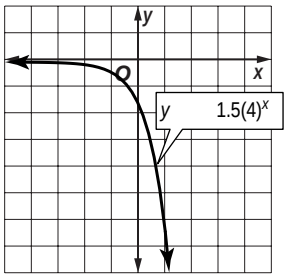
B: 16

Answers

Chapter 10 Assessment Answer Key

Form 3

Page 619

1. 

D = $\{x \mid x$ is any real number.$\}$; R = $\{y \mid y < 0\}$

2. decay

3. $y = -0.3(6)^x$

4. $-\frac{1}{4}$

5.

6. $8x - 1$

7. $5x$

8. 1

9. $a > 4$

10. 0.8212

11. 0.1133

12. 8

13. $-3, 8$

Page 620

14. $\frac{x^3}{3}$

15. ± 0.5645

16. -5.4763

17. $t > 0.9958$

18. $\frac{3 \log 2.1}{\log 5} \approx 1.3830$

19. x^4

20. 1

21. $\{x \mid 0 < x < 2 \text{ or } x > 5\}$

22. $y = 663{,}906e^{-0.0038t}$

23. about 592,375 people

24. 4.7%

25. about 123 children

B: 100; 1

Chapter 10 Assessment Answer Key

Page 621, Open-Ended Assessment
Scoring Rubric

Score	General Description	Specific Criteria
4	**Superior** A correct solution that is supported by well-developed, accurate explanations	• Shows thorough understanding of the concepts of *graphing exponential functions, solving exponential and logarithmic equations and inequalities, using properties of logarithms and the Change of Base Formula, evaluating common and natural logarithms,* and *solving applied problems involving exponential growth and decay.* • Uses appropriate strategies to solve problems. • Computations are correct. • Written explanations are exemplary. • Goes beyond requirements of some or all problems.
3	**Satisfactory** A generally correct solution, but may contain minor flaws in reasoning or computation	• Shows an understanding of the concepts of *graphing exponential functions, solving exponential and logarithmic equations and inequalities, using properties of logarithms and the Change of Base Formula, evaluating common and natural logarithms,* and *solving applied problems involving exponential growth and decay.* • Uses appropriate strategies to solve problems. • Computations are mostly correct. • Written explanations are effective. • Satisfies all requirements of problems.
2	**Nearly Satisfactory** A partially correct interpretation and/or solution to the problem	• Shows an understanding of most of the concepts of *graphing exponential functions, solving exponential and logarithmic equations and inequalities, using properties of logarithms and the Change of Base Formula, evaluating common and natural logarithms,* and *solving applied problems involving exponential growth and decay.* • May not use appropriate strategies to solve problems. • Computations are mostly correct. • Written explanations are satisfactory. • Satisfies the requirements of most of the problems.
1	**Nearly Unsatisfactory** A correct solution with no supporting evidence or explanation	• Final computation is correct. • No written explanations or work is shown to substantiate the final computation. • Satisfies minimal requirements of some of the problems.
0	**Unsatisfactory** An incorrect solution indicating no mathematical understanding of the concept or task, or no solution is given	• Shows little or no understanding of most of the concepts of *graphing exponential functions, solving exponential and logarithmic equations and inequalities, using properties of logarithms and the Change of Base Formula, evaluating common and natural logarithms,* and *solving applied problems involving exponential growth and decay.* • Does not use appropriate strategies to solve problems. • Computations are incorrect. • Written explanations are unsatisfactory. • Does not satisfy requirements of problems. • No answer may be given.

Answers

Chapter 10 Assessment Answer Key

Page 621, Open-Ended Assessment
Sample Answers

In addition to the scoring rubric found on page A25, the following sample answers may be used as guidance in evaluating open-ended assessment items.

1a. Sample answer: Choose $a = 2$, so $y = 2(1)^x$. All values for y are 2. Students should indicate that $y = a$ (regardless of their choice of a) is a constant function, not an exponential one, since the value of y never changes.

1b. Sample answer: Choose $a = 2$ and $b = -2$, so $y = 2(-2)^x$. The table of values for y are $-\frac{1}{4}, \frac{1}{2}, -1, 2, -4, 8, -16$. Students should indicate that this function is not exponential. Since values of y alternate sign, the function does not continuously increase or decrease.

2a. Students should rewrite 9 as 3^2, then demonstrate the appropriate steps to show that $x = 4$.

2b. Students should take the common logarithm of both sides of the equation, then write all steps to show that $x = \frac{6 \log 9}{5 \log 3 - \log 9}$ or 4.

2c. Sample answer: The first method is easier when both sides of an equation can be written as a power of the same base.

2d. Sample answer: $2^x = 16^{x+1}$; $x = -\frac{4}{3}$

3a. Sample answer: All three are logarithmic equations which can be written in exponential form as $(\text{base})^2 = x$. The bases of the three logarithms differ: the first equation has base 3, the second has base 10, and the third has base e.

3b. 9; 100; $e^2 \approx 7.3891$; Sample answer: $\log_5 x = 2$; 25

4a. Students should show all appropriate steps in reaching the final two steps:
$z(2 \log 2 - \log 12) \geq \log 12$
$z \geq \frac{\log 12}{2 \log 2 - \log 12} \approx -2.2619$

4b. Students should state that Ruby's checks show that her solution is incorrect.

4c. In the last step of her solution, Ruby should have changed the direction of the inequality symbol to obtain the solution $\{z \mid z \leq -2.2619\}$.

5a. Jones Corporation:
\$10,000,000; \$9,043,821
Davis Company:
\$8,000,000; \$8,836,977

5b. Students should indicate that since annual profits have been increasing for the Davis Company, stock in this company would most likely increase in value also.

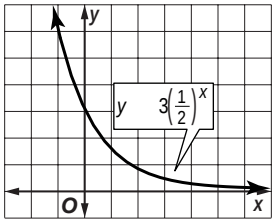
5c. Students should demonstrate an understanding that the Jones Corporation's profit equation represents exponential decay since $b = 0.99 < 1$, while the Davis Company's profit equation represents exponential growth since $b = 1.01 > 1$.

Chapter 10 Assessment Answer Key

Vocabulary Test/Review
Page 622

1. natural logarithm
2. exponential function
3. exponential decay
4. natural logarithmic function
5. exponential growth
6. Change of Base Formula
7. logarithmic function
8. exponential equations
9. common logarithm
10. rate of growth
11. Sample answer: A logarithm is the power to which you must raise a number to get a specified number.
12. Sample answer: The natural base, *e*, is the base of natural logarithms. This is an irrational number that is approximately 2.71828.

Quiz (Lessons 10–1 and 10–2)
Page 623

1. 

D = {x | x is any real number.}, R = {y | $y > 0$}
2. $y = -5\left(\frac{1}{2}\right)^x$; decay
3. $3^{3\sqrt{5}}$
4. $-\frac{3}{2}$
5.
6. $\log_{81} 9 = \frac{1}{2}$
7. $216^{\frac{2}{3}} = 36$
8. $\frac{3}{2}$
9. $\frac{1}{4}$
10.

Quiz (Lesson 10–3)
Page 623

1. 0.7481
2. 1.9747
3. $\frac{9}{2}$
4. $\frac{5}{32}$
5. 7

Quiz (Lessons 10–4 and 10–5)
Page 624

1. 0.1761
2. 1.4110
3. 1.7556
4. 13.6972
5.
6. 15.0855
7.
8. $\frac{\log 25}{\log 3} \approx 2.9299$
9. $3 = \ln 2x$
10. 0.3

Quiz (Lesson 10–6)
Page 624

1. 277.3 min
2. 2.8 yr
3. C
4. $231,366
5. $k \approx 0.0277$; $y = ae^{0.0277t}$

Answers

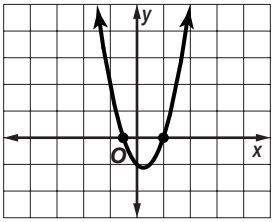
Chapter 10 Assessment Answer Key

Mid-Chapter Test
Page 625

1. C
2. D
3. B
4. A
5. A
6. D
7. $y = -3(2^x)$
8.
9. 25
10. 4
11. 9
12. 6.3924

Cumulative Review
Page 626

1. Z, Q, R
2. $t \geq 0$, $r \geq 0$, $t + r \leq 90$ $25t + 40r \leq 3000$; (0, 0), (0, 75), (40, 50), (90, 0)
3. $f(t, r) = 14t + 20r$ 40 tent sites, 50 RV sites; \$1560
4. $3x\sqrt[3]{y}$
5. between 0 and −1; 1



6. $3, 1 + i, 1 - i$
7. $x^2 + 4x - 16$; $x^2 - 14$; $2x^3 + 3x^2 - 32x + 15$; $\frac{x^2 + 2x - 15}{2x - 1}$, $x \neq \frac{1}{2}$
8. $\frac{y^2}{25} - \frac{x^2}{9} = 1$
9. $(2\sqrt{3}, 1), (2\sqrt{3}, -1), (-2\sqrt{3}, 1), (-2\sqrt{3}, -1)$
10. $\frac{2n^2s^4}{5m}$
11. $\frac{7}{x - 5}$
12. greatest integer
13. $y = 4\left(\frac{1}{5}\right)^x$
14. 81
15. $\{n \mid n > 12.2009\}$

Chapter 10 Assessment Answer Key

Standardized Test Practice

Page 627

1. **A** B C D

2. E F **G** H

3. A B C **D**

4. E F G **H**

5. A **B** C D

6. E F **G** H

7. **A** B C D

8. E F G **H**

9. A B **C** D

Page 628

10.

4	2		
.	/ .	/ .	.
	0	0	0
1	1	1	1
2	**2**	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

11.

4	/	5	
.	/ .	/ .	.
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	**5**	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

12.

1	0	8	
.	/ .	/ .	.
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	**8**	8
9	9	9	9

13.

4	8		
.	/ .	/ .	.
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	**8**	8	8
9	9	9	9

14. A B **C** D

15. A **B** C D

16. A B C **D**

17. **A** B C D

18. A **B** C D

Answers

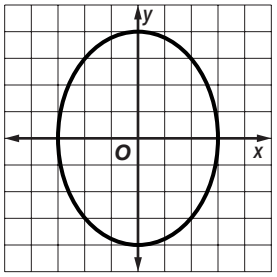
Chapter 10 Assessment Answer Key

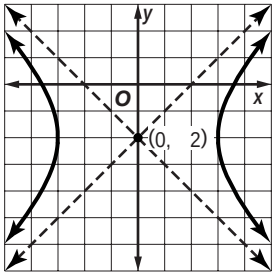
Unit 3 Test

Page 629

1. $\left(-\frac{3}{2}, 5\right)$

2. $y = -\frac{1}{4}(x - 4)^2 + 1$

3. 

4. 

5. $(x + 4)^2 + (y - \sqrt{7})^2 = 25$

6. $\frac{x^2}{49} + \frac{(y - 1)^2}{16} = 1$

7. $\frac{(x - 5)^2}{25} - \frac{(y + 2)^2}{36} = 1$; hyperbola

8. $(2\sqrt{3}, \sqrt{3})$, $(-2\sqrt{3}, -\sqrt{3})$

9. $\frac{yz^4}{16x^3}$

10. $\frac{7d - 15}{2(d - 3)(d + 3)}$

11. $(m + 1)(m - 5)(m + 7)$

12. hole: $x = 2$

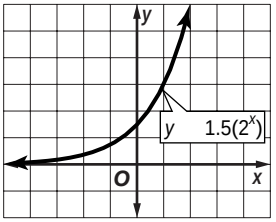
Page 630

13. 144

14. square root

15. inverse variation

16. 7

17. 

$D = \{x \mid x$ is all real numbers$\}$; $R = \{y \mid y > 0\}$

18. decay

19. $x^{5 + \pi}$

20. -1

21. 25

22. 4

23. 5.7071

24. $x \geq 1.0148$

25. 9

26. $-3x$

27. 1.544

28. $\frac{\log 19}{\log 6} \approx 1.6433$

29. \$131,635.03